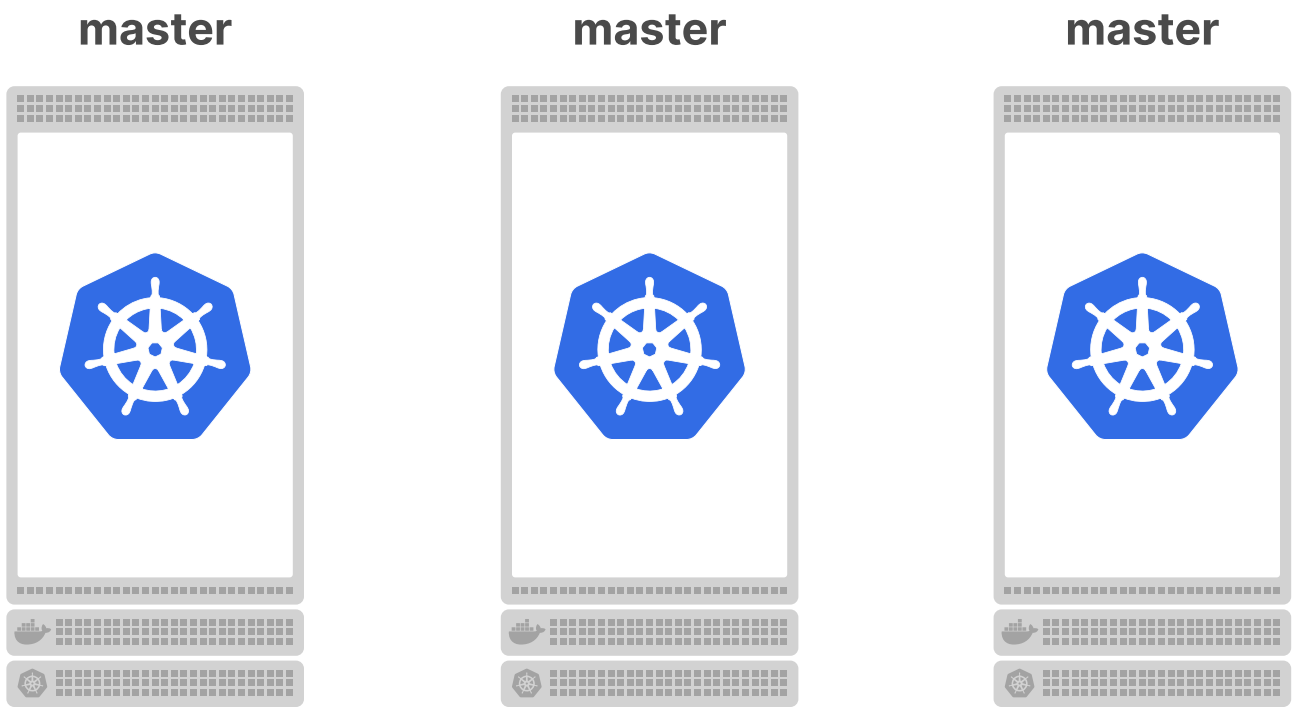
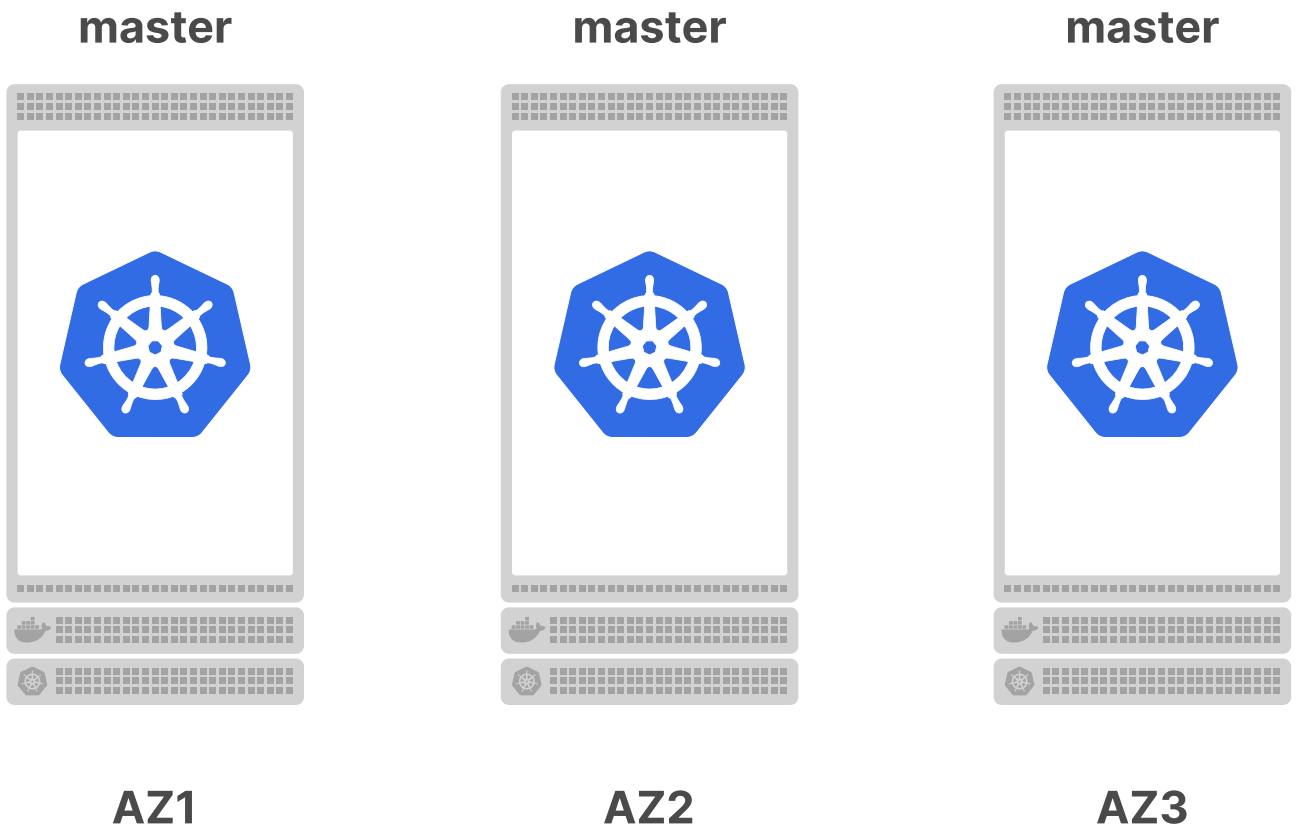
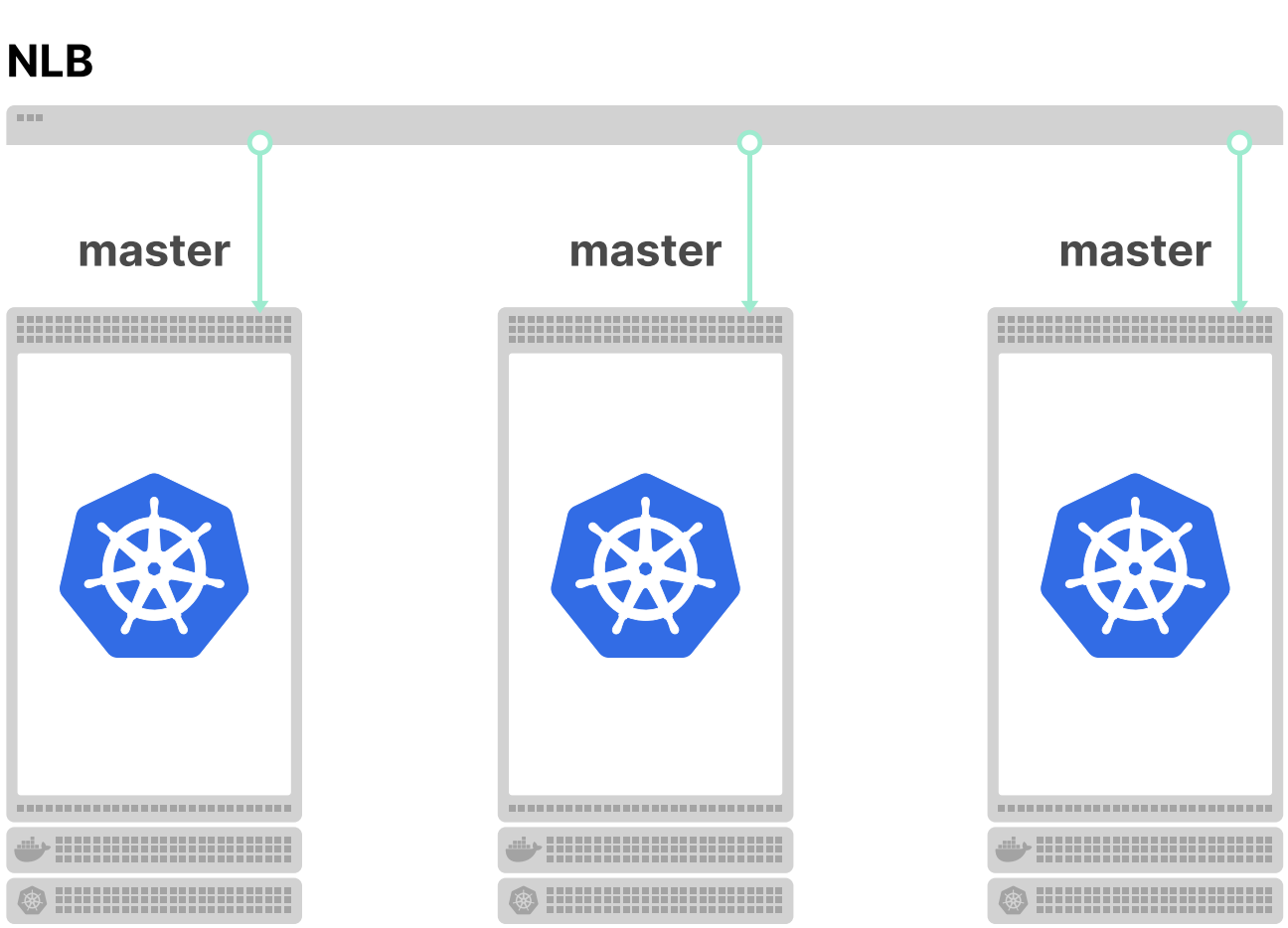

Amazon EKS

Control plane

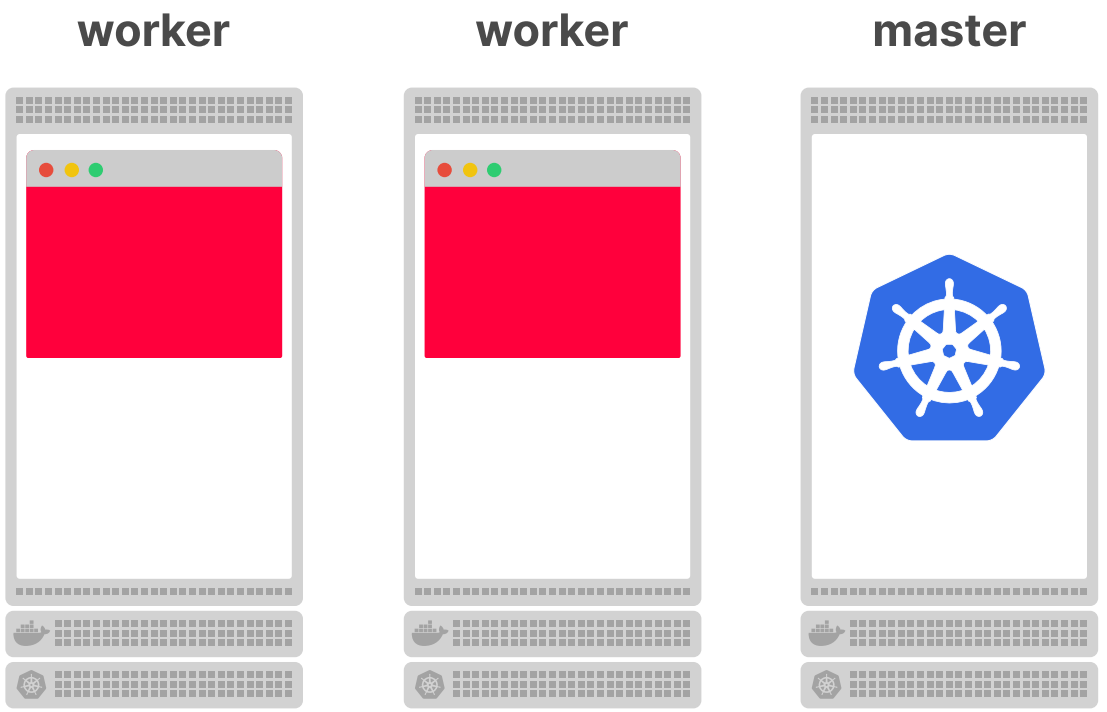

master
master
master

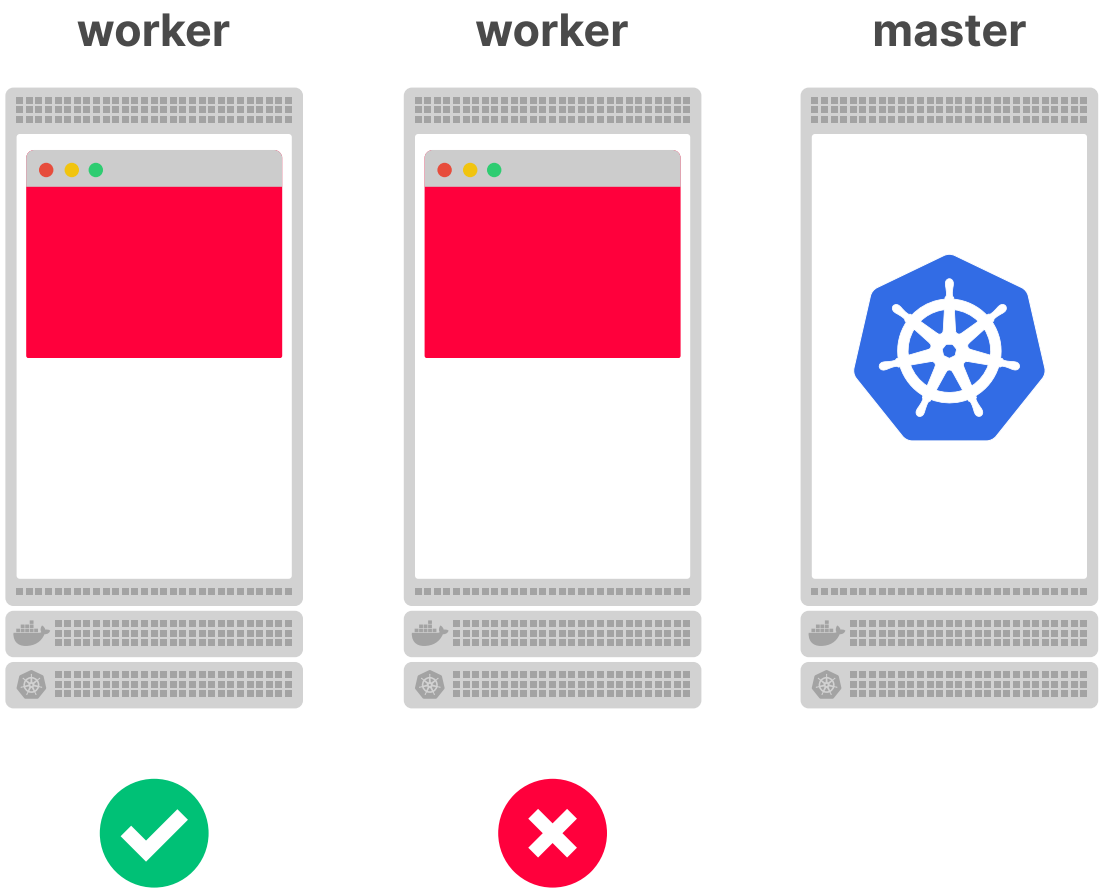

master
master
master
AZ1
AZ2
AZ3

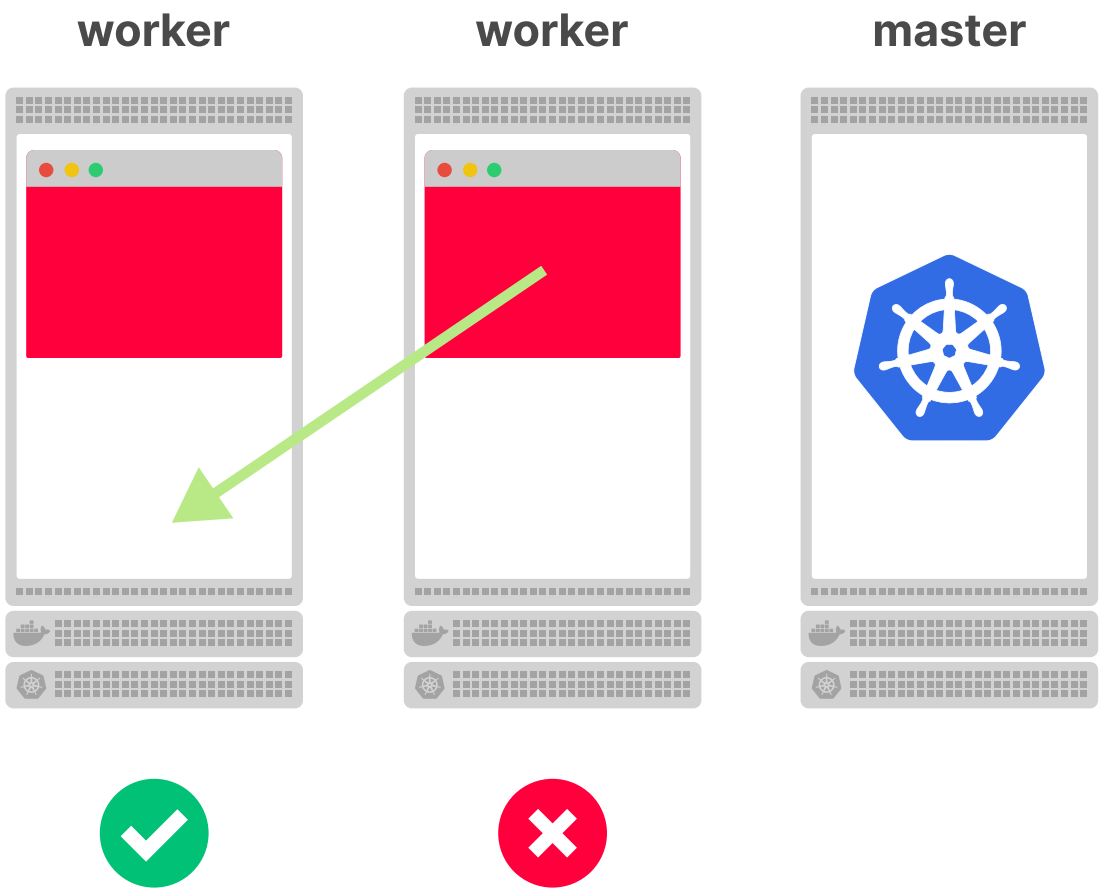

NLB
master
master
master

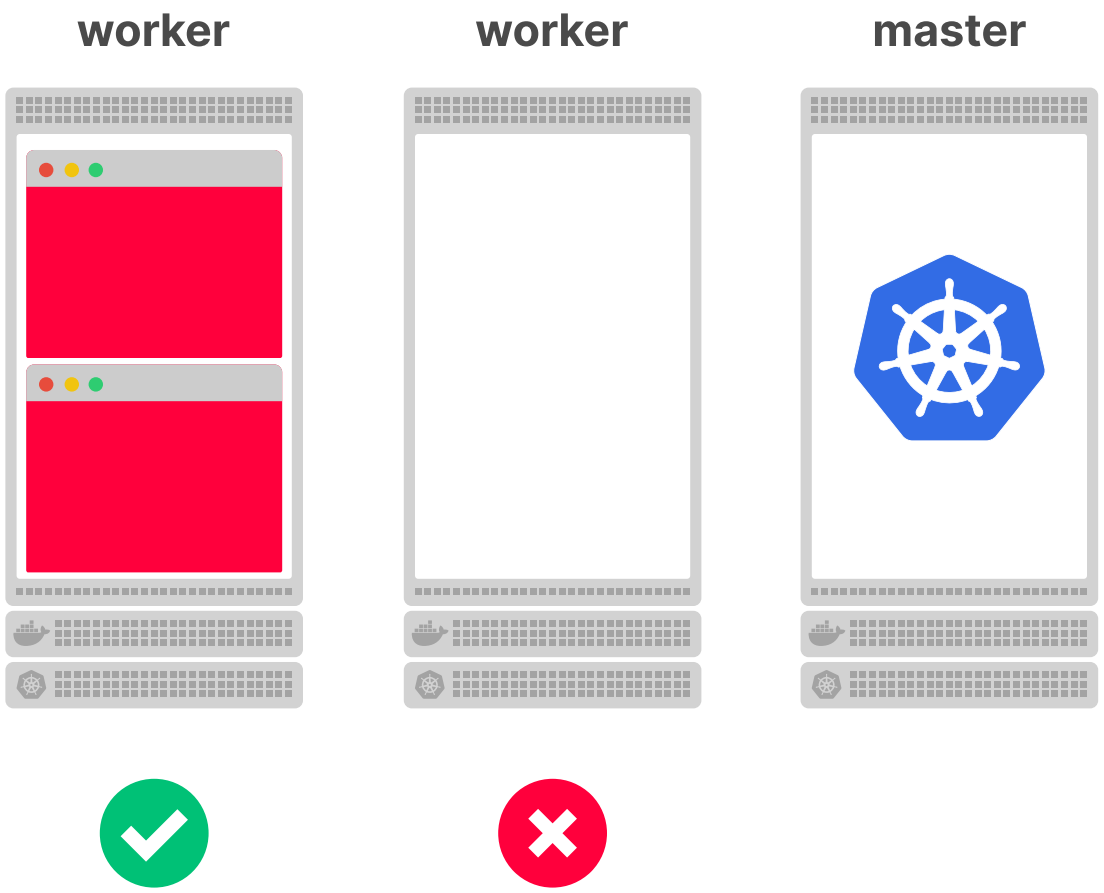
Control plane

Controller manager configuration


worker
worker
master

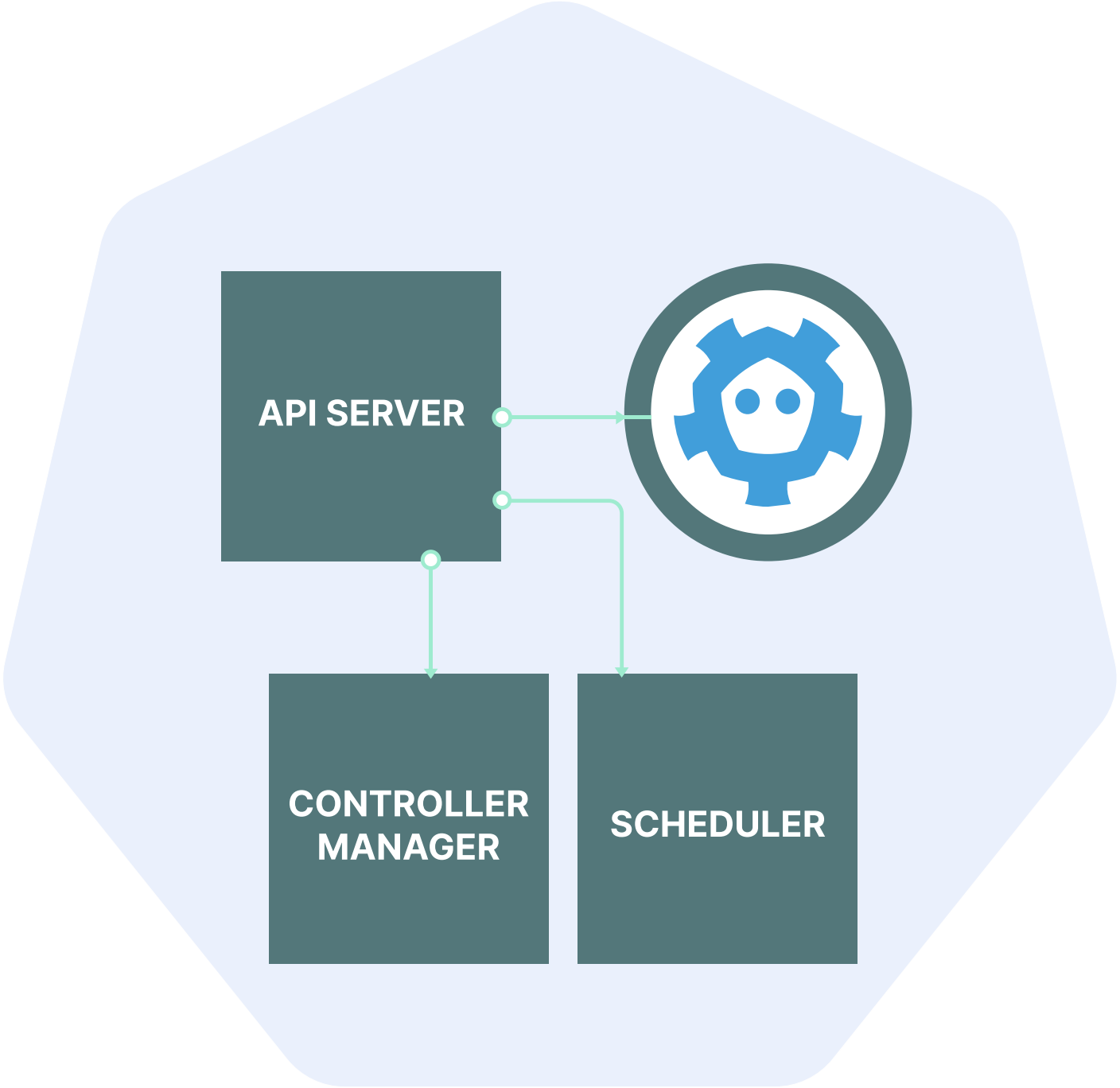

worker
worker
master


worker
worker
master


worker
worker
master

`--pod-eviction-timeout`

Controller manager configuration


API SERVER
CONTROLLER MANAGER
SCHEDULER

[EKS] [request]: Ability to configure pod-eviction-timeout #159

github.com/aws/containers-roadmap/issues/159

```
~$ aws eks update-cluster-config \
    --region <region-code> \
    --name <prod> \
    --logging '{"clusterLogging":[{"types":["audit"],"enabled":true}]}'
```

Control plane

Limits

Limits

Max number of clusters per region	100*

Limits

Max number of clusters per region	100*
Max node pools	**30**

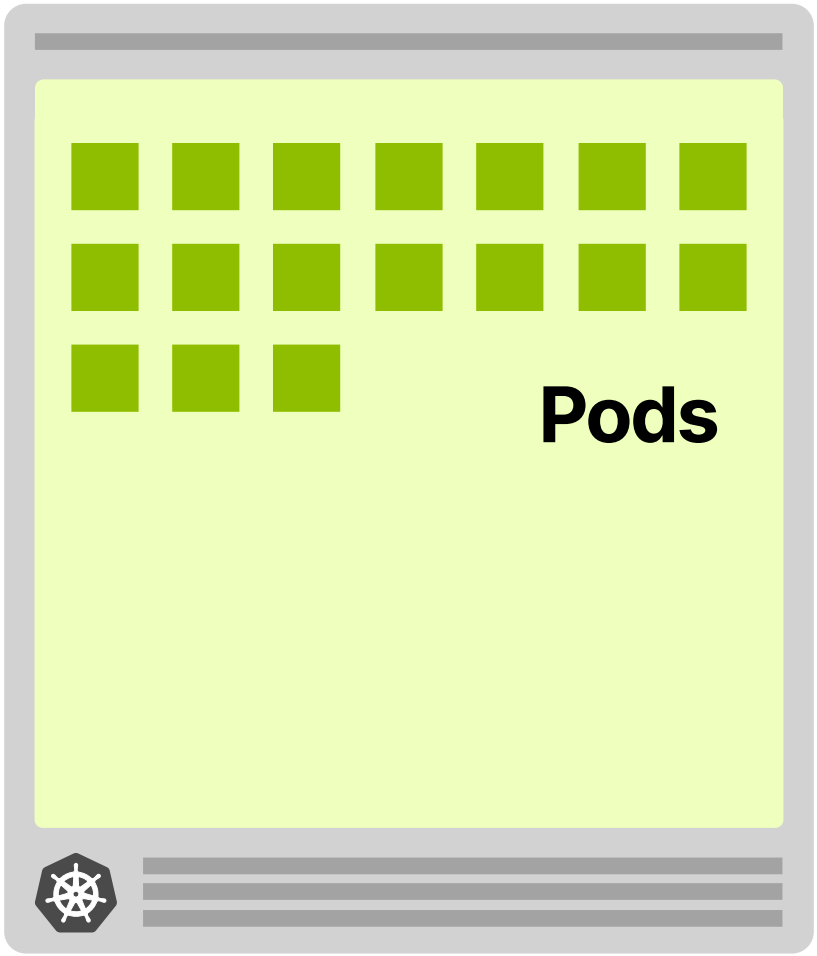
Limits

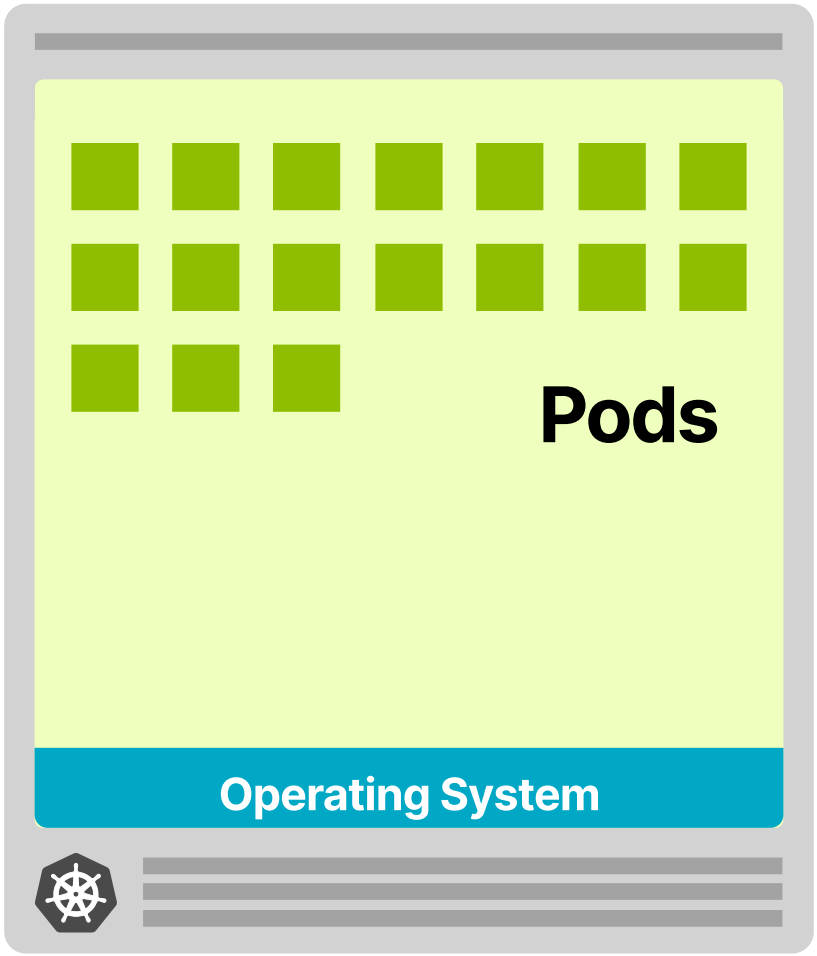
Max number of clusters per region	100*
Max node pools	30
Max nodes per node pool	**450**

Limits

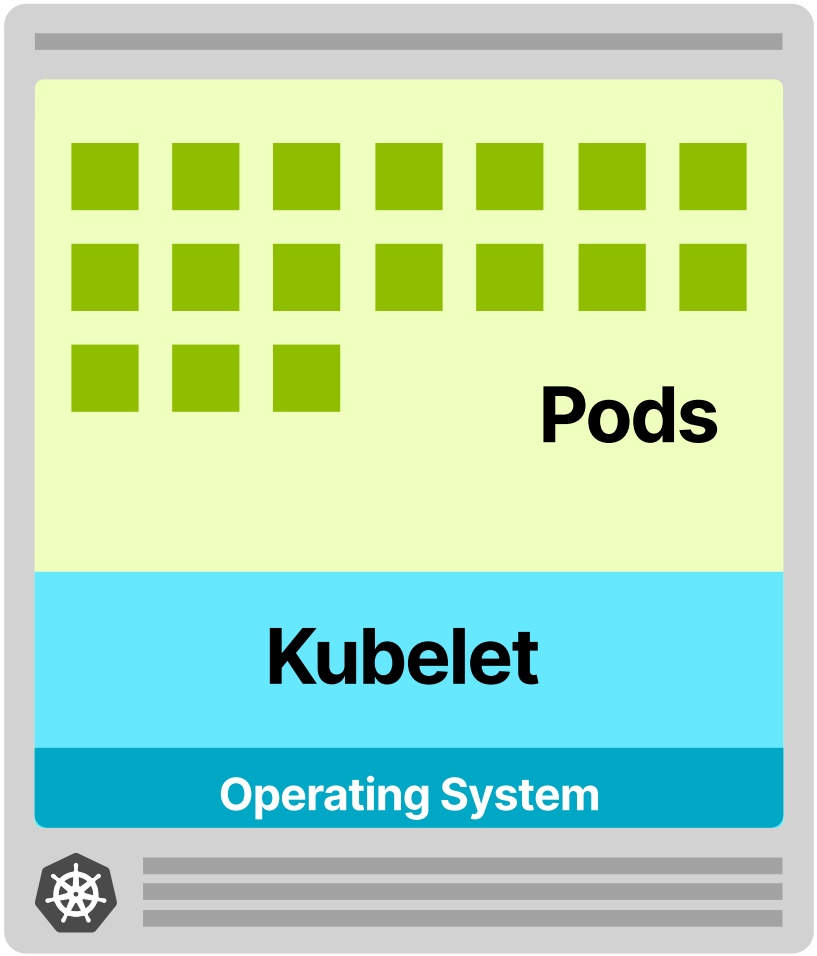
Max number of clusters per region	100*
Max node pools	30
Max nodes per node pool	450
Max nodes per cluster	**13500**

Node instances


Pods

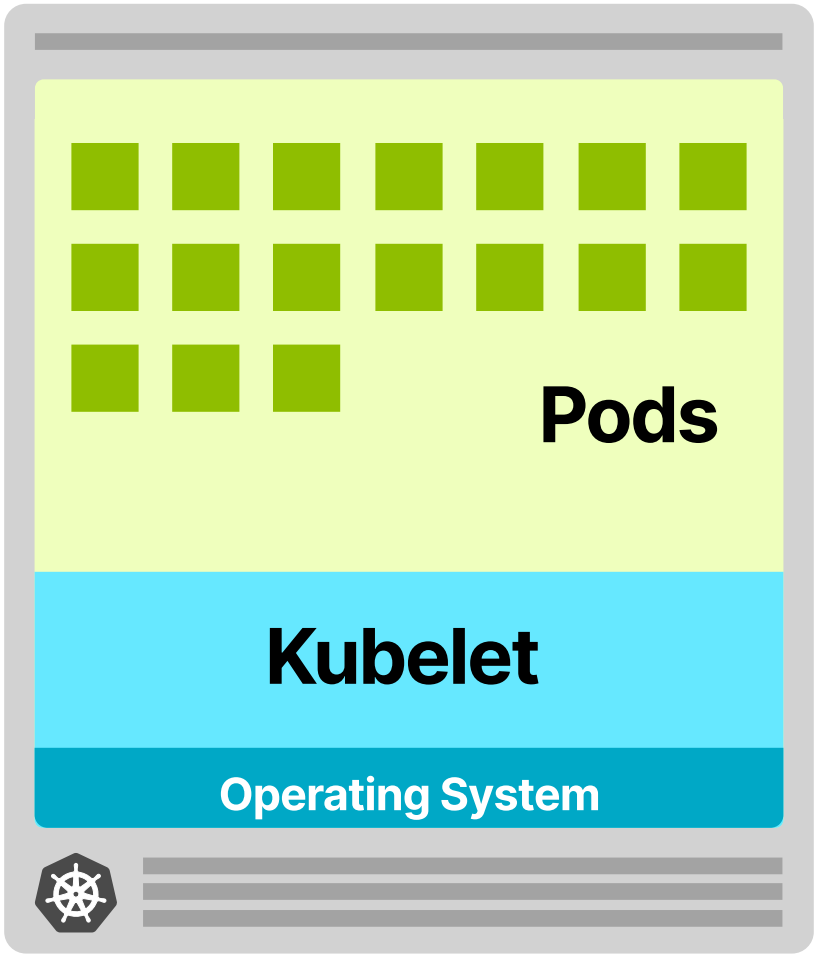


4 **Memory and CPU reserved to the OS**



3 Memory and CPU reserved to the kubelet

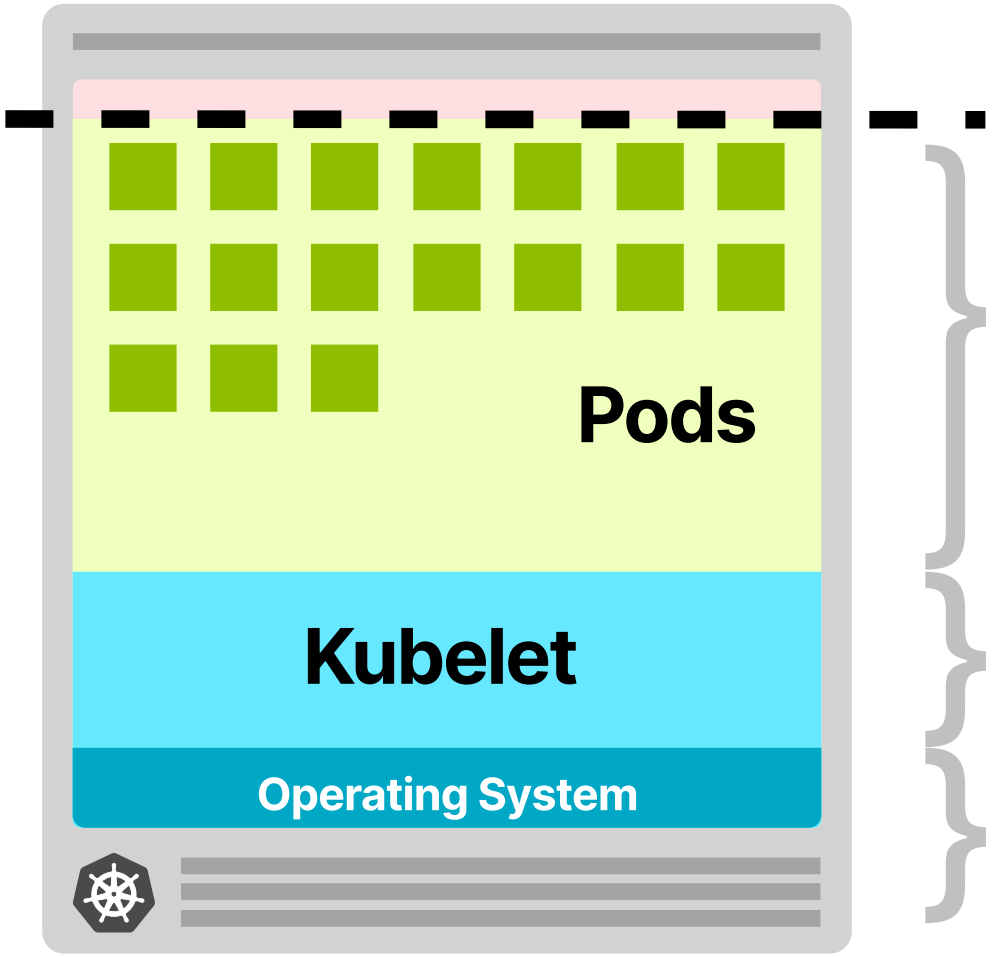
4 Memory and CPU reserved to the OS



2 Memory and CPU left to Pods

3 Memory and CPU reserved to the kubelet

4 Memory and CPU reserved to the OS



1. Eviction threshold
2. Memory and CPU left to Pods
3. Memory and CPU reserved to the kubelet
4. Memory and CPU reserved to the OS

Kubelet reserved memory

+ 255MiB

Kubelet reserved memory

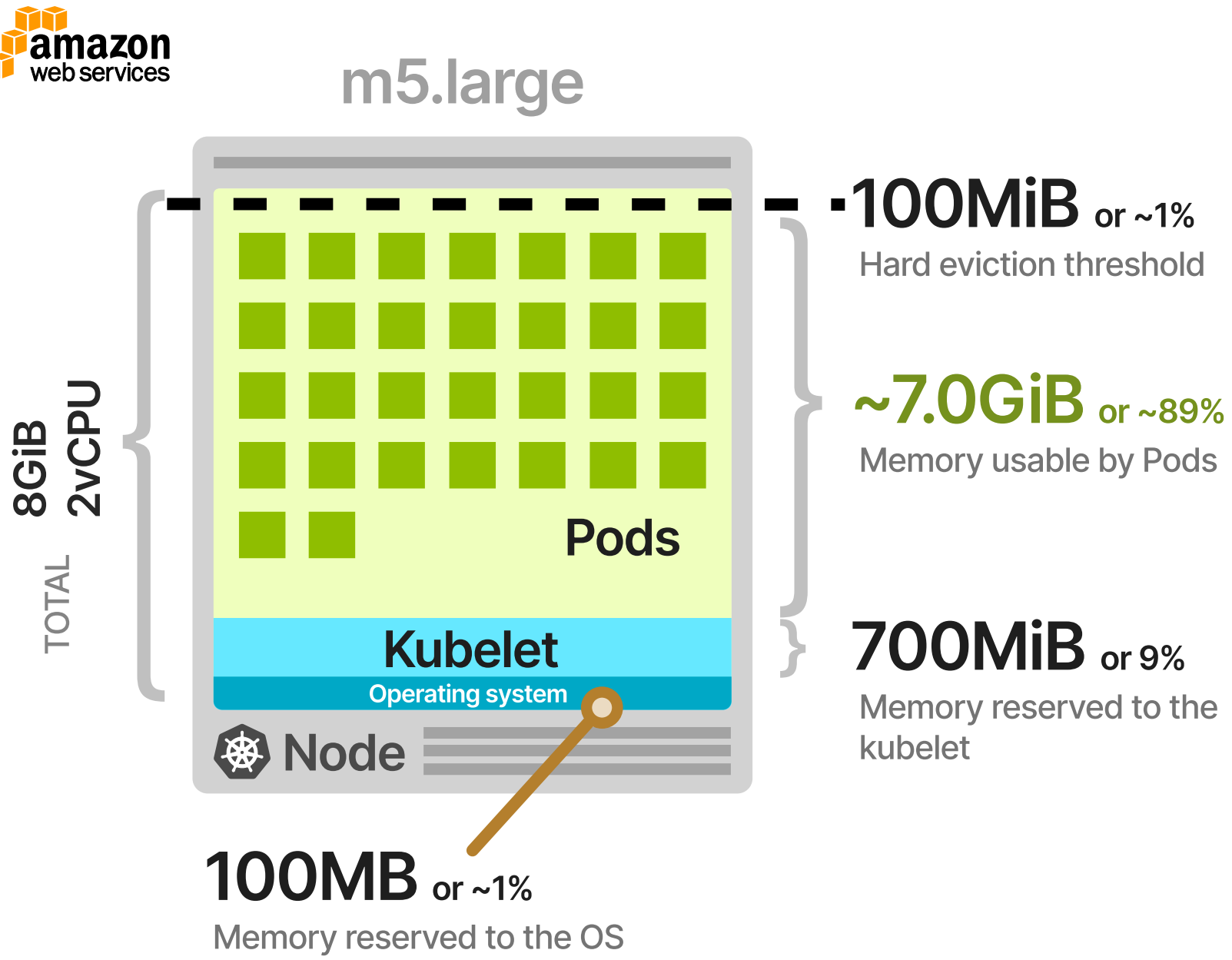
+ 255MiB
+ 11MiB * MAX_PODS

Kubelet reserved memory

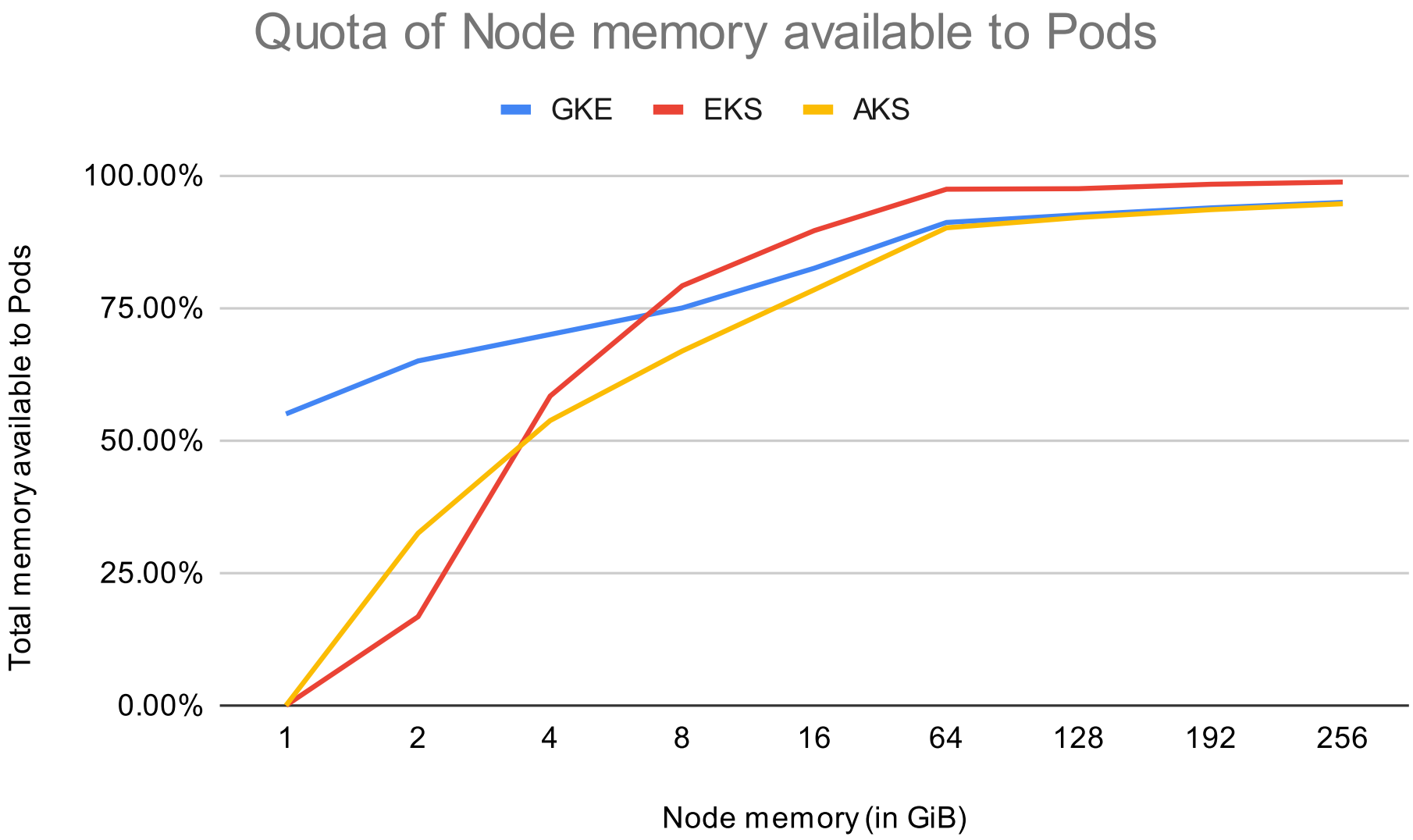
+ 255MiB
+ 11MiB * MAX_PODS

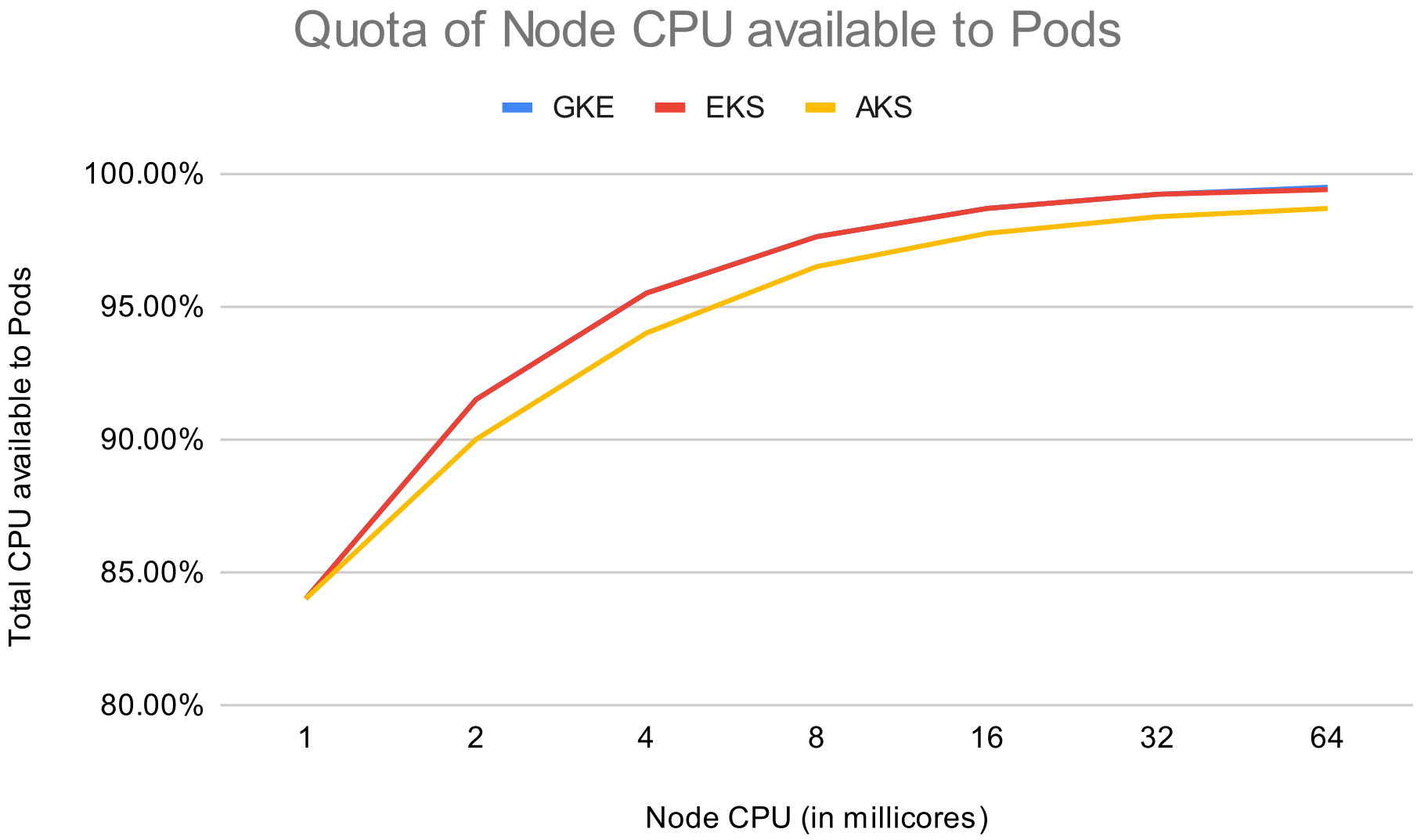
= Reserved memory

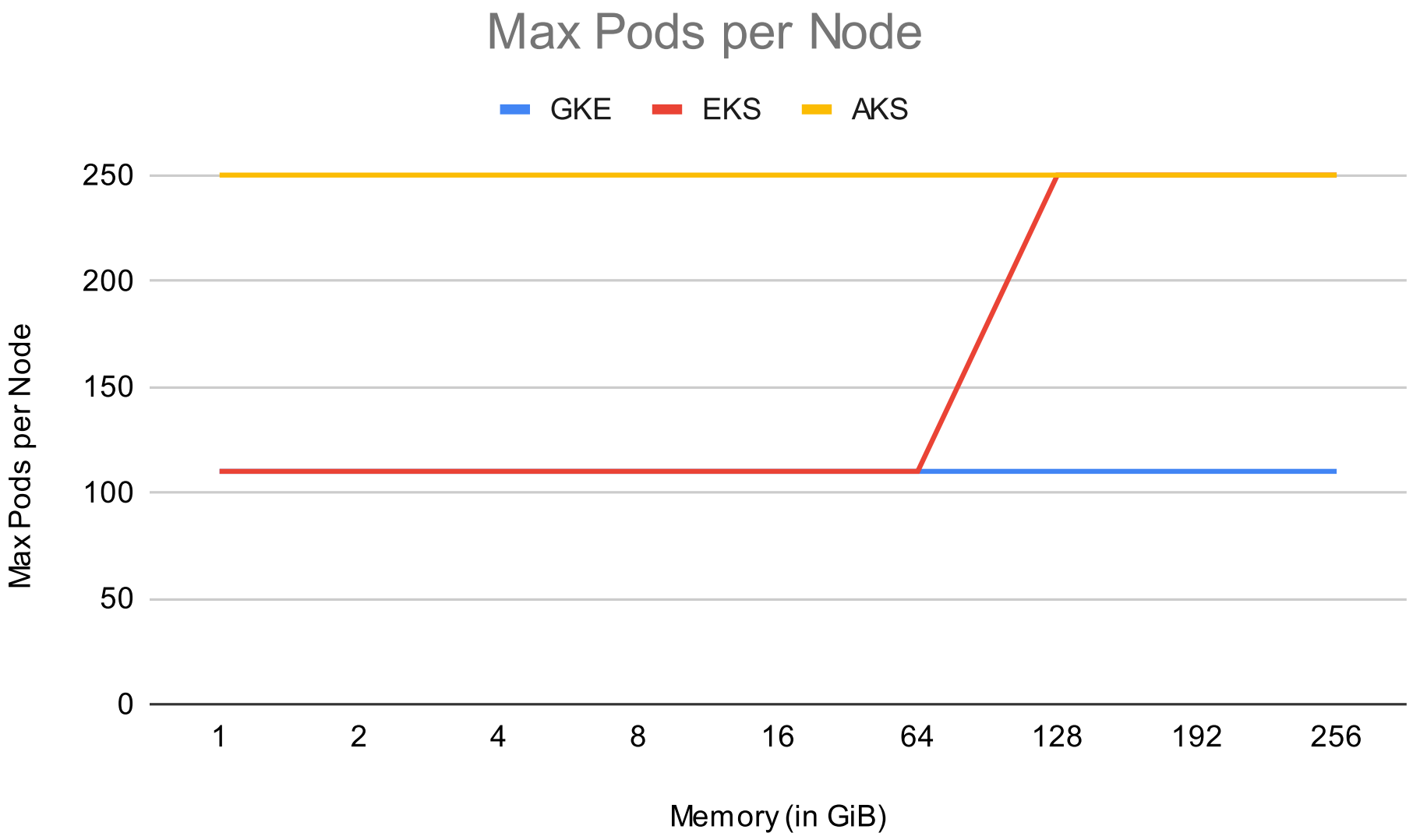
Instance type	Maximum network interfaces	Private IPv4 addresses per interface	IPv6 addresses per interface
a1.medium	2	4	4
a1.large	3	10	10
a1.xlarge	4	15	15
a1.2xlarge	4	15	15
a1.4xlarge	8	30	30
a1.metal	8	30	30
c1.medium	2	6	IPv6 not supported
c1.xlarge	4	15	IPv6 not supported
c3.large	3	10	10
c3.xlarge	4	15	15
c3.2xlarge	4	15	15
c3.4xlarge	8	30	30
c3.8xlarge	8	30	30
c4.large	3	10	10
c4.xlarge	4	15	15
c4.2xlarge	4	15	15
c4.4xlarge	8	30	30
c4.8xlarge	8	30	30
c5.large	3	10	10
c5.xlarge	4	15	15

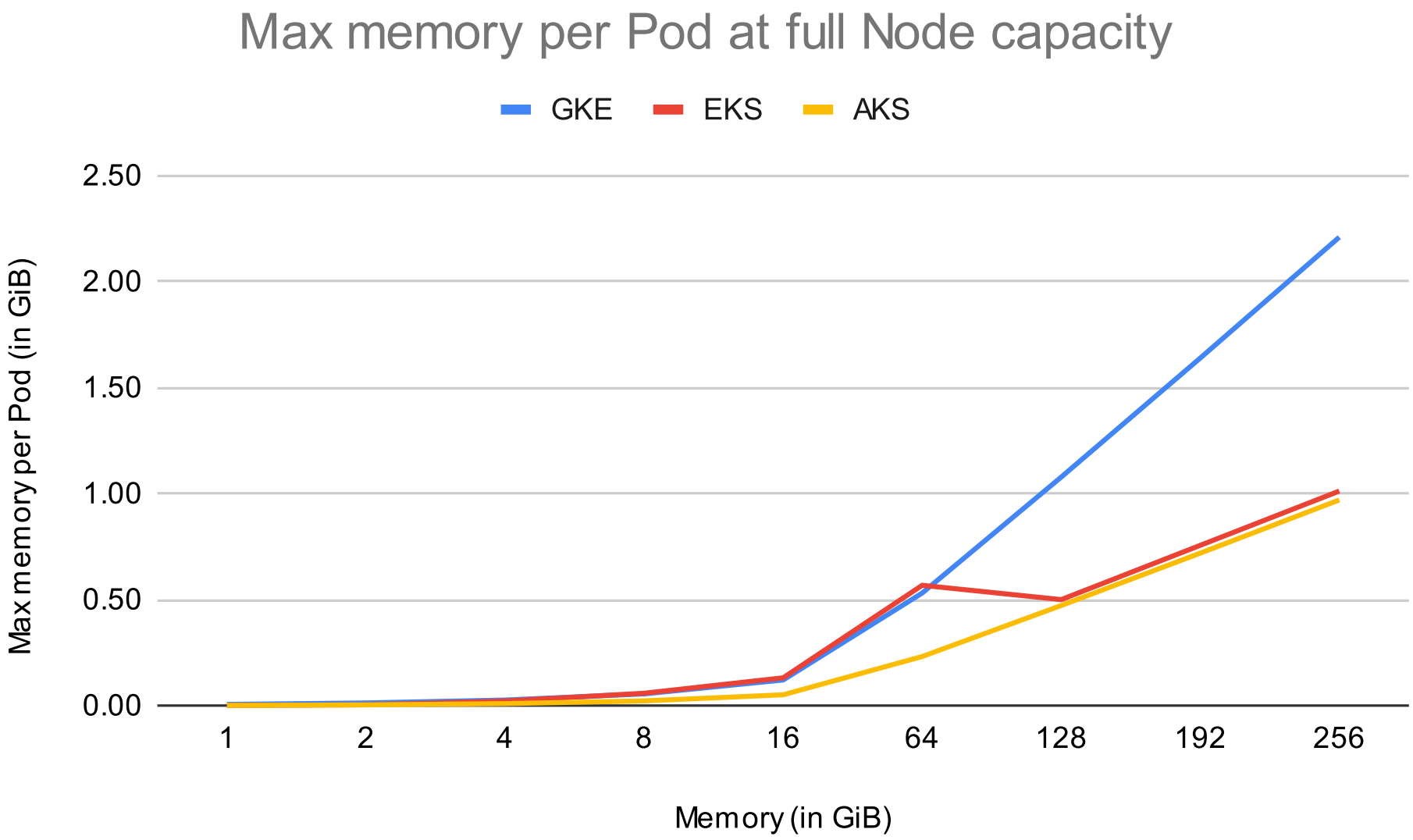

amazon
web services
m5.large
8GiB
2vCPU
TOTAL
Pods
Kubelet
Operating system
Node
100MiB or ~1%
Hard eviction threshold
~7.0GiB or ~89%
Memory usable by Pods
700MiB or 9%
Memory reserved to the kubelet
100MB or ~1%
Memory reserved to the OS

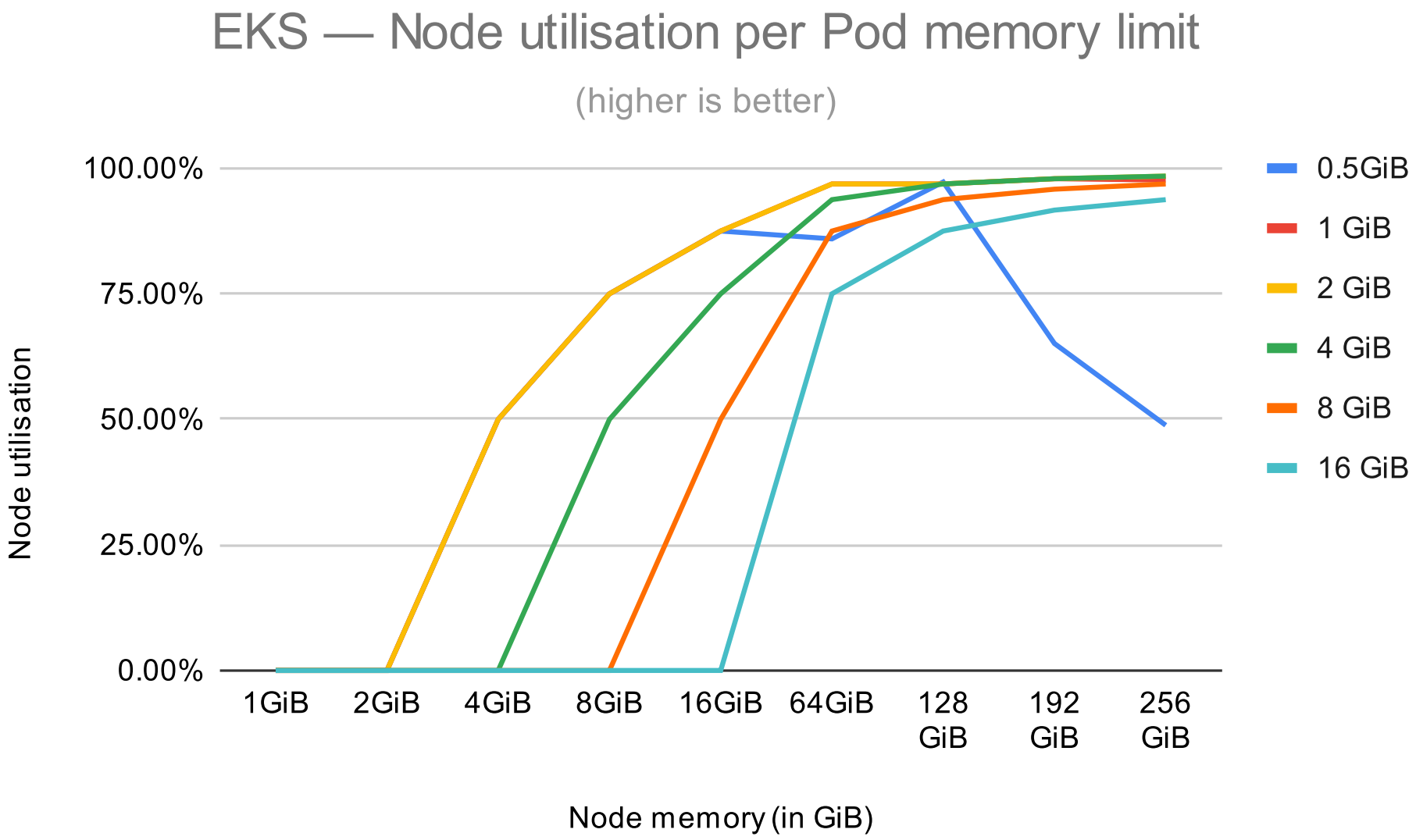
Node instances utilization


Quota of Node memory available to Pods
GKE
EKS
AKS
100.00%
75.00%
50.00%
25.00%
0.00%
Total memory available to Pods
1
2
4
8
16
64
128
192
256
Node memory (in GiB)


Quota of Node CPU available to Pods
GKE
EKS
AKS
100.00%
95.00%
90.00%
85.00%
80.00%
Total CPU available to Pods
1
2
4
8
16
32
64
Node CPU (in millicores)


Max Pods per Node
GKE
EKS
AKS
250
200
150
100
50
0
Max Pods per Node
1
2
4
8
16
64
128
192
256
Memory (in GiB)


Max memory per Pod at full Node capacity
GKE
EKS
AKS
2.50
2.00
1.50
1.00
0.50
0.00
Max memory per Pod (in GiB)
1
2
4
8
16
64
128
192
256
Memory (in GiB)


EKS — Node utilisation per Pod memory limit
(higher is better)
100.00%
75.00%
50.00%
25.00%
0.00%
Node utilisation
1GiB
2GiB
4GiB
8GiB
16GiB
64GiB
128 GiB
192 GiB
256 GiB
Node memory (in GiB)
0.5GiB
1 GiB
2 GiB
4 GiB
8 GiB
16 GiB

Worker nodes

Worker nodes

Self managed ASGs

Managed groups

Fargate

Worker nodes

Self managed ASGs

Managed groups

Fargate

Worker nodes

Self managed ASGs

Managed groups

Fargate

Worker nodes

1. Self managed ASGs

Self managed ASGs

SecurityGroups & IAM

Kubelet

AMIs

Bootstrap

Self managed ASGs

SecurityGroups & IAM

Kubelet

AMIs

Bootstrap

Self managed ASGs

SecurityGroups & IAM

Kubelet

AMIs

Bootstrap

Self managed ASGs

SecurityGroups & IAM

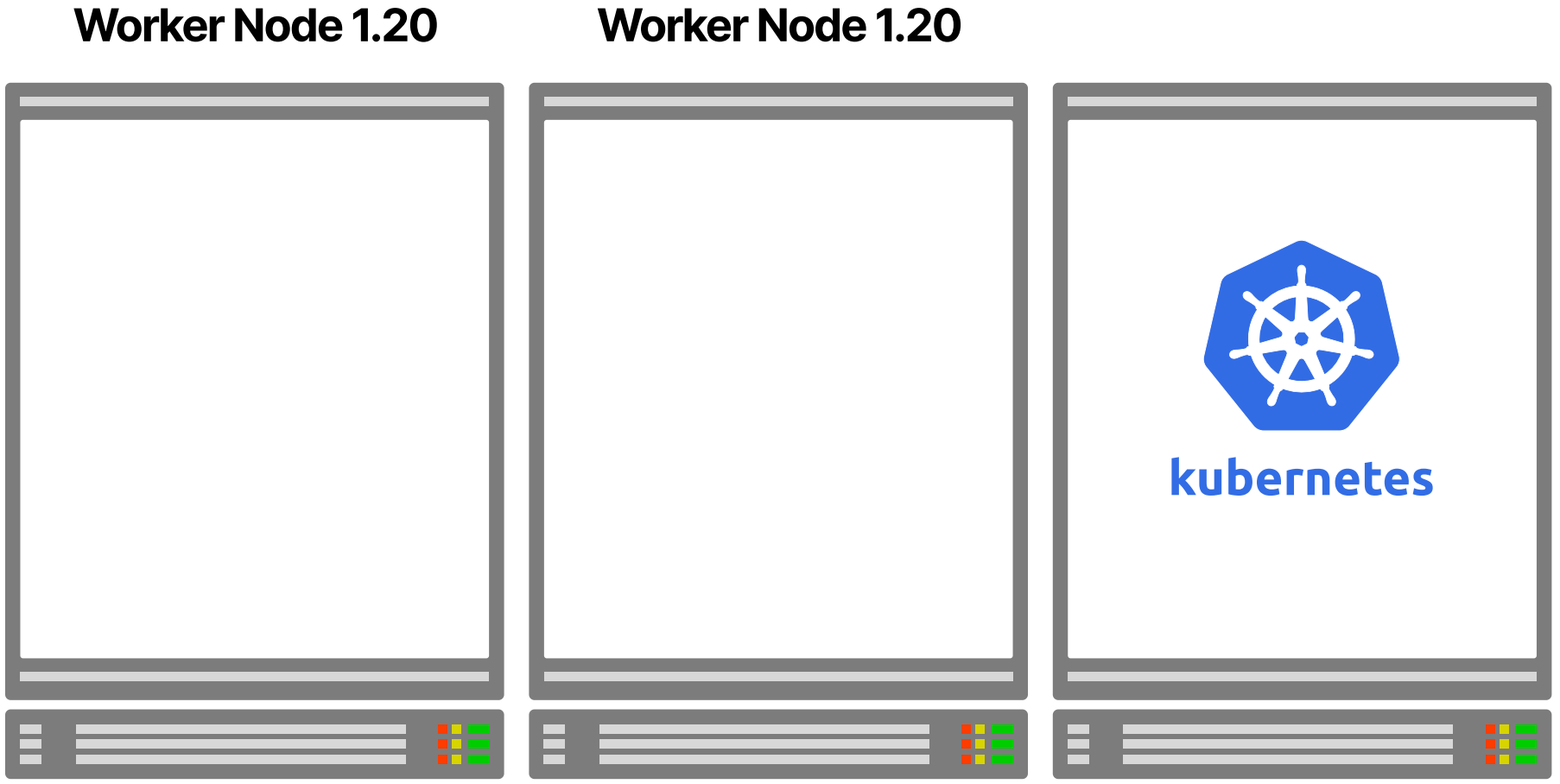
Kubelet

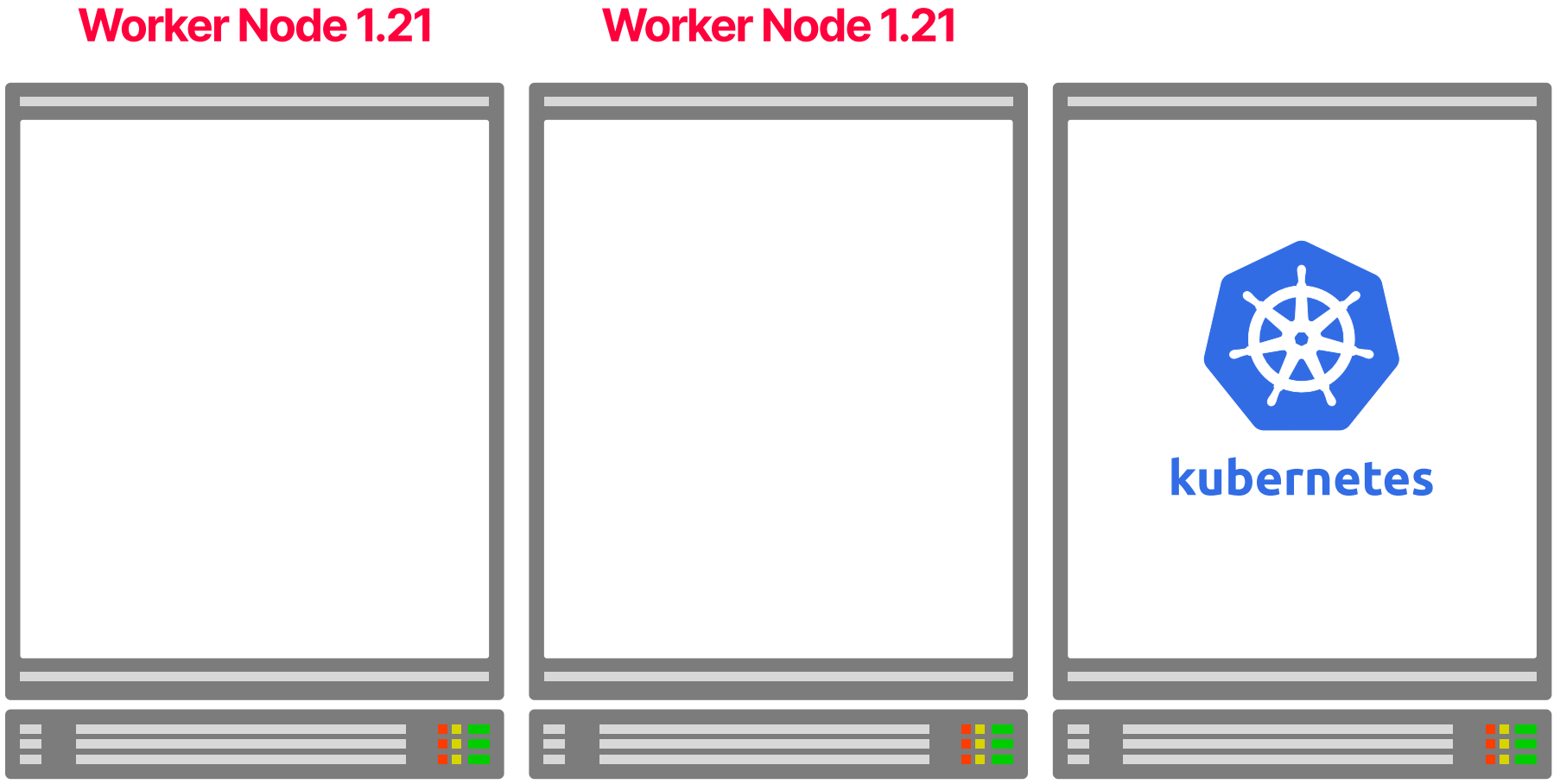
AMIs

Bootstrap

Worker nodes

1. Self managed ASGs — operations


Worker Node 1.20
Worker Node 1.20
kubernetes


Worker Node 1.21
Worker Node 1.21
kubernetes

Cordon the node

Remove the node

Add the new node

Uncordon

Self managed ASGs — operations

Cordon the node

Remove the node

Add the new node

Uncordon

Self managed ASGs — operations

Cordon the node

Remove the node

Add the new node

Uncordon

Self managed ASGs — operations

Cordon the node

Remove the node

Add the new node

Uncordon

Worker nodes

2. Managed groups

Managed nodes

Amazon ASGs / EC2

Standard bootstrap

Standard AMI / SGs

Standard Kubelet config

Managed nodes

Amazon ASGs / EC2

Standard bootstrap

Standard AMI / SGs

Standard Kubelet config

Managed nodes

Amazon ASGs / EC2

Standard bootstrap

Standard AMI / SGs

Standard Kubelet config

Managed nodes

Amazon ASGs / EC2

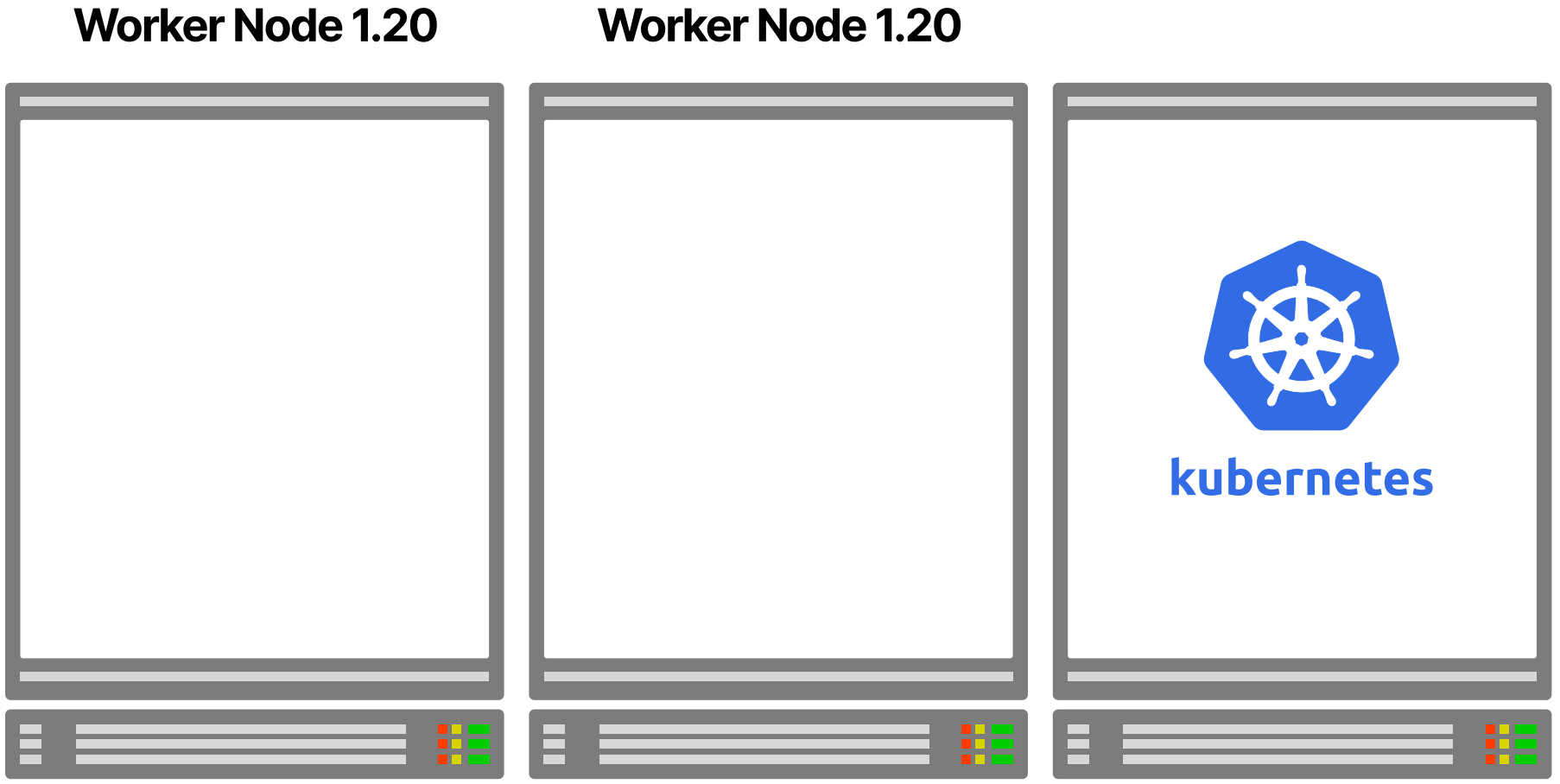
Standard bootstrap

Standard AMI / SGs

Standard Kubelet config

Managed nodes

2. Managed groups — operations


Worker Node 1.20
Worker Node 1.20
kubernetes

```
~$ eksctl upgrade nodegroup \
  --name=nodegroup \
  --cluster=eksworkshop-eksctl \
  --kubernetes-version=1.21


~$ _
```

Worker Node 1.20

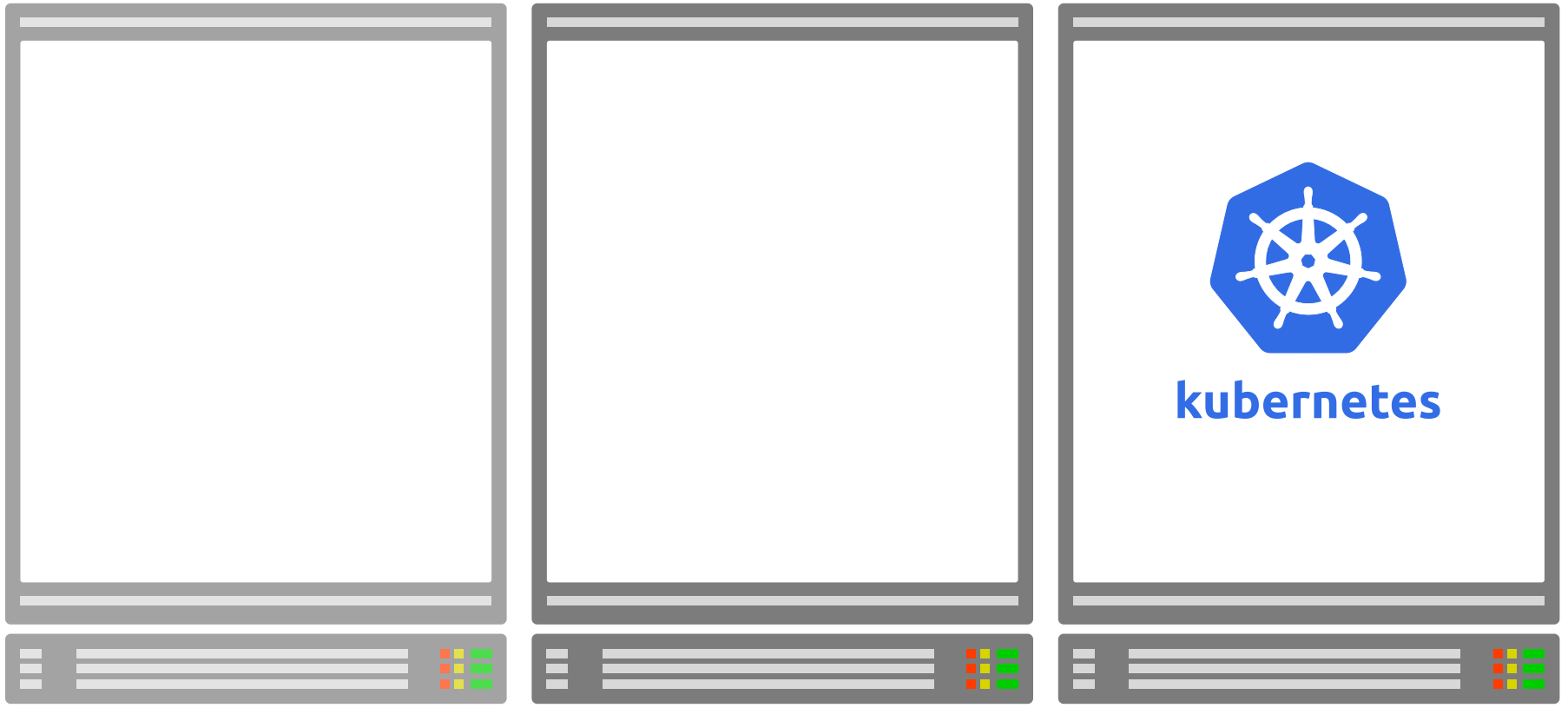
Worker Node 1.20



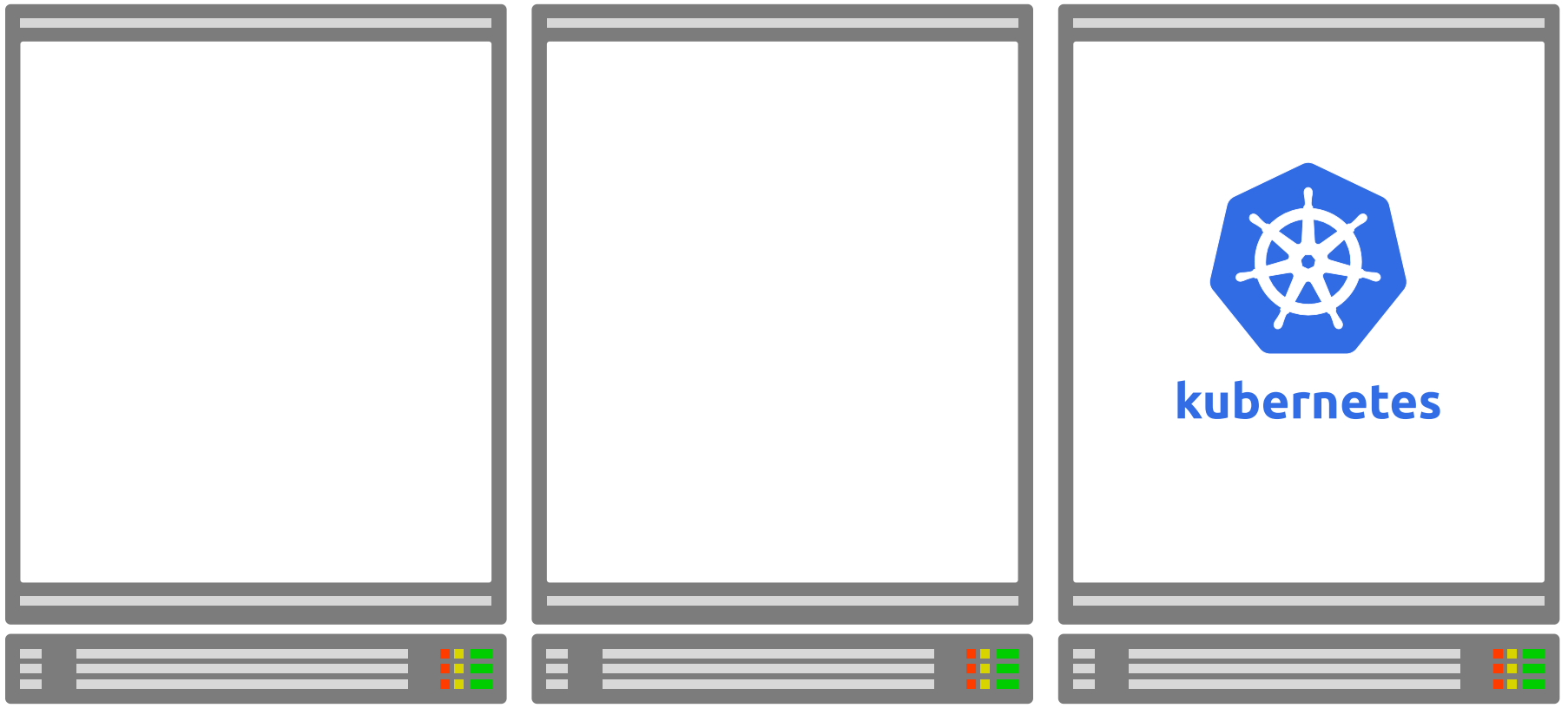
CORDON

Worker Node 1.21

Worker Node 1.20

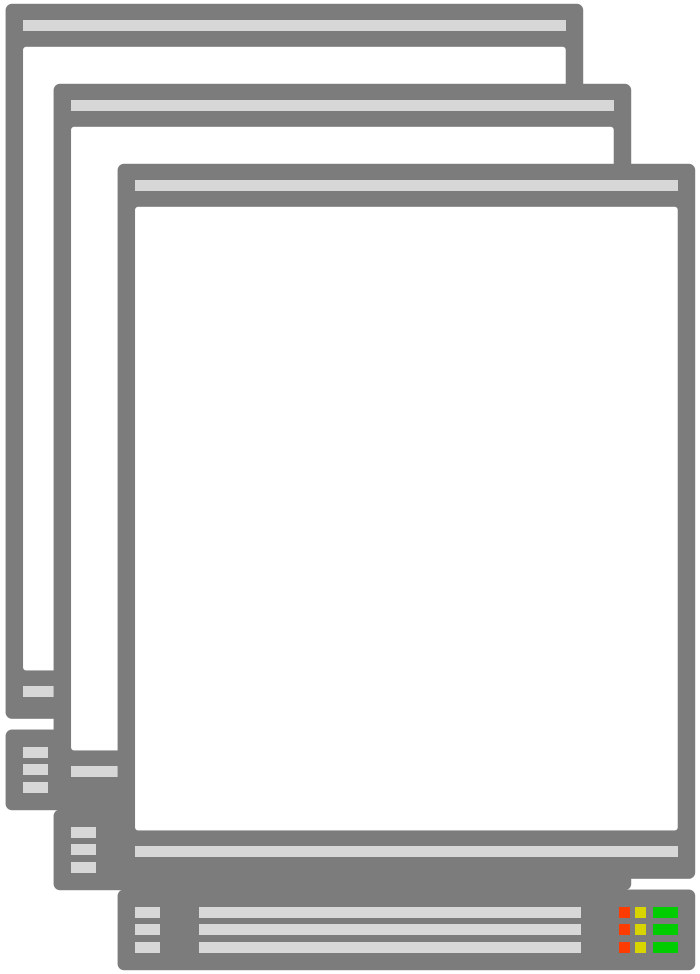


UNCORDON

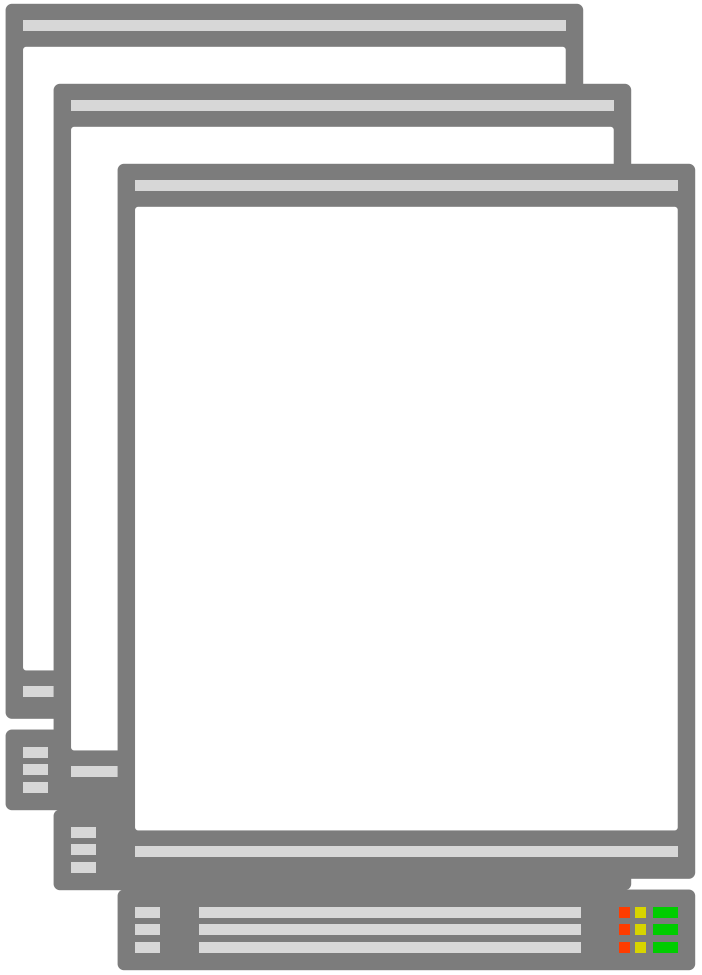

Worker Node 1.21
Worker Node 1.20
kubernetes

Managed nodes

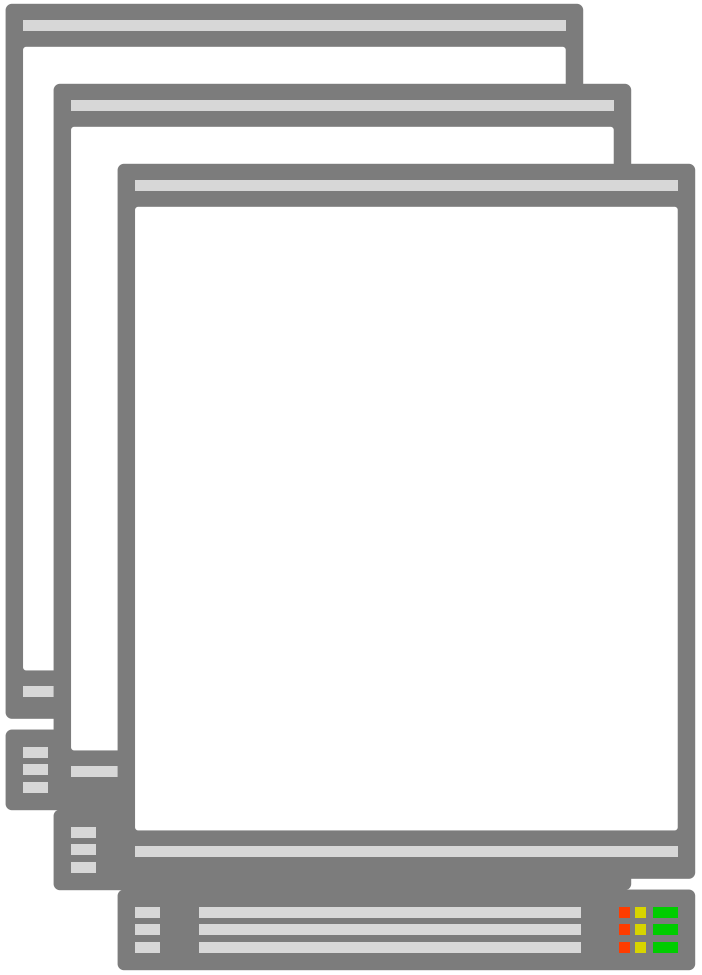
2. Managed groups — launch templates



Custom AMI



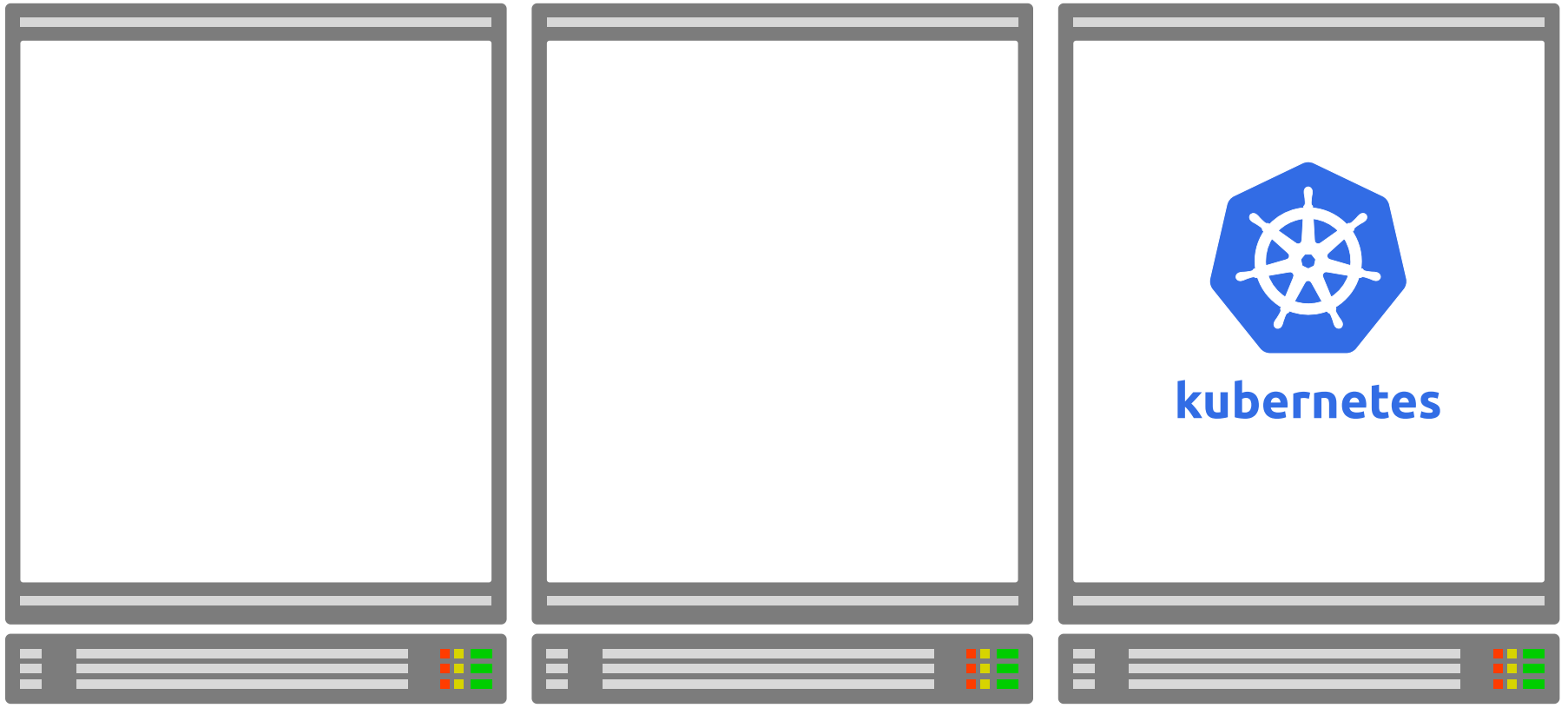
Custom AMI
Custom SGs

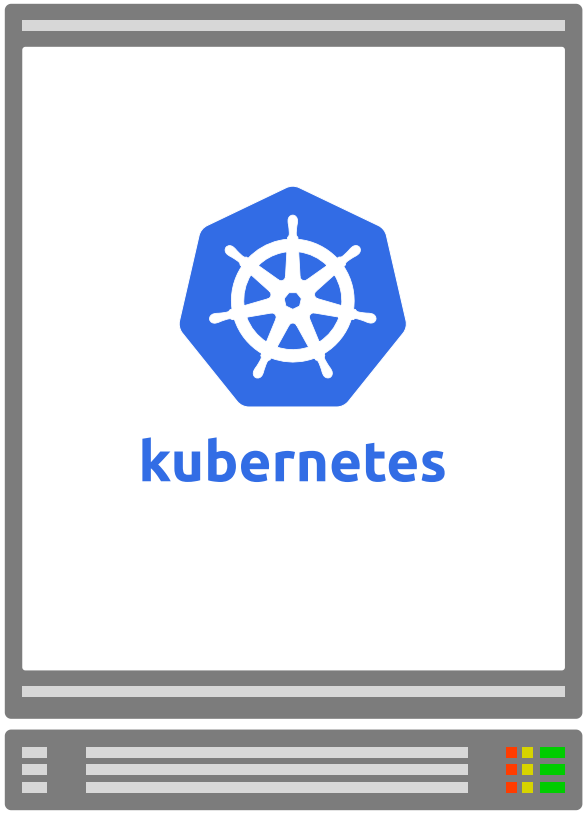


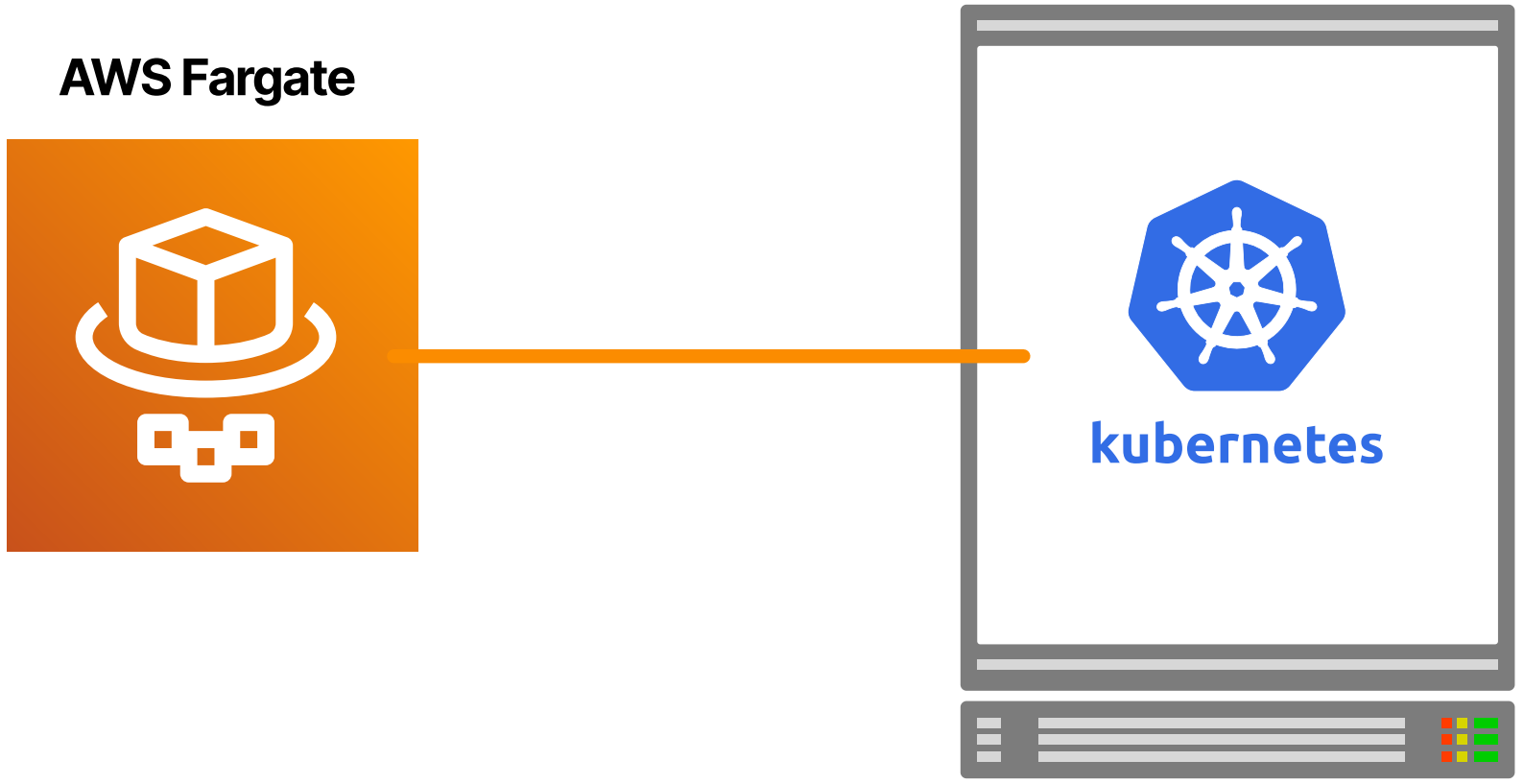
Custom AMI
Custom SGs
Custom User data

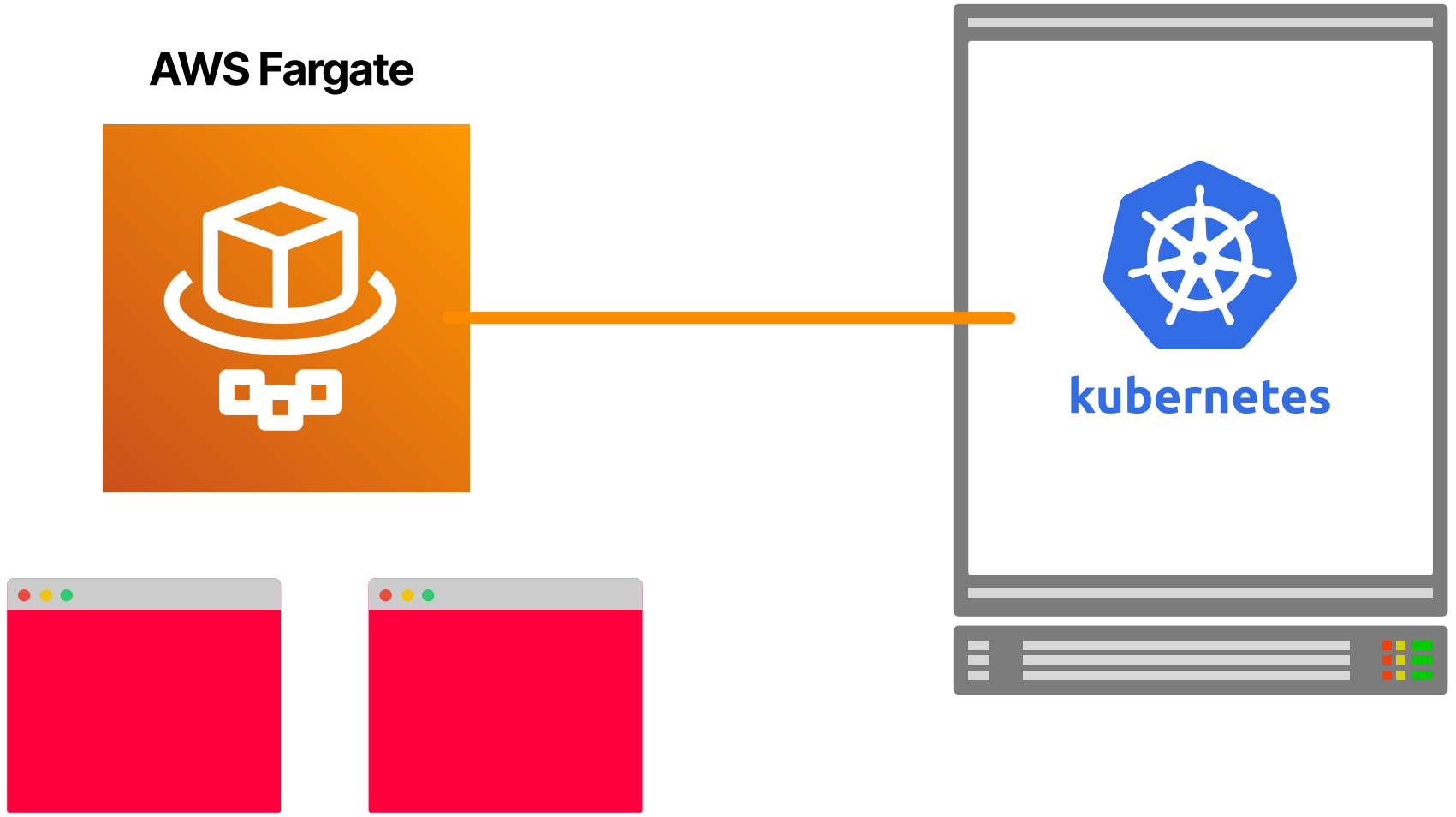
Worker nodes

3. Fargate


kubernetes

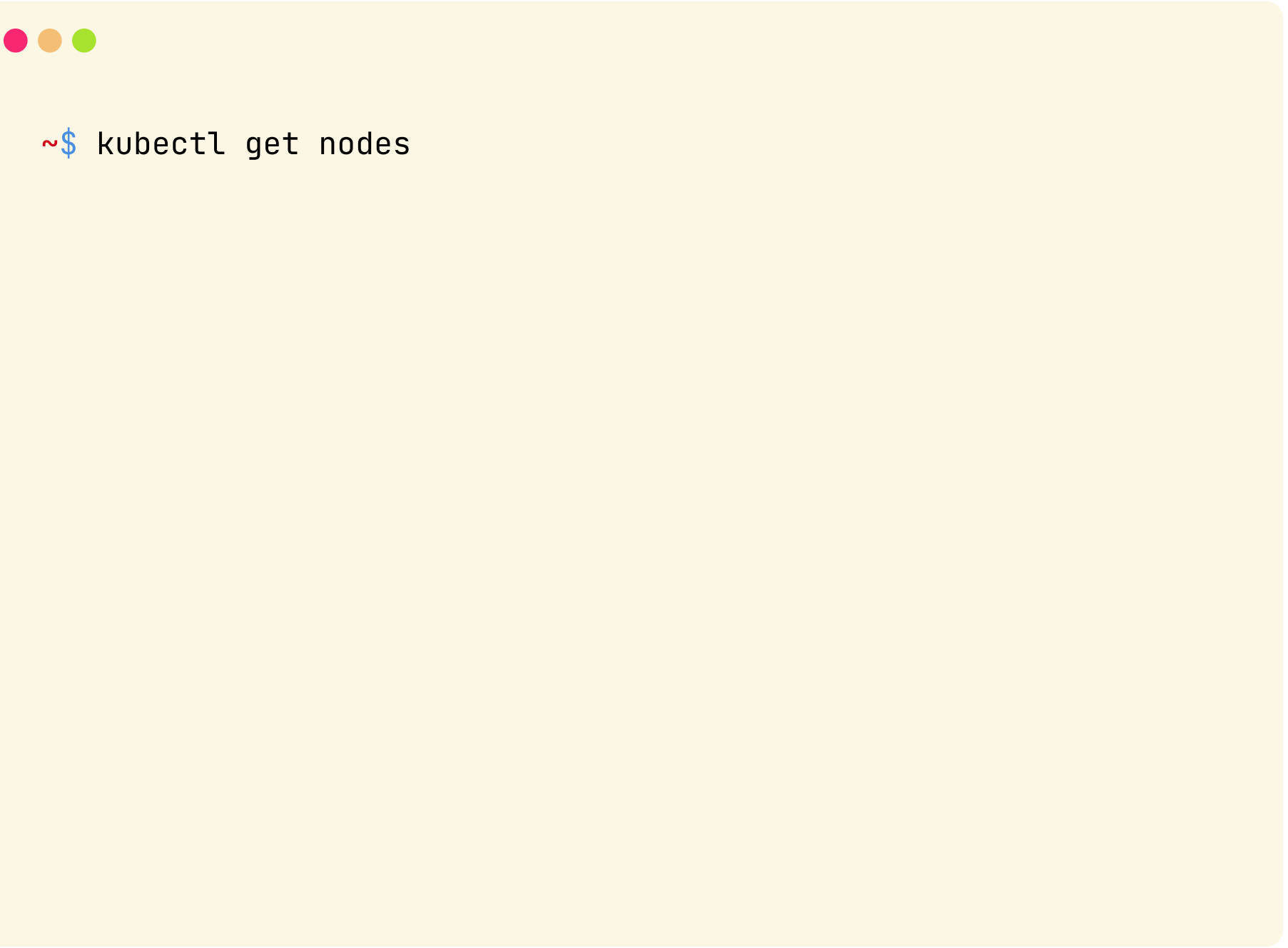

kubernetes


AWS Fargate
kubernetes


AWS Fargate
kubernetes

Worker nodes

3. Fargate — severs in serverless

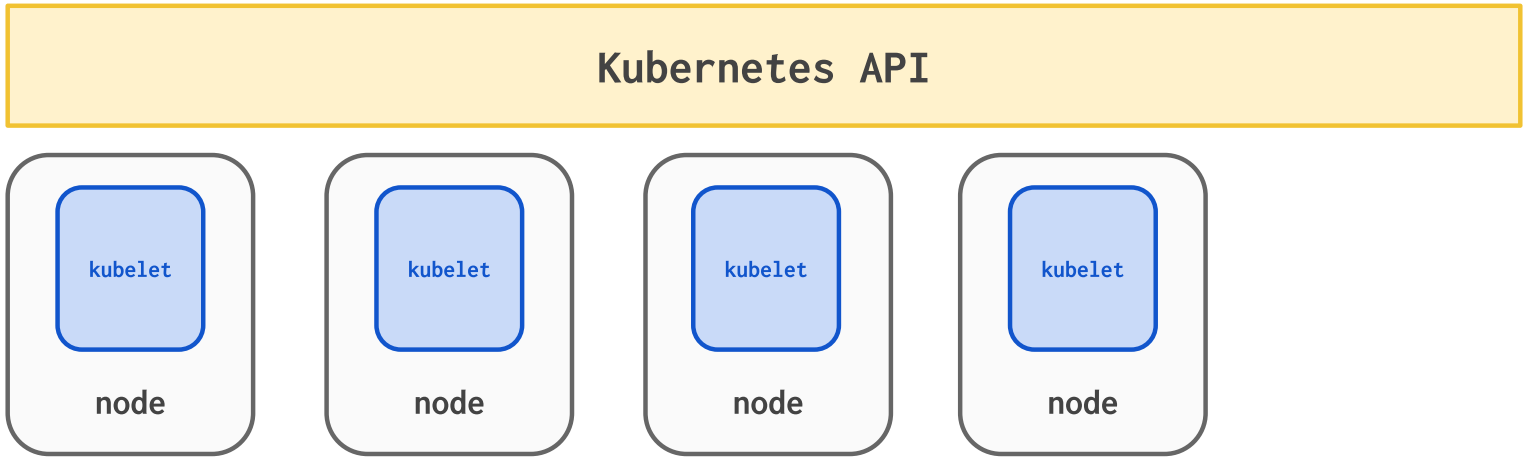

~$ kubectl get nodes

```
~$ kubectl get nodes

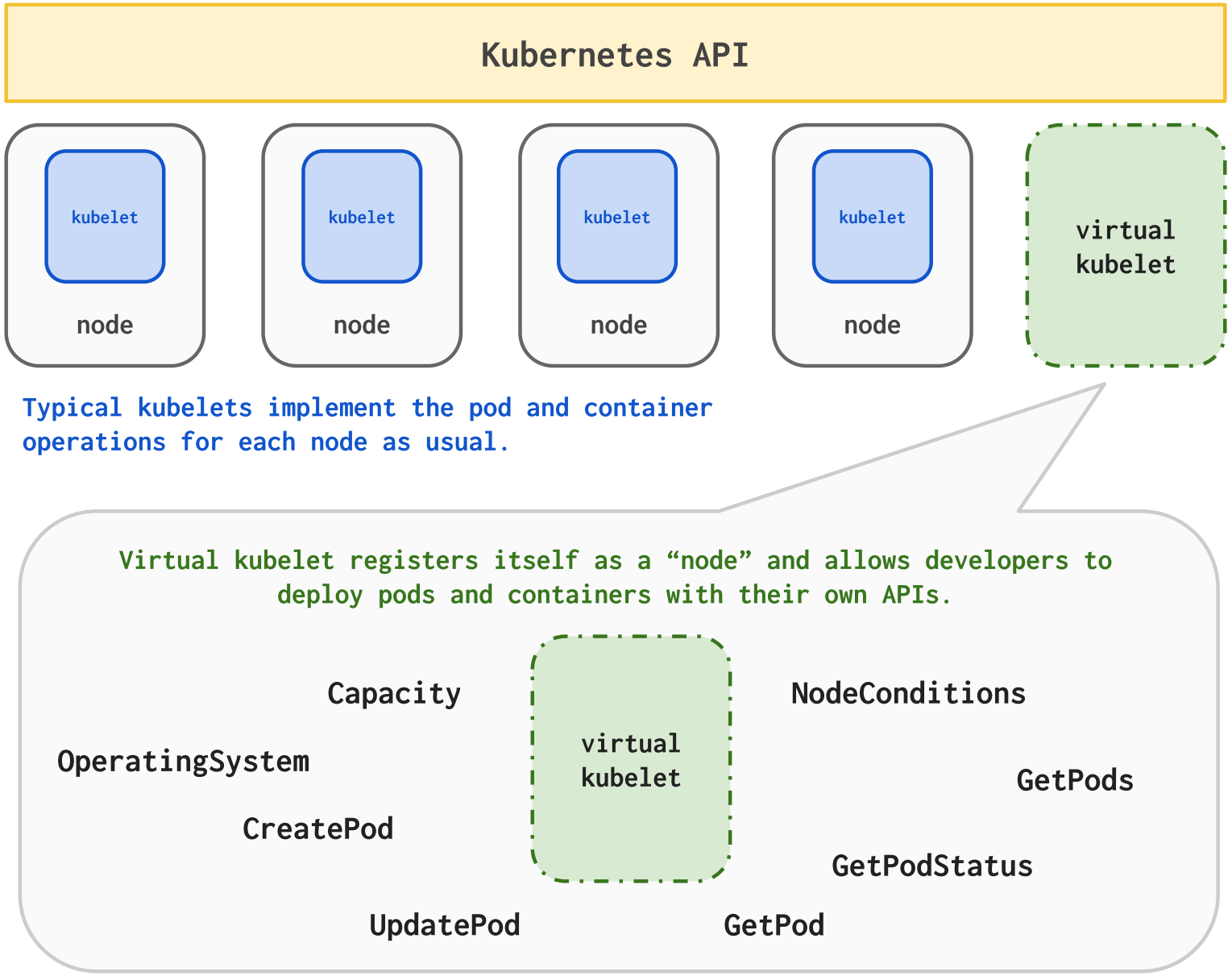
NAME                                          STATUS   VERSION
fargate-ip-19...7.compute.internal   Ready    v1.19.6-eks-e91815
fargate-ip-19...8.compute.internal   Ready    v1.19.6-eks-e91815
```

virtual-kubelet/virtual-kubelet

Fargate — severs in serverless



Typical kubelets implement the pod and container operations for each node as usual.


Kubernetes API
kubelet
node
kubelet
node
kubelet
node
kubelet
node
virtual kubelet
Typical kubelets implement the pod and container operations for each node as usual.
Virtual kubelet registers itself as a “node” and allows developers to deploy pods and containers with their own APIs.
Capacity
NodeConditions
virtual kubelet
OperatingSystem
GetPods
CreatePod
GetPodStatus
UpdatePod
GetPod

Worker nodes

3. Fargate — Limitations

Fargate — Limitations

No NodePort / LoadBalancer

No DaemonSet

Only EFS

Random EC2

Fargate — Limitations

No NodePort / LoadBalancer

No DaemonSet

Only EFS

Random EC2

Fargate — Limitations

No NodePort / LoadBalancer

No DaemonSet

Only EFS

Random EC2

Fargate — Limitations

No NodePort / LoadBalancer

No DaemonSet

Only EFS

Random EC2

Worker nodes

Comparison

	Self Managed ASGs	Managed Node Groups	Fargate
Automated Updates	No	Limited support	Yes
Auto Scaling	Limited support	Limited support	Yes
Graceful Scaling	No	Yes	Yes
ELB Support	Yes	Yes	Limited support
Stateful workloads (Persistent Volumes support)	Yes	Yes	Limited support (only EFS is supported)
Log aggregation	Yes	Yes	Limited support (only fluent-bit is supported)
Consistent Performance	Yes	Yes	No
Custom Instance Types (e.g GPU)	Yes	Yes	No
Custom AMI	Yes	Yes	No
Custom Security Group Rules	Yes	Yes	No
SSH Access	Yes	Yes	No
Custom SSH Auth	Yes	Yes	No

	Self Managed ASGs	Managed Node Groups	argate
Automated Updates	No	Limited support	es
Auto Scaling	Limited support	Limited support	es
Graceful Scaling	No	Yes	es
ELB Support	Yes	Yes	imited support
Stateful workloads (Persistent Volumes support)	Yes	Yes	imited support (only EFS is upported)
Log aggregation	Yes	Yes	imited support (only fluent-it is supported)
Consistent Performance	Yes	Yes	o
Custom Instance Types (e.g GPU)	Yes	Yes	o
Custom AMI	Yes	Yes	o
Custom Security Group Rules	Yes	Yes	o
SSH Access	Yes	Yes	o
Custom SSH Auth	Yes	Yes	o

Identities

Identities

Normal users

Service accounts

Identities

Normal users

Service accounts

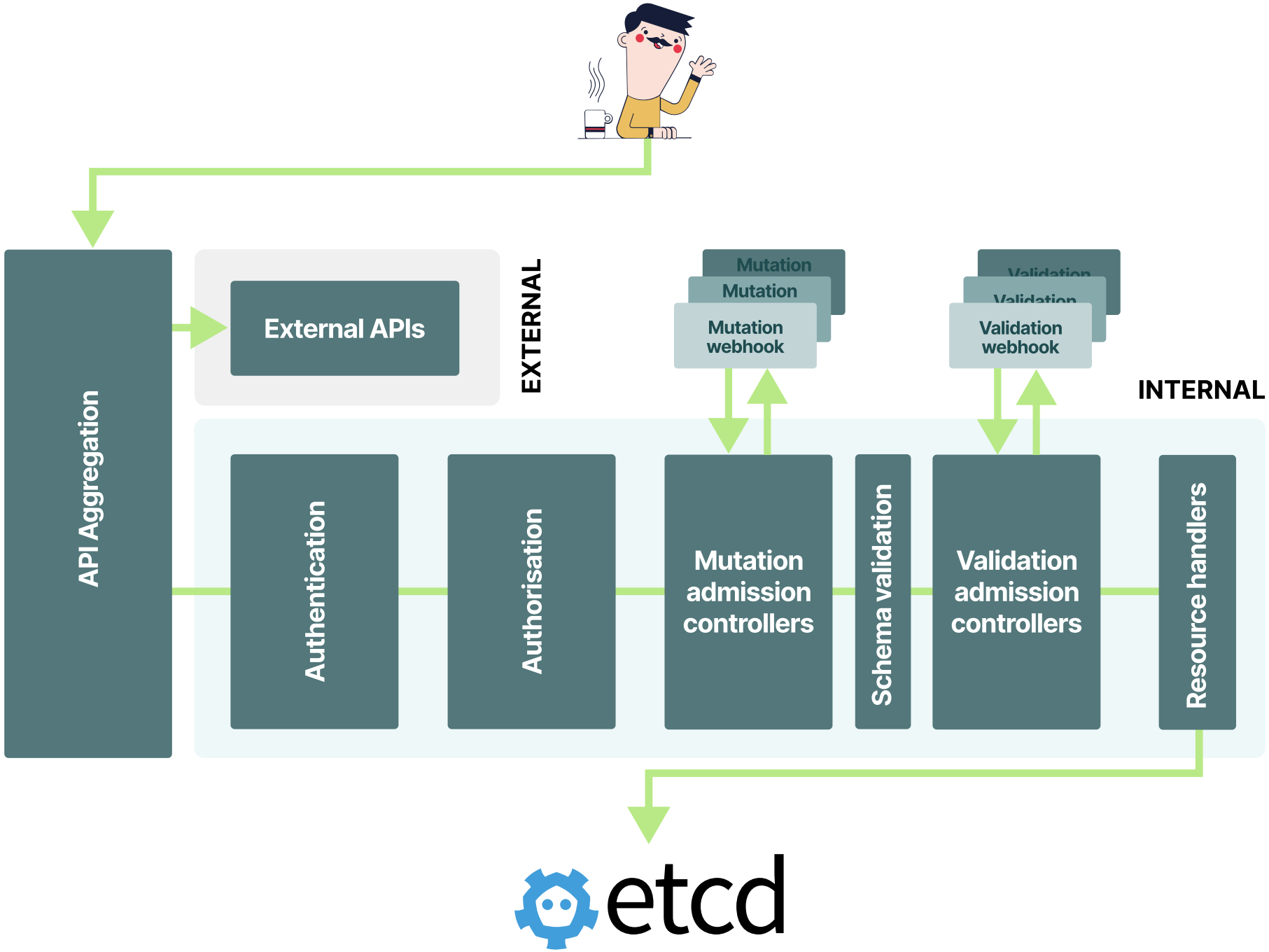
Identities

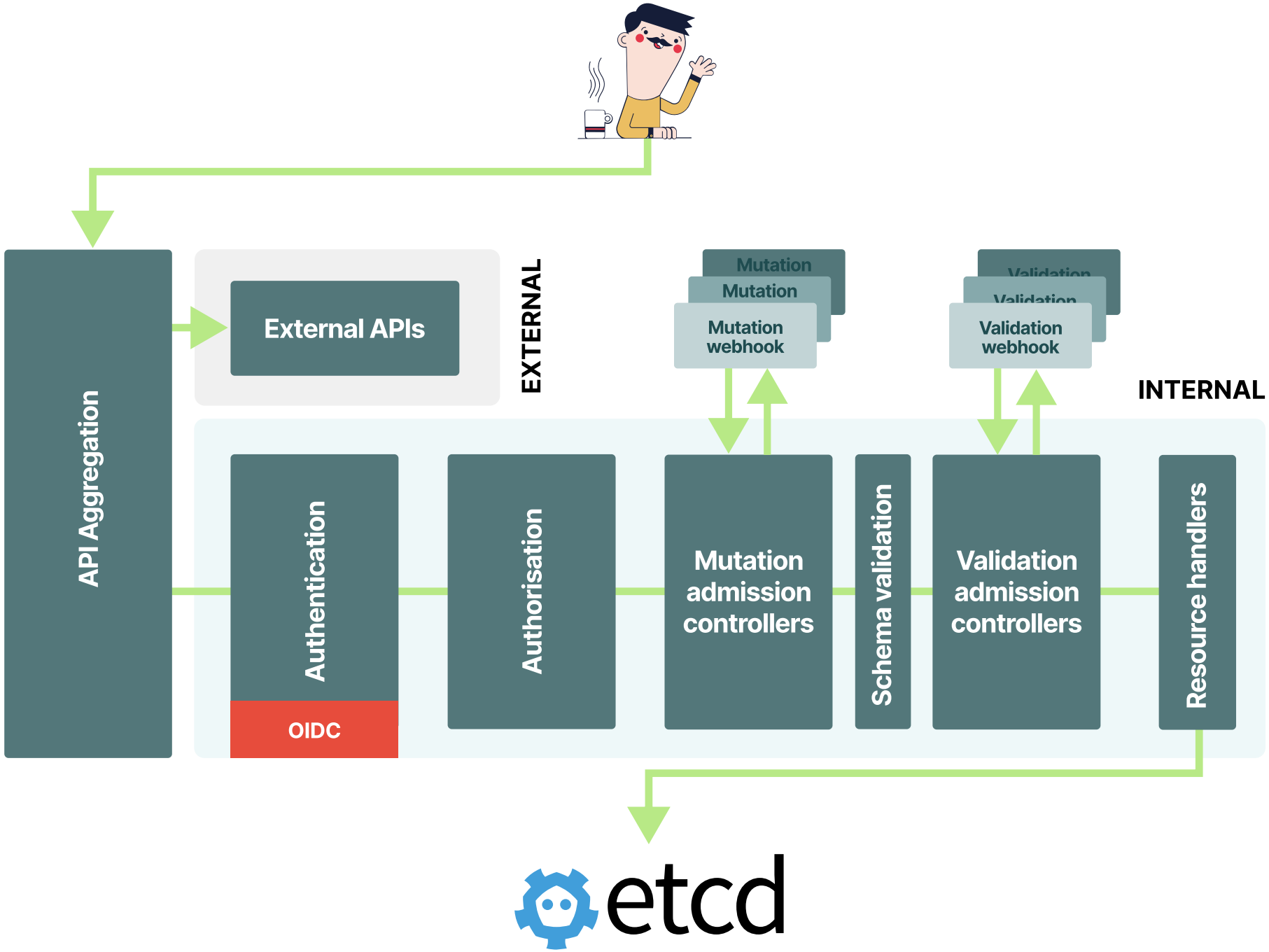
Users

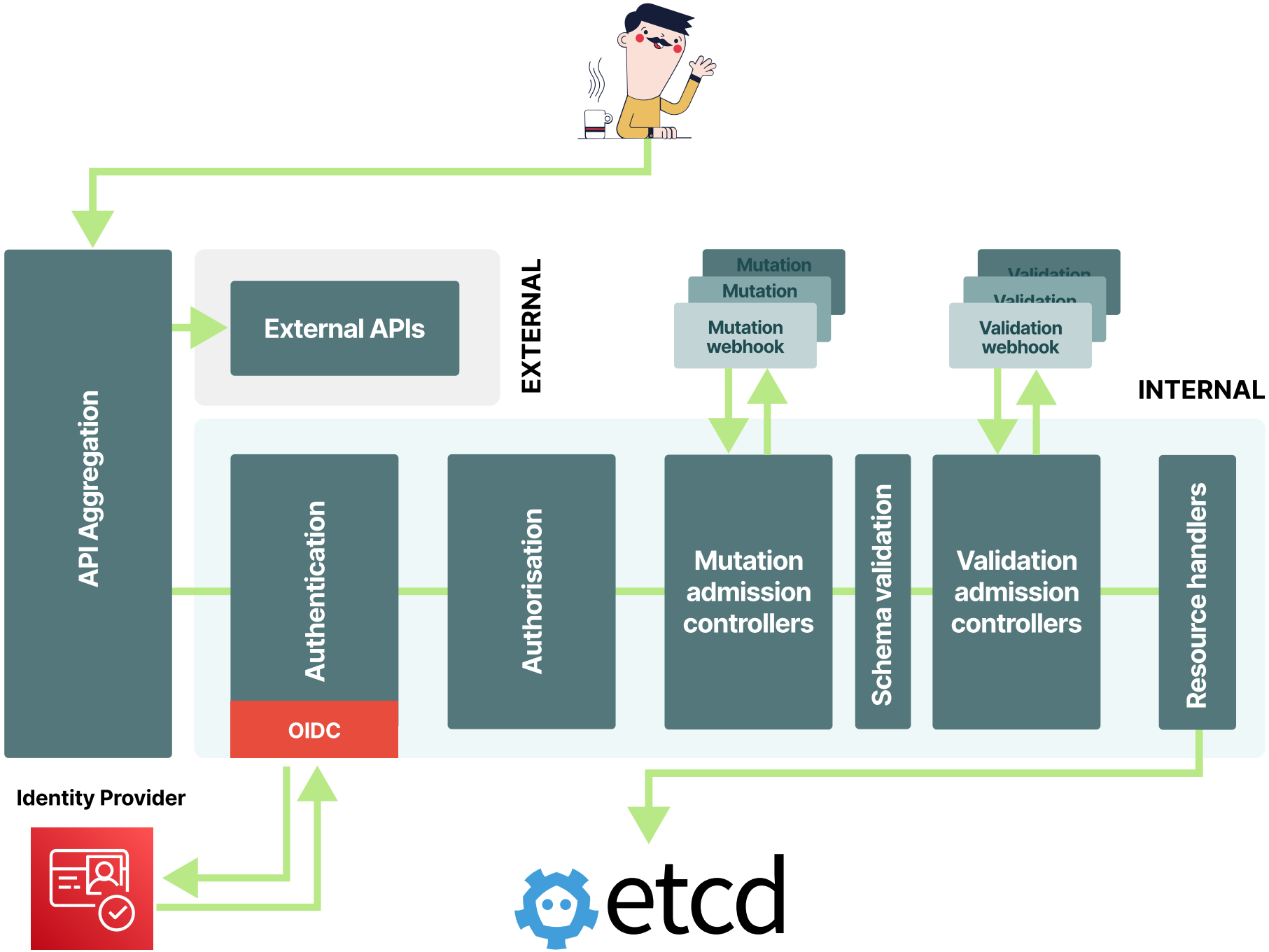


API SERVER

etcd

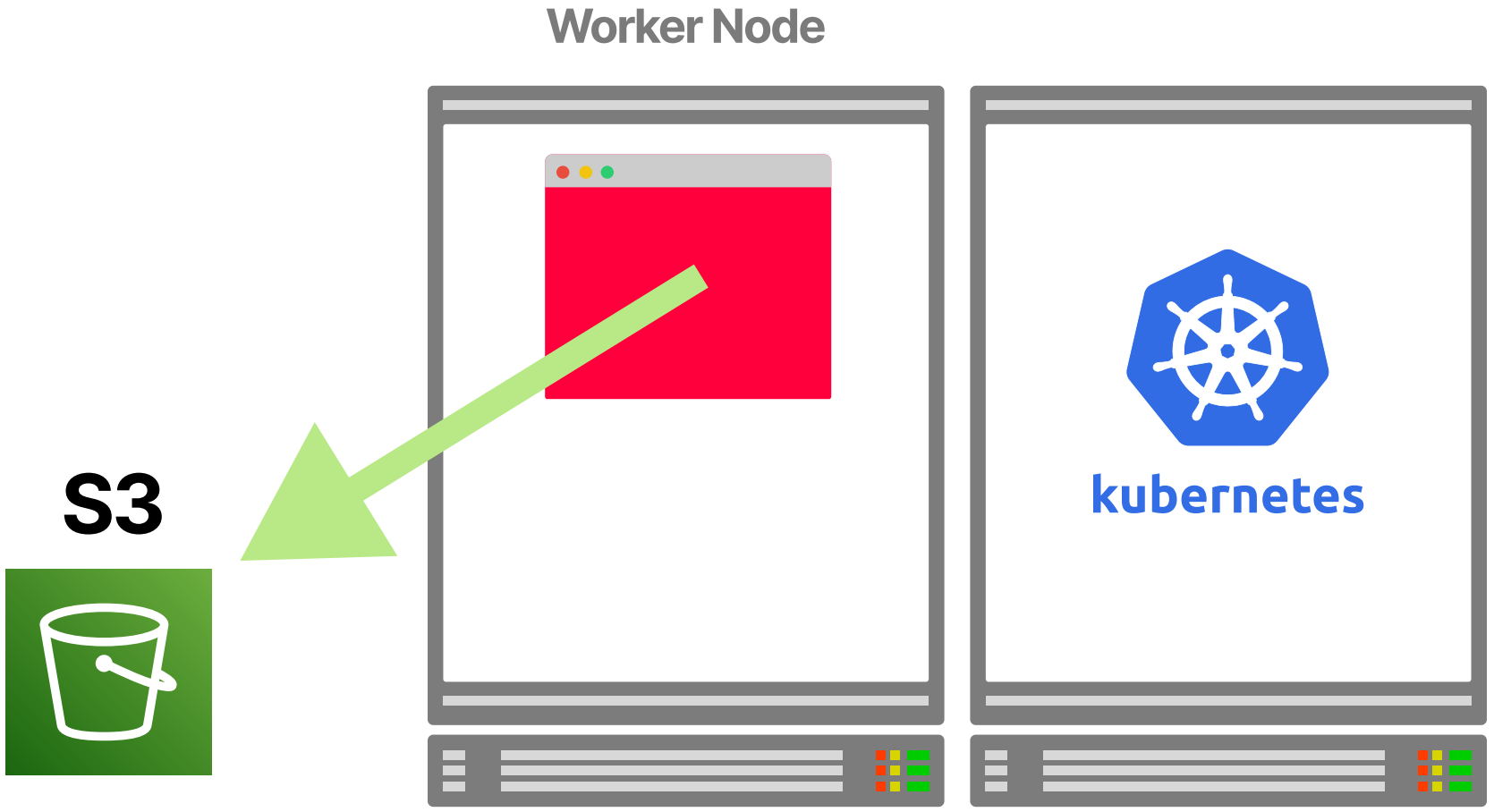

API Aggregation
External APIs
EXTERNAL
Mutation
Mutation
Mutation webhook
Validation
Validation
Validation webhook
INTERNAL
Authentication
Authorisation
Mutation admission controllers
Schema validation
Validation admission controllers
Resource handlers
etcd

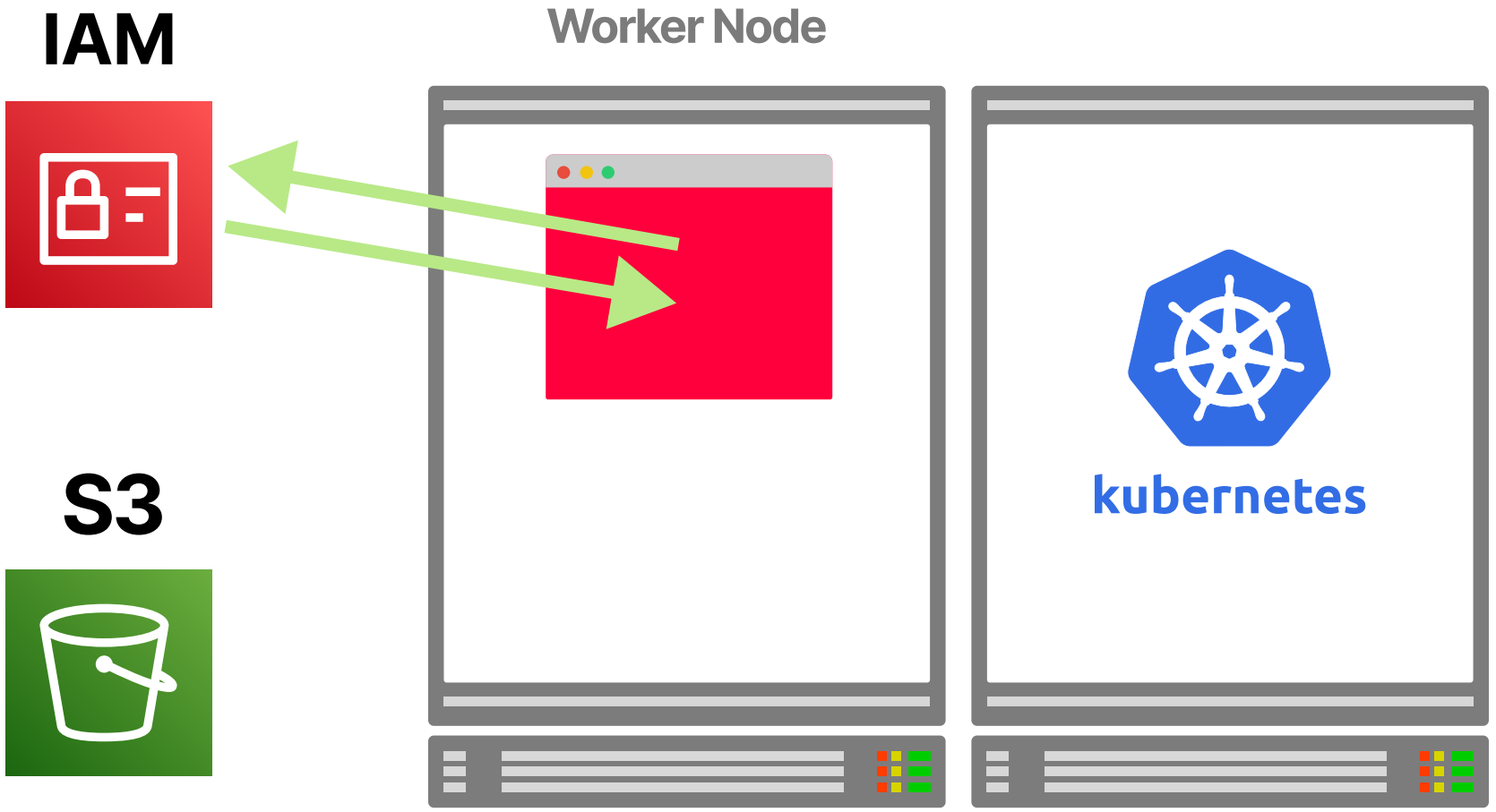

API Aggregation
External APIs
EXTERNAL
Mutation
Mutation
Mutation webhook
Validation
Validation
Validation webhook
INTERNAL
Authentication
OIDC
Authorisation
Mutation admission controllers
Schema validation
Validation admission controllers
Resource handlers
etcd

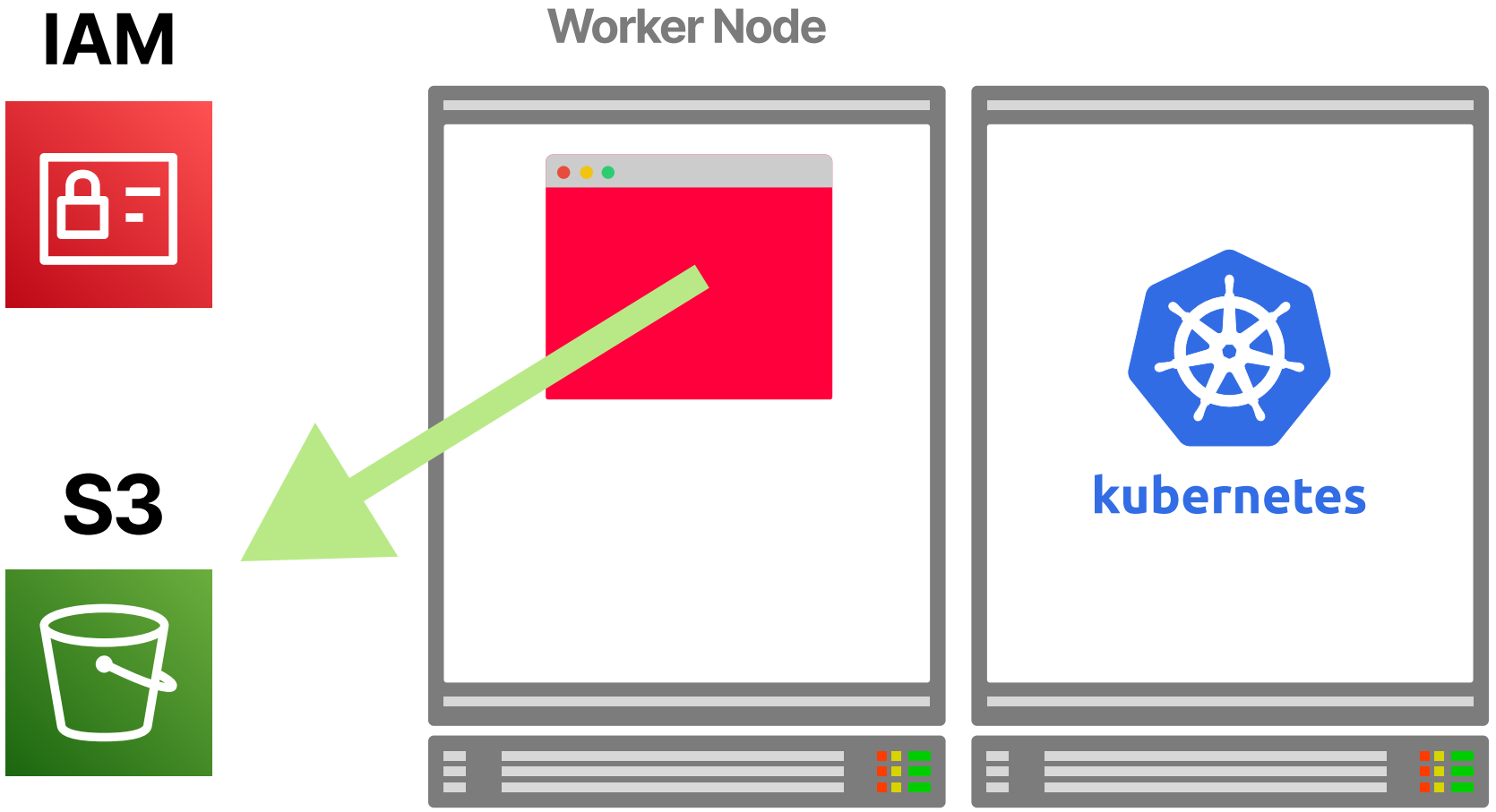

API Aggregation
External APIs
EXTERNAL
Mutation
Mutation
Mutation webhook
Validation
Validation
Validation webhook
INTERNAL
Authentication
OIDC
Authorisation
Mutation admission controllers
Schema validation
Validation admission controllers
Resource handlers
Identity Provider
etcd

Identities

IAM roles for Service Accounts


Worker Node
S3
kubernetes


IAM
S3
Worker Node
kubernetes


IAM
S3
Worker Node
kubernetes

IAM roles for Service Accounts

Create IAM OIDC provider

Create an IAM role

Associate an IAM role with the service account

IAM roles for Service Accounts

Create IAM OIDC provider

Create an IAM role

Associate an IAM role with the service account

IAM roles for Service Accounts

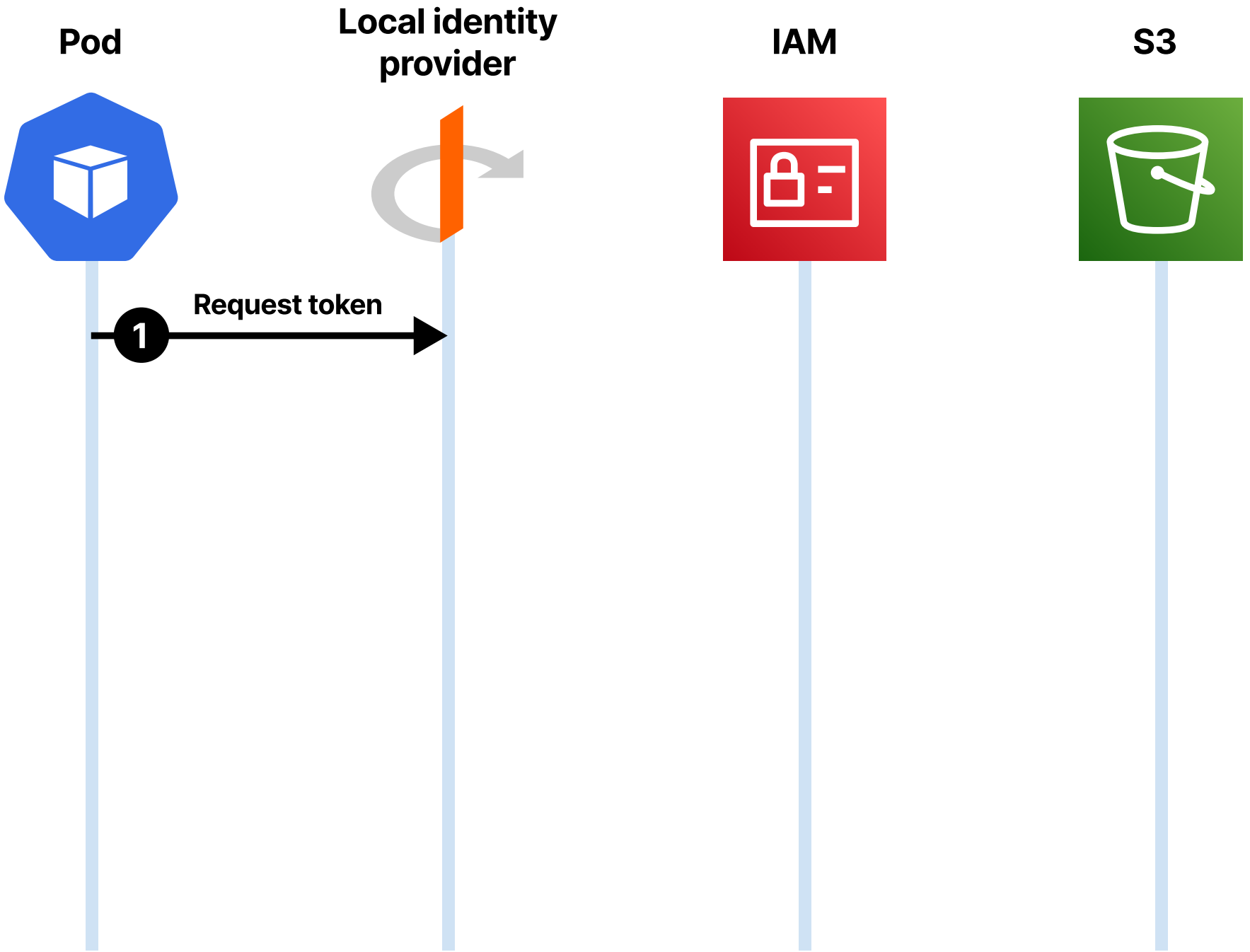
Create IAM OIDC provider

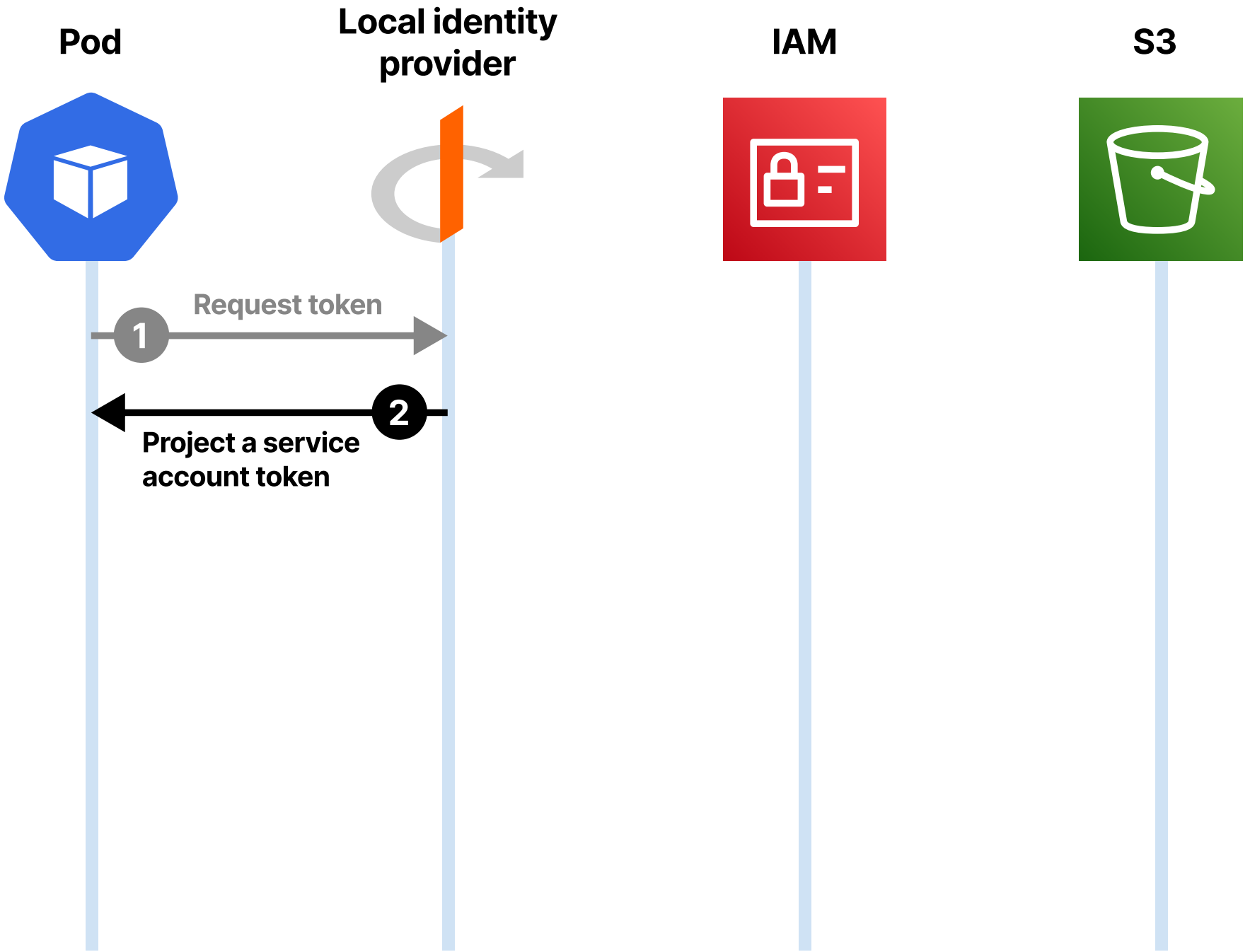
Create an IAM role

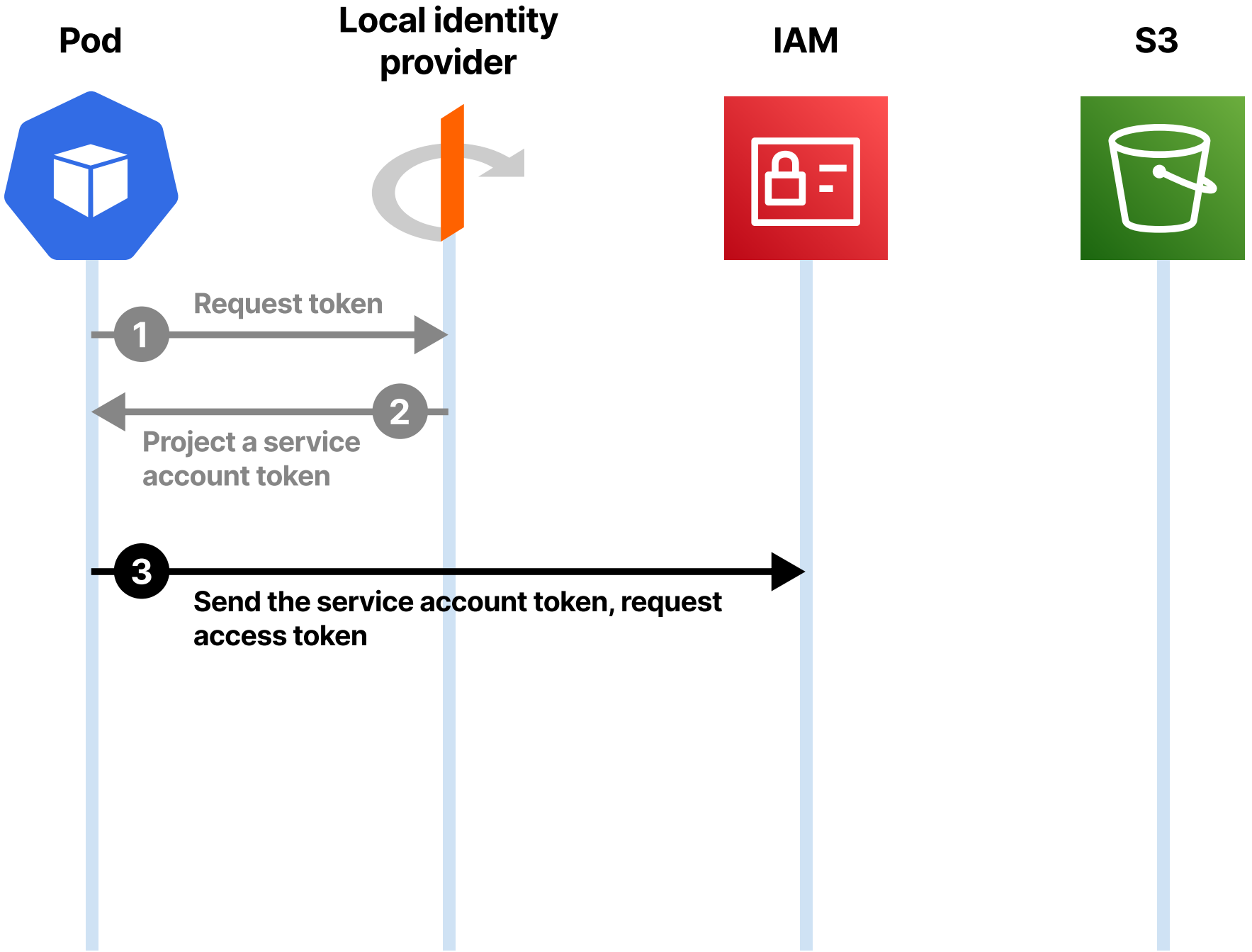
Associate an IAM role with the service account

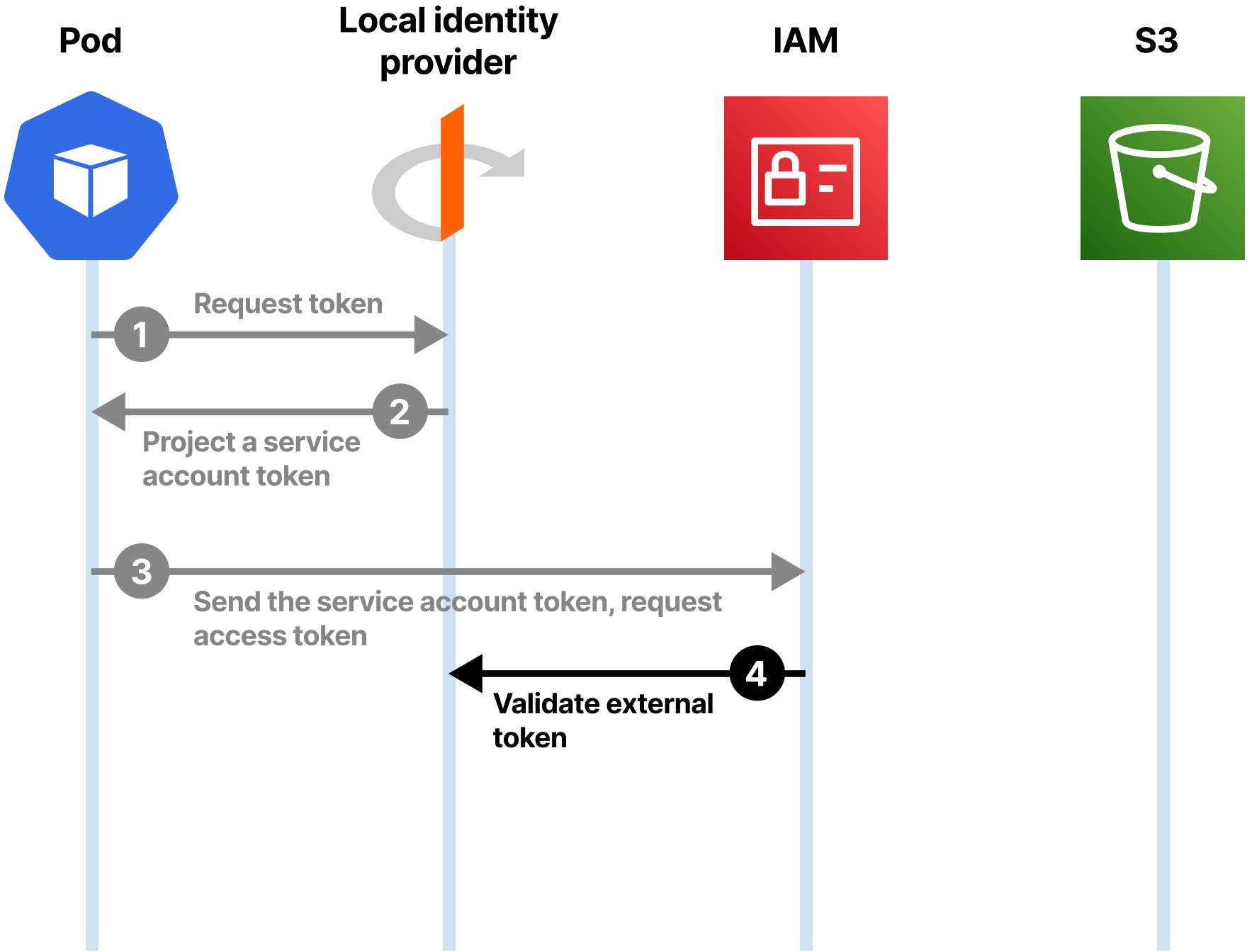
Federated identies

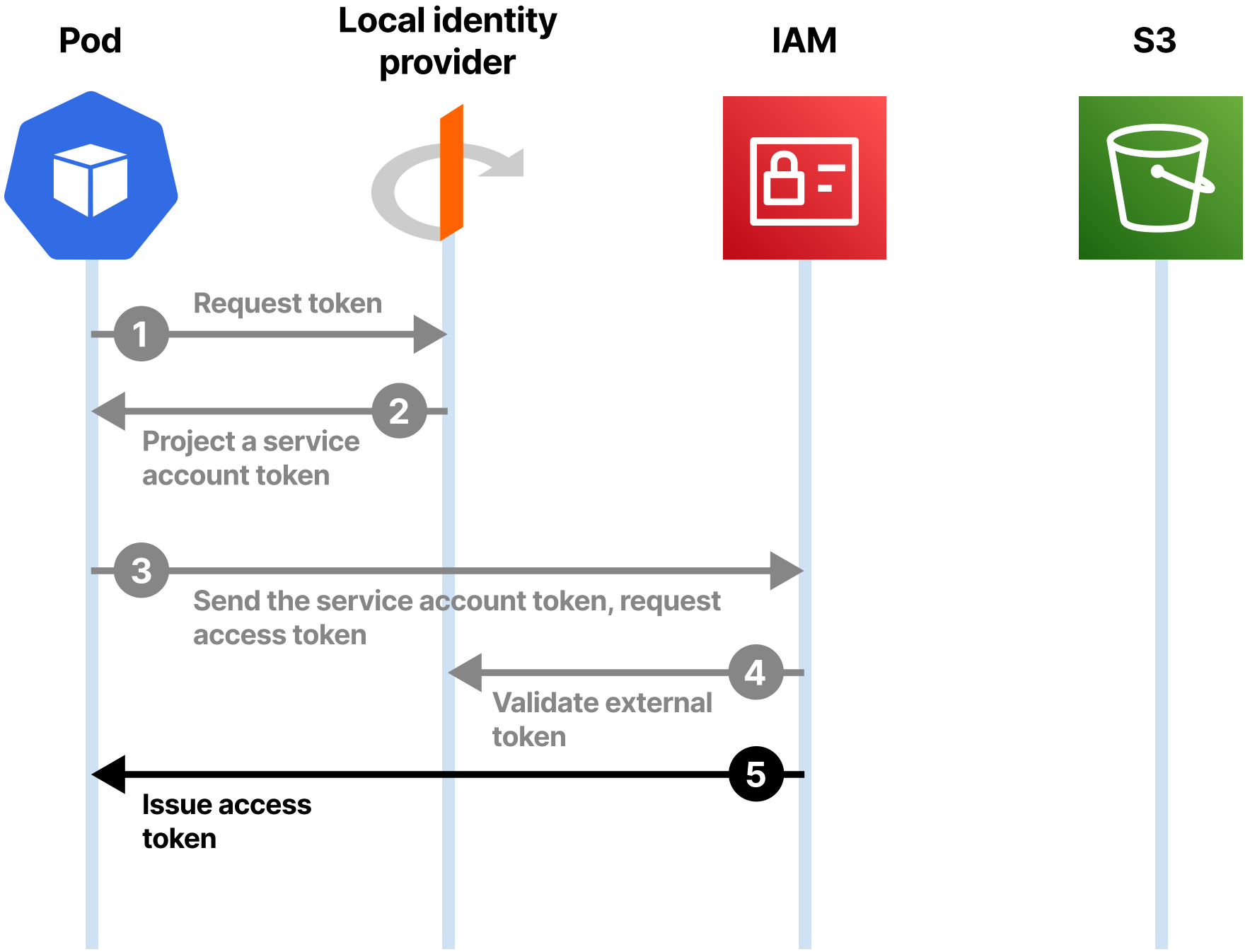
IAM roles for Service Accounts

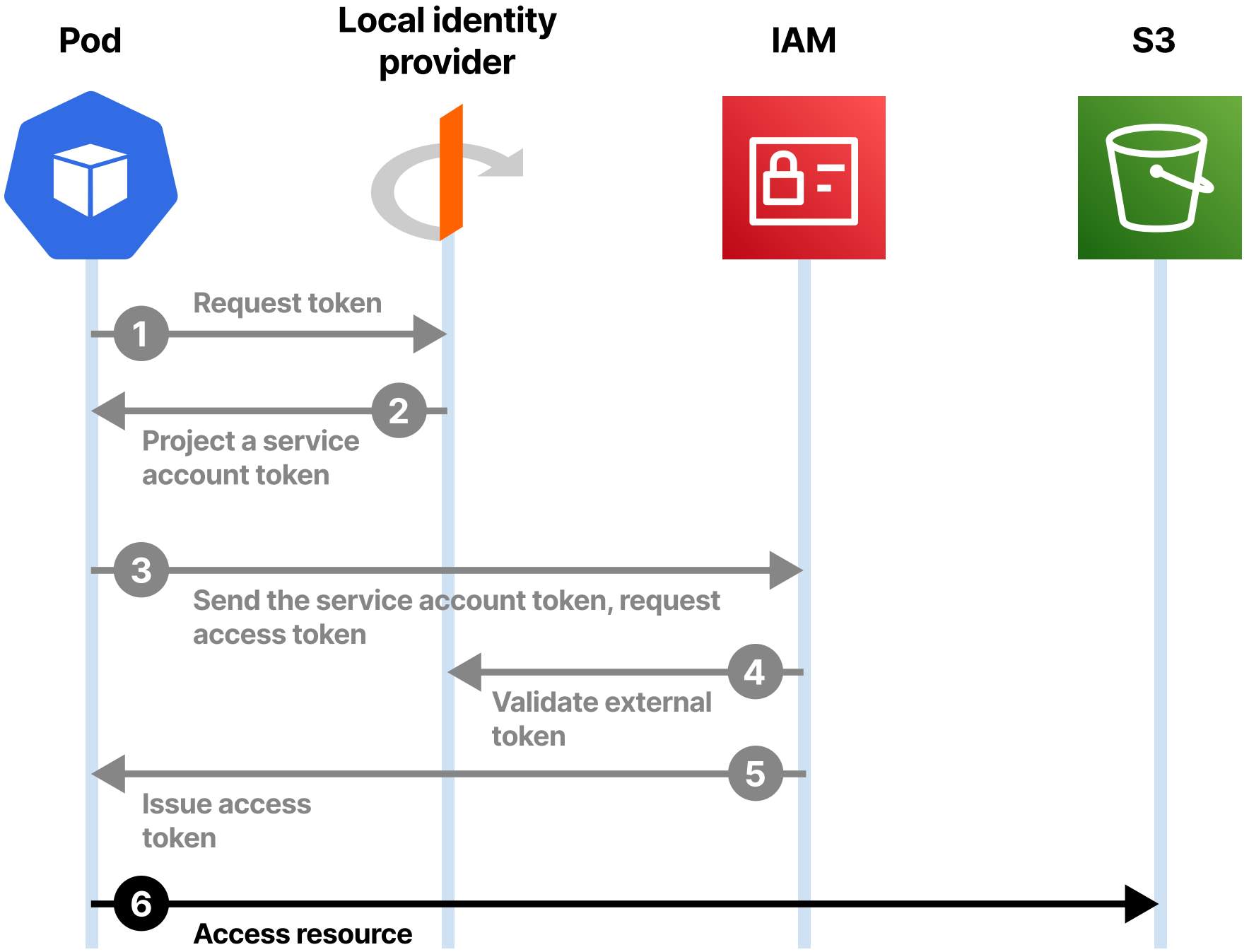

Pod
Local identity provider
IAM
S3
Request token
1

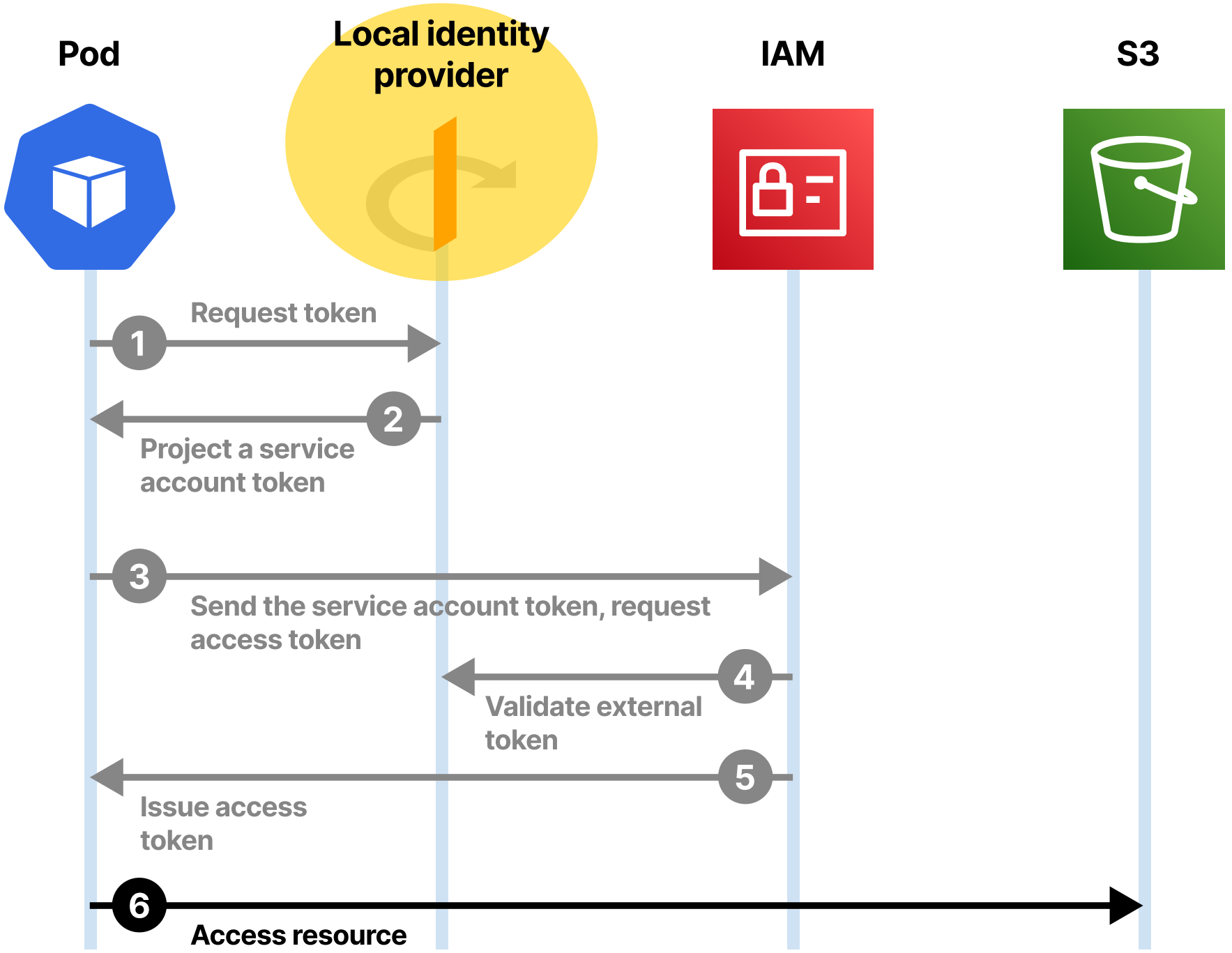

Pod
Local identity provider
IAM
S3
Request token
1
2
Project a service account token

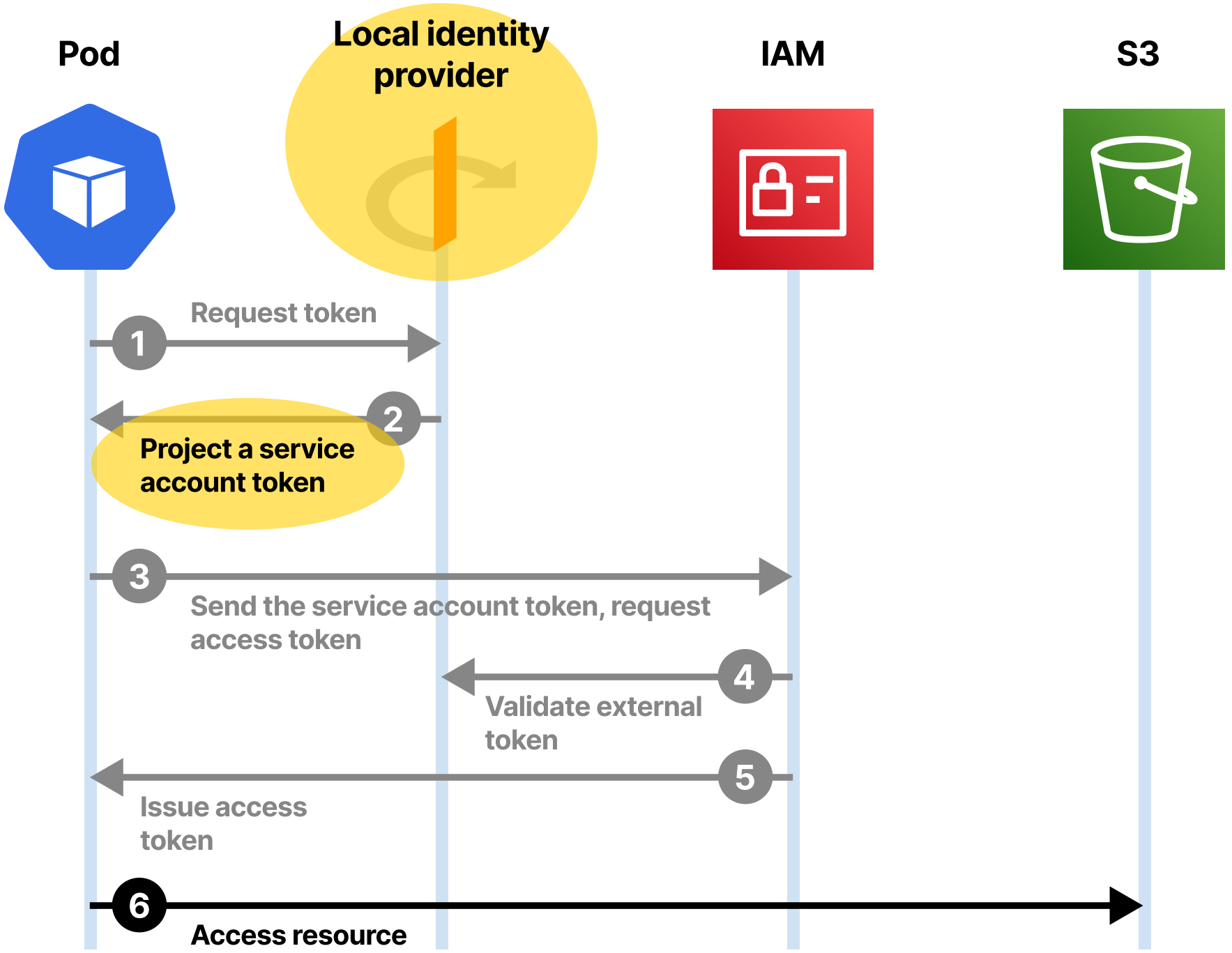

Pod
Local identity provider
IAM
S3
Request token
1
2
Project a service account token
3
Send the service account token, request access token

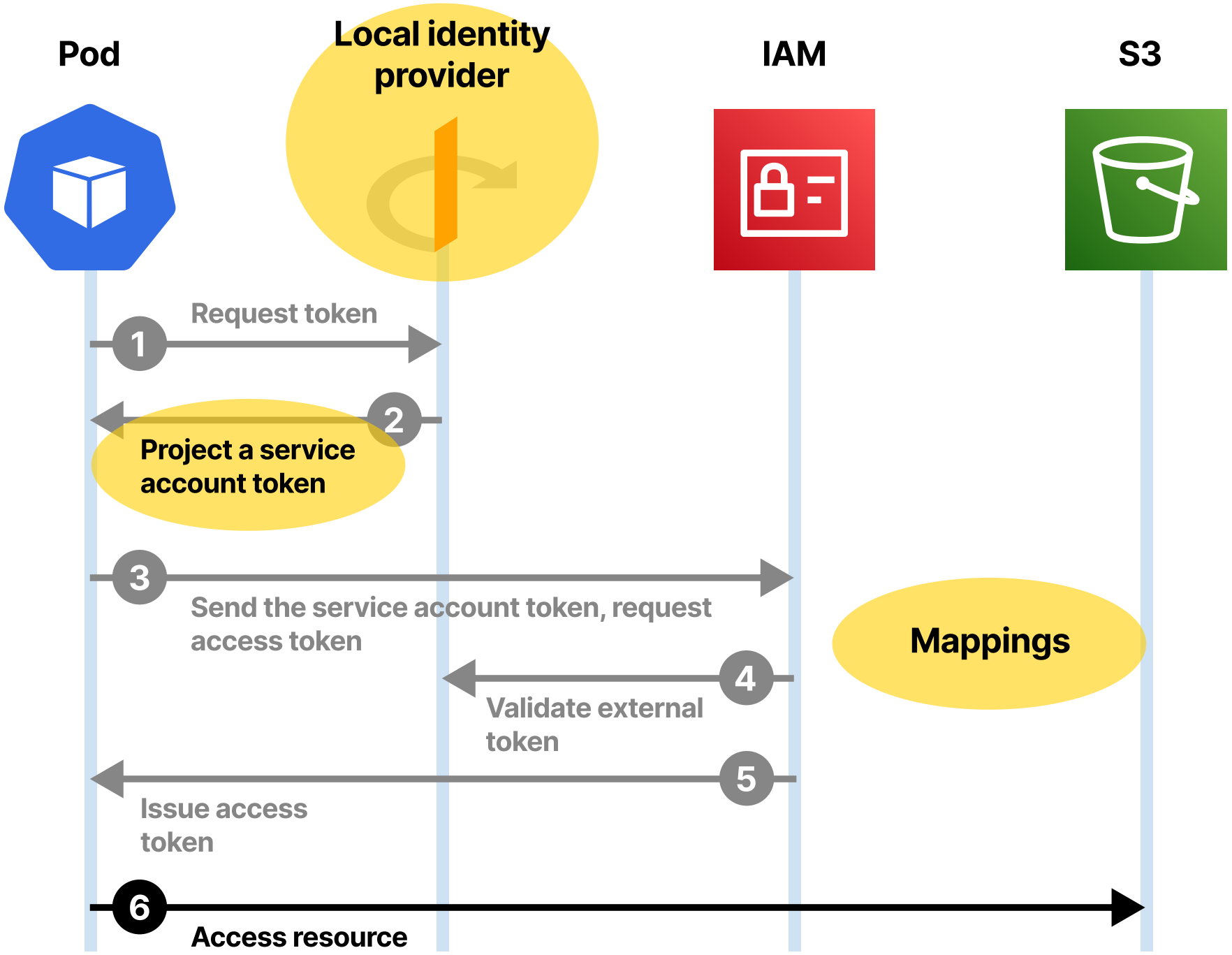

Pod
Local identity provider
IAM
S3
Request token
1
2
Project a service account token
3
Send the service account token, request access token
4
Validate external token


Pod
Local identity provider
IAM
S3
Request token
1
2
Project a service account token
3
Send the service account token, request access token
4
Validate external token
5
Issue access token


Pod
Local identity provider
IAM
S3
1
Request token
2
Project a service account token
3
Send the service account token, request access token
4
Validate external token
5
Issue access token
6
Access resource

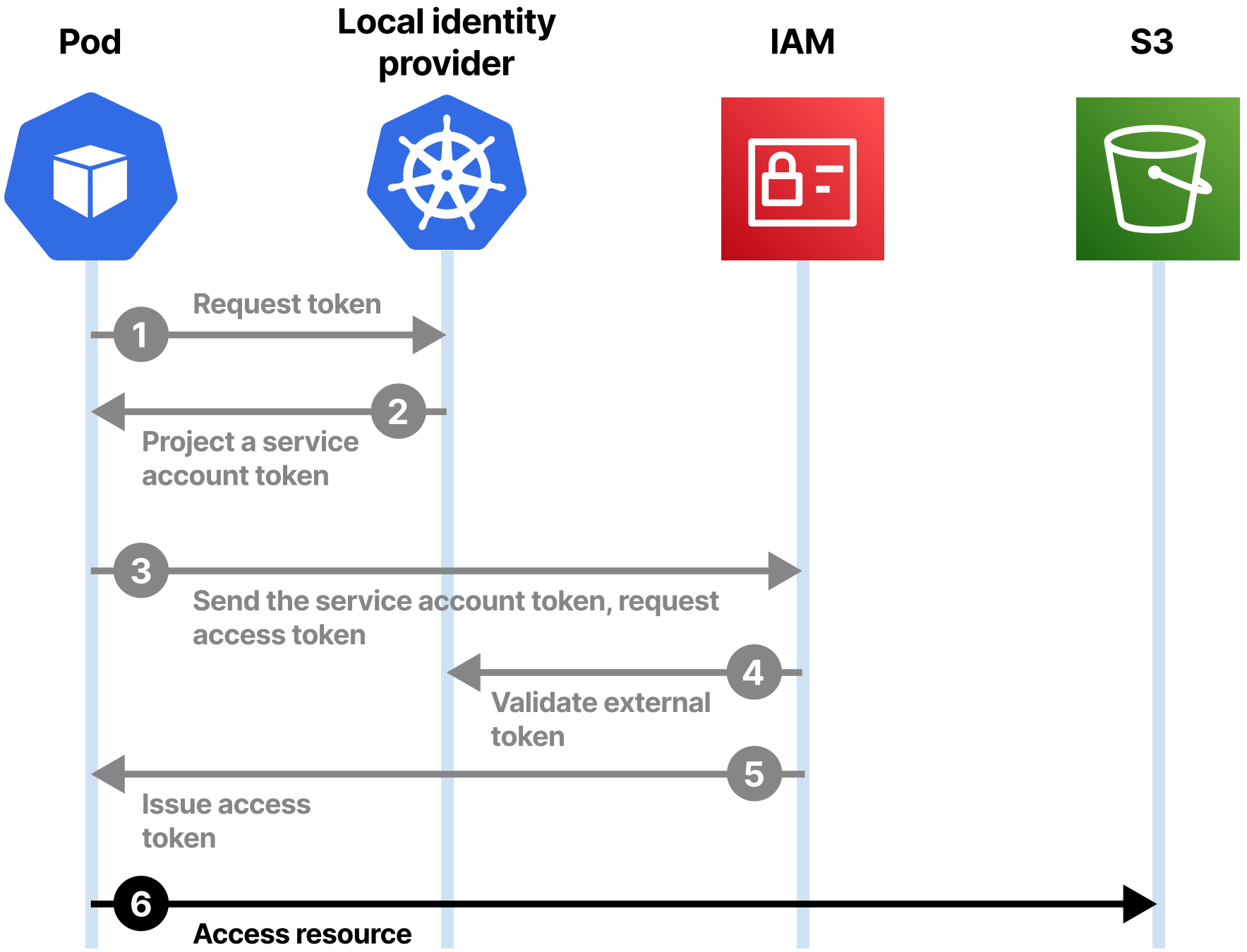

Pod
Local identity provider
IAM
S3
Request token
1
2
Project a service account token
3
Send the service account token, request access token
4
Validate external token
5
Issue access token
6
Access resource


Pod
Local identity provider
IAM
S3
Request token
1
2
Project a service account token
3
Send the service account token, request access token
4
Validate external token
5
Issue access token
6
Access resource


Pod
Local identity provider
IAM
S3
Request token
1
2
Project a service account token
3
Send the service account token, request access token
Mappings
4
Validate external token
5
Issue access token
6
Access resource

1. Local identity provider

IAM roles for Service Accounts


Pod
Local identity provider
IAM
S3
Request token
1
2
Project a service account token
3
Send the service account token, request access token
4
Validate external token
5
Issue access token
6
Access resource

Local identity provider

Kubernetes OIDC

Generate identities

Verify identities

API endpoint(s)

Local identity provider

Kubernetes OIDC

Generate identities

Verify identities

API endpoint(s)

Local identity provider

Kubernetes OIDC

Generate identities

Verify identities

API endpoint(s)

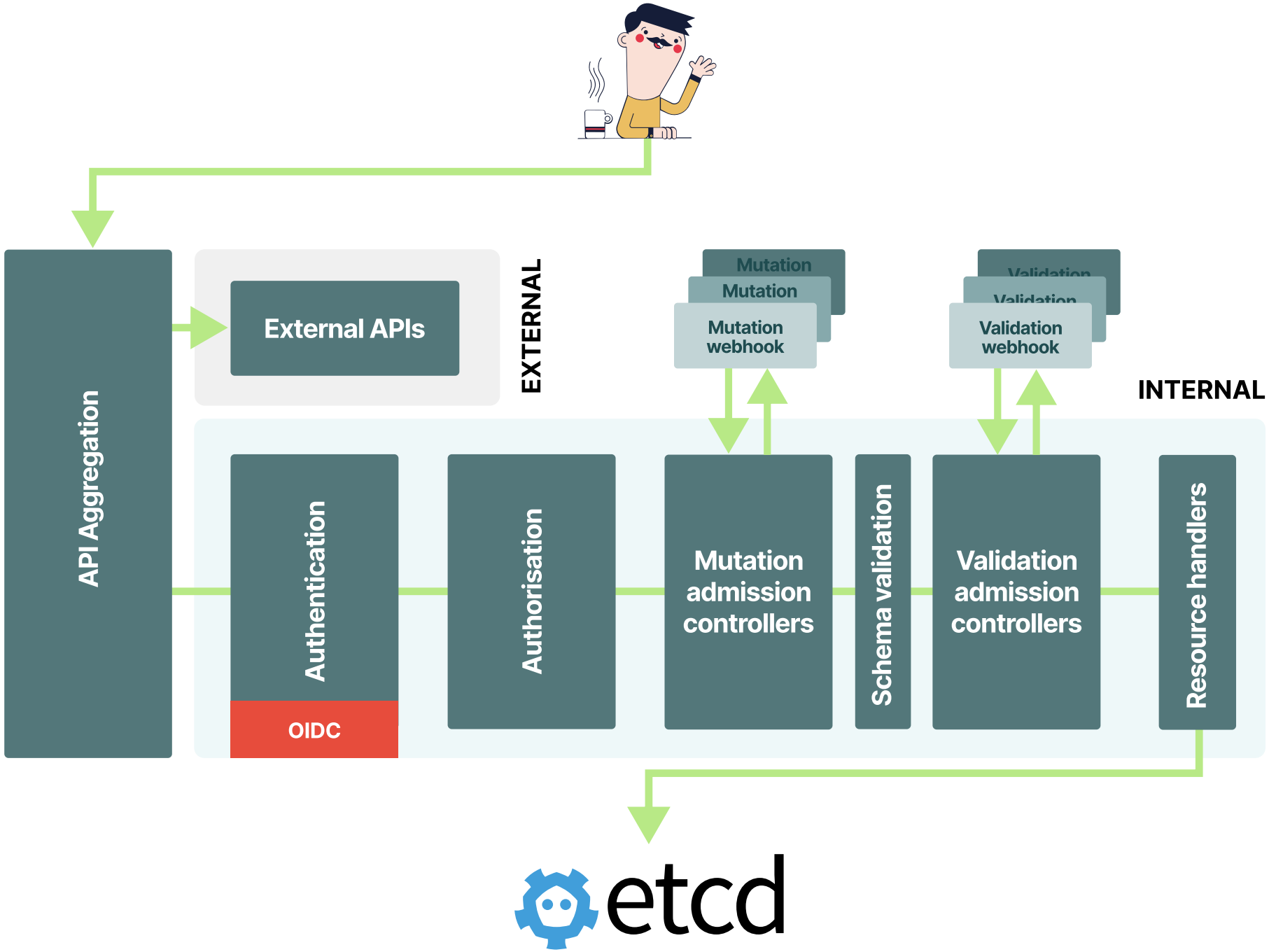
Local identity provider

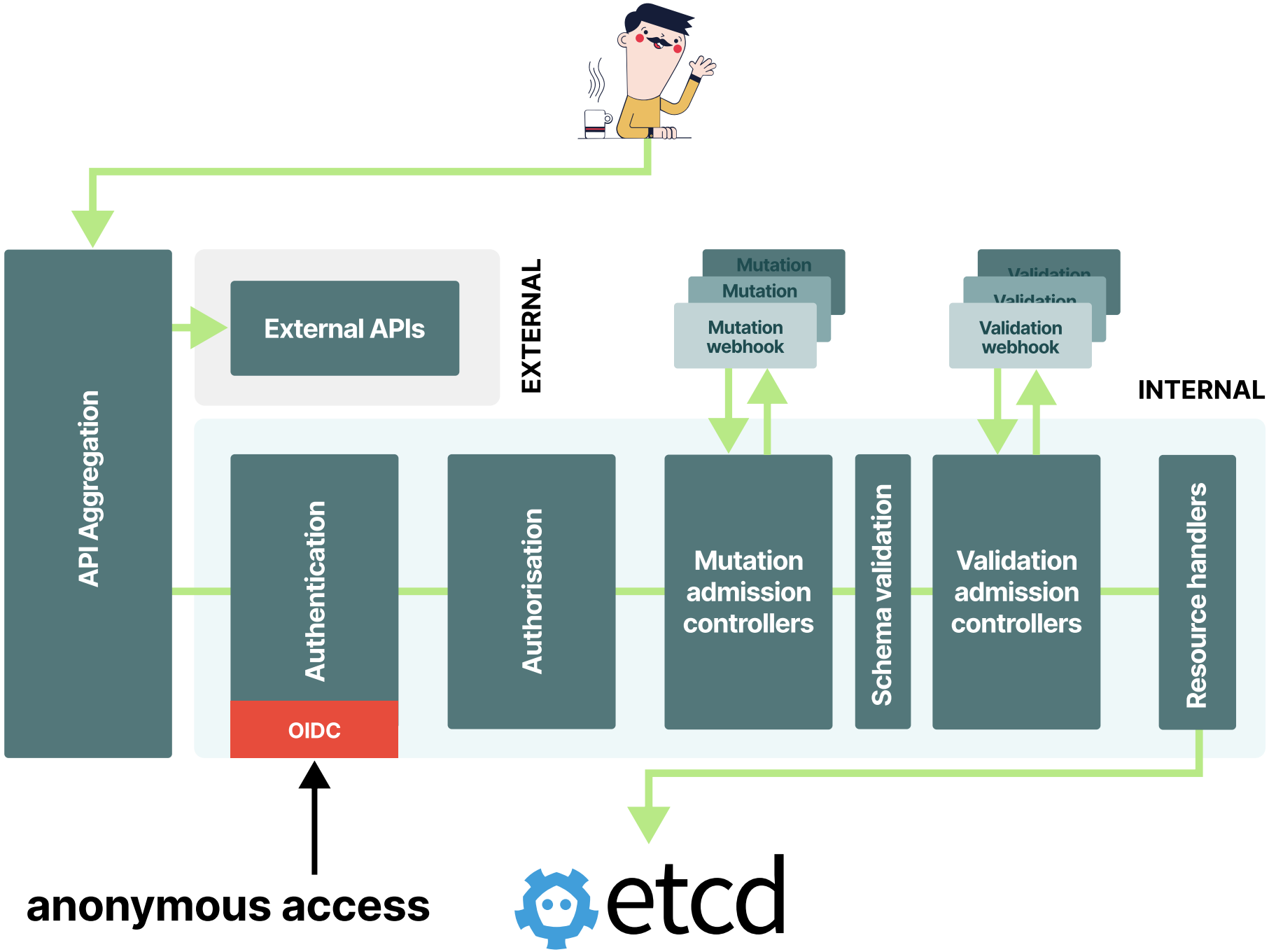
Kubernetes OIDC

Generate identities

Verify identities

API endpoint(s)


API Aggregation
External APIs
EXTERNAL
Mutation
Mutation
Mutation webhook
Validation
Validation
Validation webhook
INTERNAL
Authentication
OIDC
Authorisation
Mutation admission controllers
Schema validation
Validation admission controllers
Resource handlers
etcd

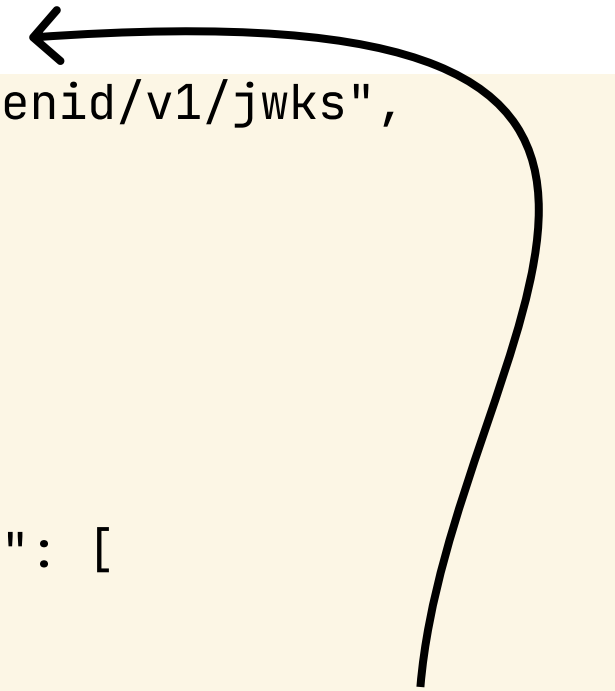

API Aggregation
External APIs
EXTERNAL
Mutation
Mutation
Mutation webhook
Validation
Validation
Validation webhook
INTERNAL
Authentication
OIDC
Authorisation
Mutation admission controllers
Schema validation
Validation admission controllers
Resource handlers
anonymous access
etcd

```
~$ curl --cacert kubernetes_ca.crt \
  https://localhost:6443/.well-known/openid-configuration | jq

{
  "issuer": "https://localhost:6443",
  "jwks_uri": "https://localhost:6443/openid/v1/jwks",
  "response_types_supported": [
    "id_token"
  ],
  "subject_types_supported": [
    "public"
  ],
  "id_token_signing_alg_values_supported": [
    "RS256"
  ]
}
```

```
~$ curl --cacert kubernetes_ca.crt \
  https://localhost:6443/.well-known/openid-configuration | jq

{
  "issuer": "https://localhost:6443",
  "jwks_uri": "https://localhost:6443/openid/v1/jwks",
  "response_types_supported": [
    "id_token"
  ],
  "subject_types_supported": [
    "public"
  ],
  "id_token_signing_alg_values_supported": [
    "RS256"
  ]
}
```



OIDC issuer URL

2. Tokens

IAM roles for Service Accounts

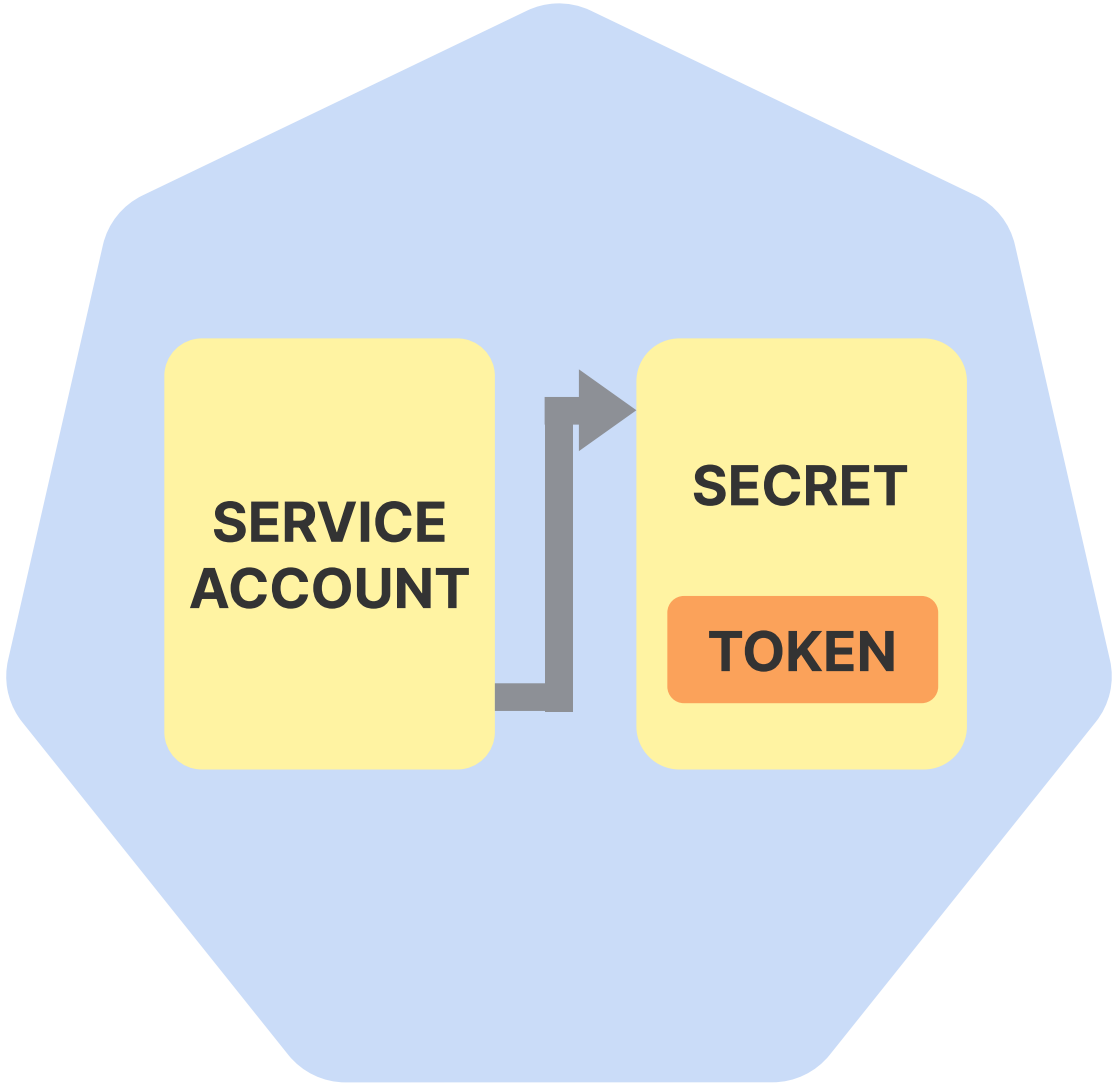


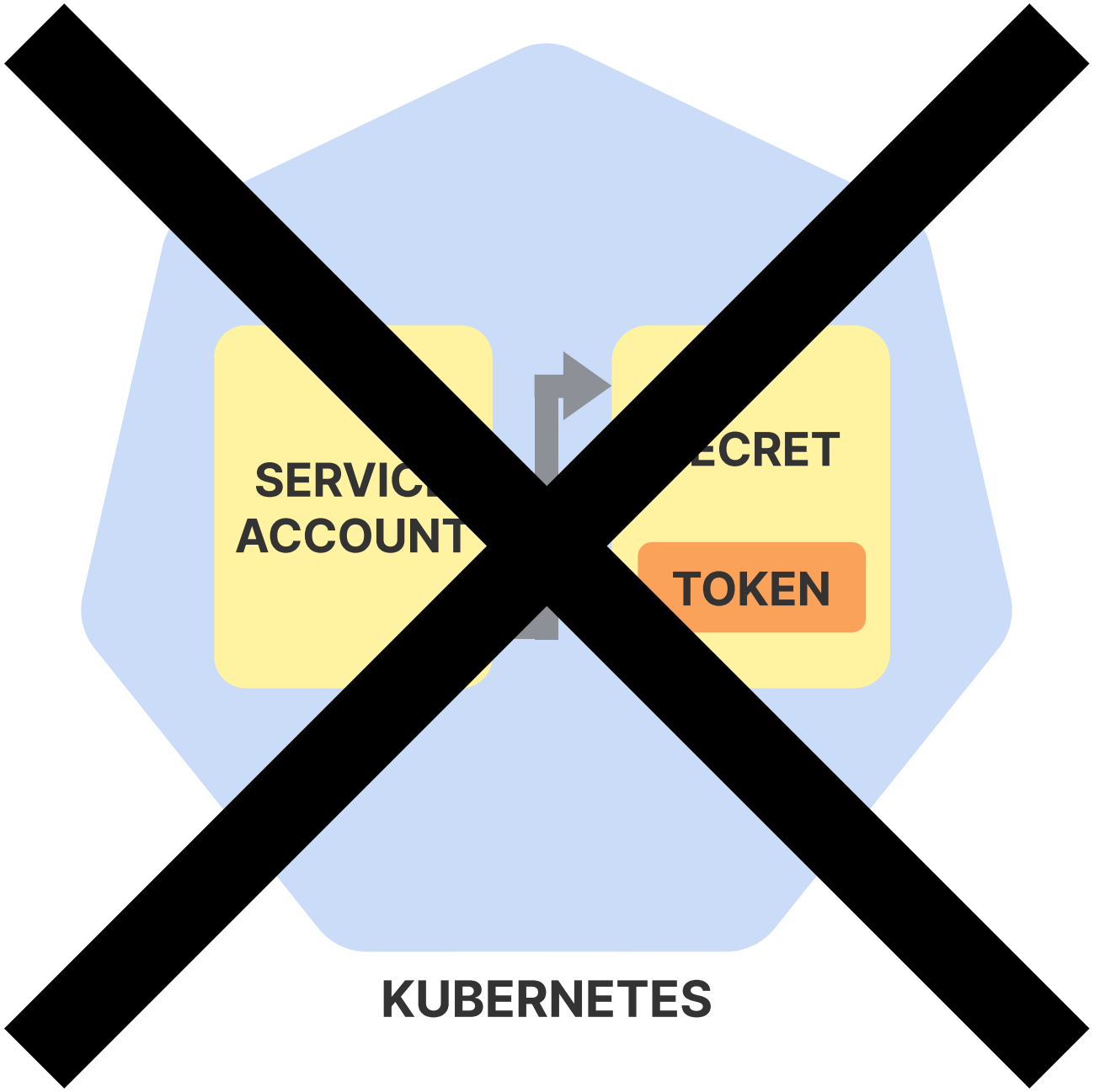
CREATE
SERVICEACCOUNT



KUBERNETES

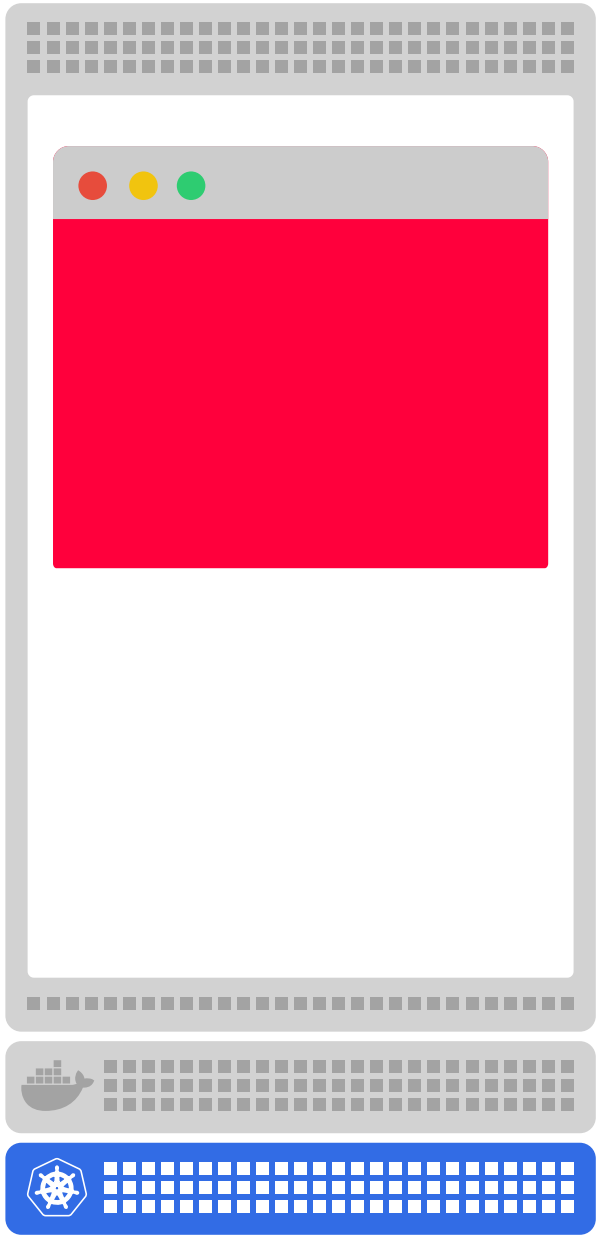
```
apiVersion: v1
kind: ServiceAccount
metadata:
 name: foo
```


SERVICE ACCOUNT
SECRET
TOKEN
KUBERNETES


SERVIC
ACCOUNT
ECRET
TOKEN
KUBERNETES

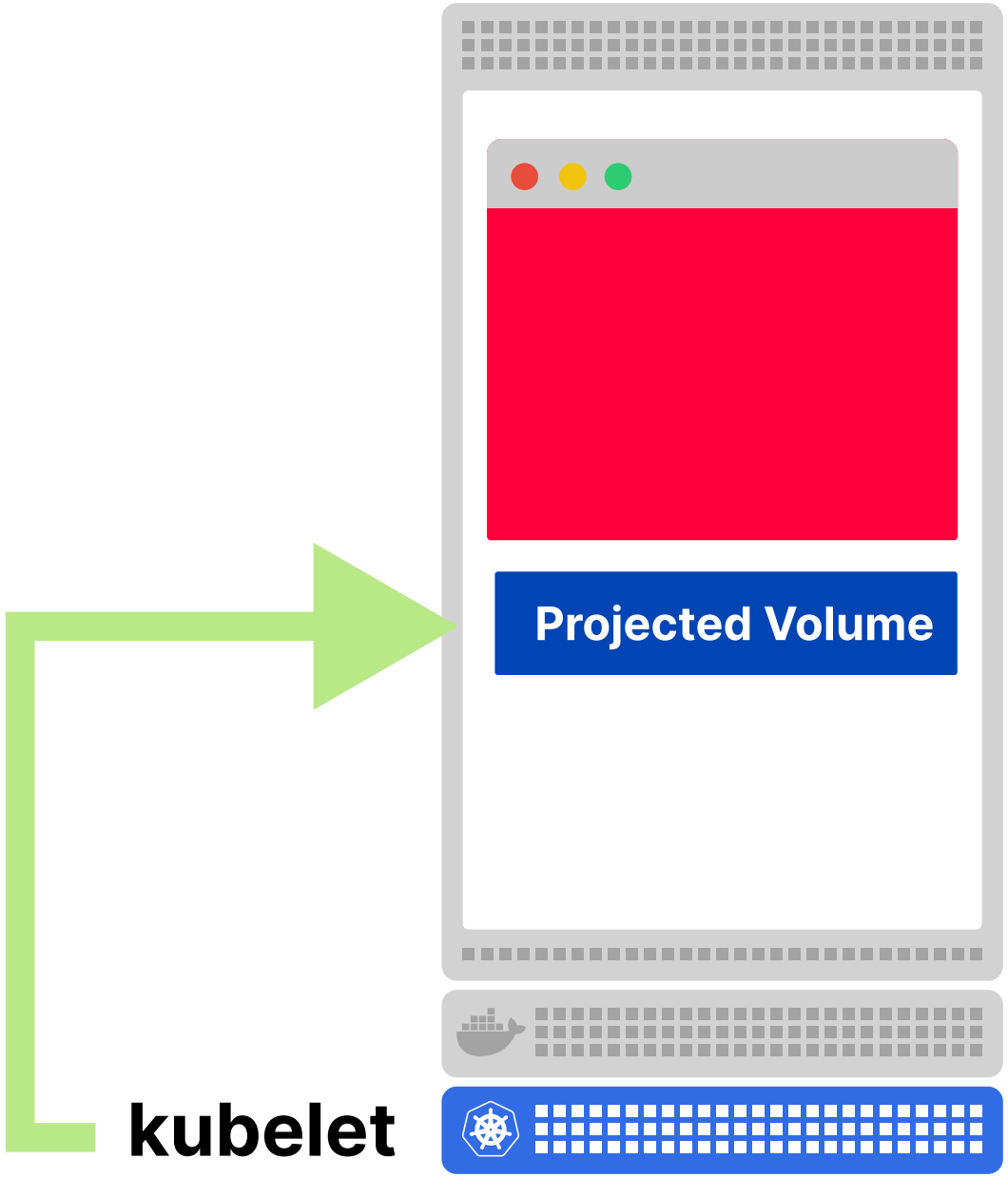
`ProjectedServiceAccountToken`

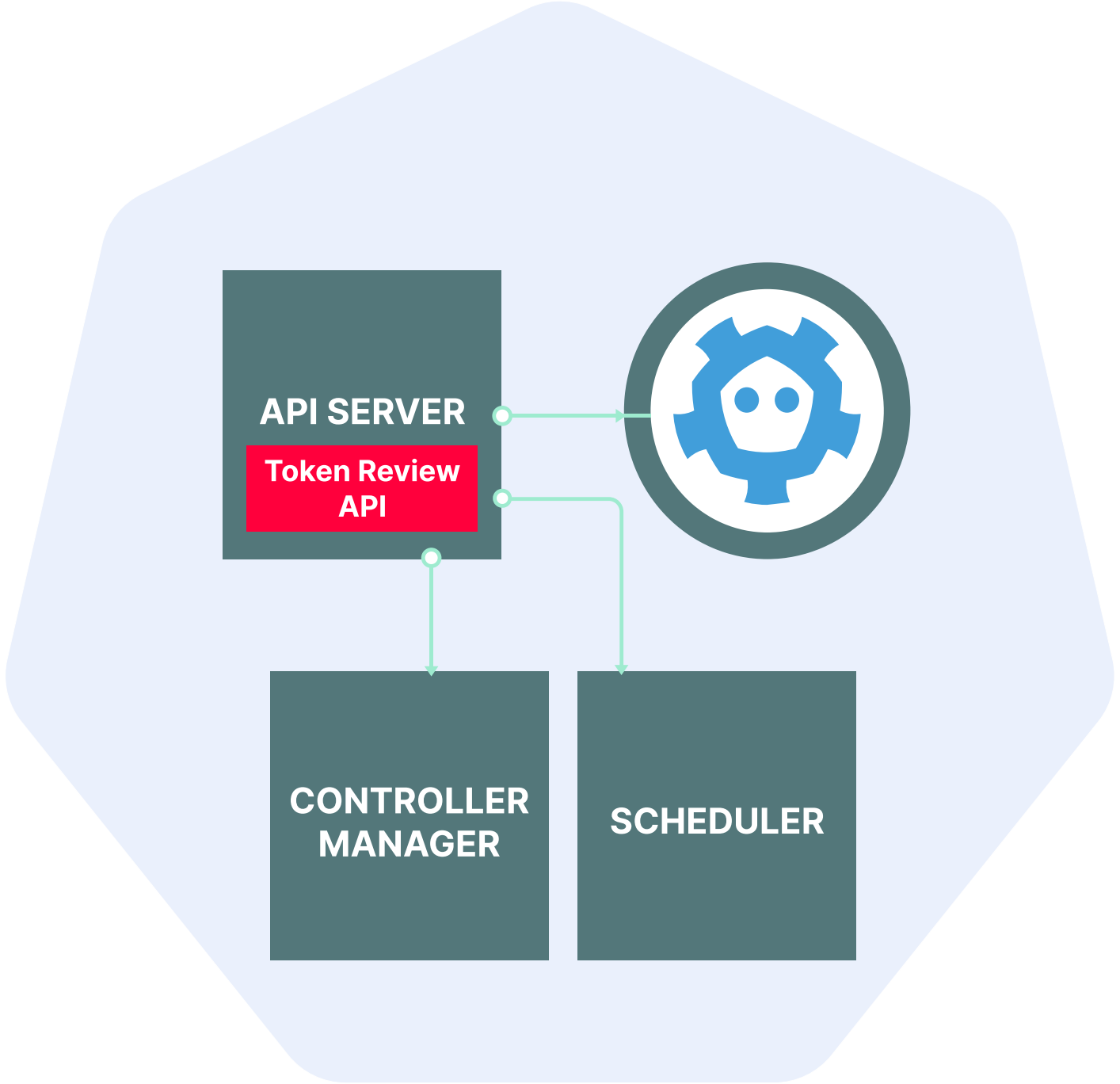
IAM roles for Service Accounts



kubelet

```
~$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: api
spec:
  serviceAccount: api
  volumes:
    - name: api-token
      projected:
        sources:
        - serviceAccountToken:
            path: api-token
            expirationSeconds: 600
            audience: data-store
  containers:
  - name: app
    image: amitsaha/k8s-sa-volume-demo-api:sa-2
    volumeMounts:
      - mountPath: /var/run/secrets/tokens
        name: api-token
```


Projected Volume
kubelet


API SERVER
Token Review API
CONTROLLER MANAGER
SCHEDULER

```
apiVersion: authentication.k8s.io/v1
kind: TokenReview
metadata:
  name: test
spec:
  token: eyJhbGciOiJS…
status:
  audiences:
  - data-store
  authenticated: true
  user:
    groups:
    - system:serviceaccounts
    - system:serviceaccounts:api
    - system:authenticated
    uid: 7df3a132-c9fe-48d1-9fa5-b654c3156977
    username: system:serviceaccount:api:api
```

```
apiVersion: authentication.k8s.io/v1
kind: TokenReview
metadata:
  name: test
spec:
  token: eyJhbGciOiJS…
status:
  audiences:
  - data-store
  authenticated: true
  user:
    groups:
    - system:serviceaccounts
    - system:serviceaccounts:api
    - system:authenticated
    uid: 7df3a132-c9fe-48d1-9fa5-b654c3156977
    username: system:serviceaccount:api:api
```

```
apiVersion: authentication.k8s.io/v1
kind: TokenReview
metadata:
  name: test
spec:
  token: eyJhbGciOiJS…
status:
  audiences:
  - data-store
  authenticated: true
  user:
    groups:
    - system:serviceaccounts
    - system:serviceaccounts:api
    - system:authenticated
    uid: 7df3a132-c9fe-48d1-9fa5-b654c3156977
    username: system:serviceaccount:api:api
```

3. Associates IAM role with the service account

IAM roles for Service Accounts

```
apiVersion: v1
kind: ServiceAccount
metadata:
  annotations:
    eks.amazonaws.com/role-arn: arn:aws:iam::<ACCOUNT_ID>:role/<IAM_ROLE_NAME>
```

End to end

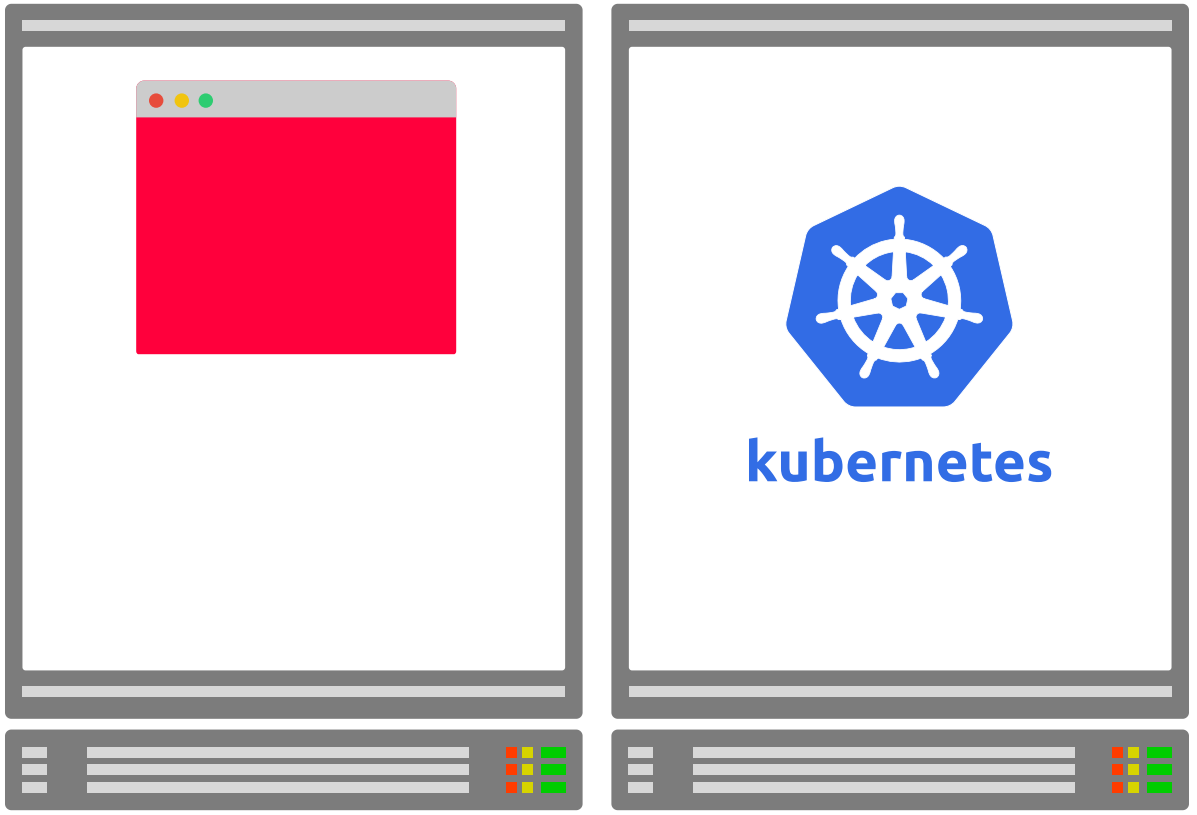
IAM roles for Service Accounts

IAM



S3



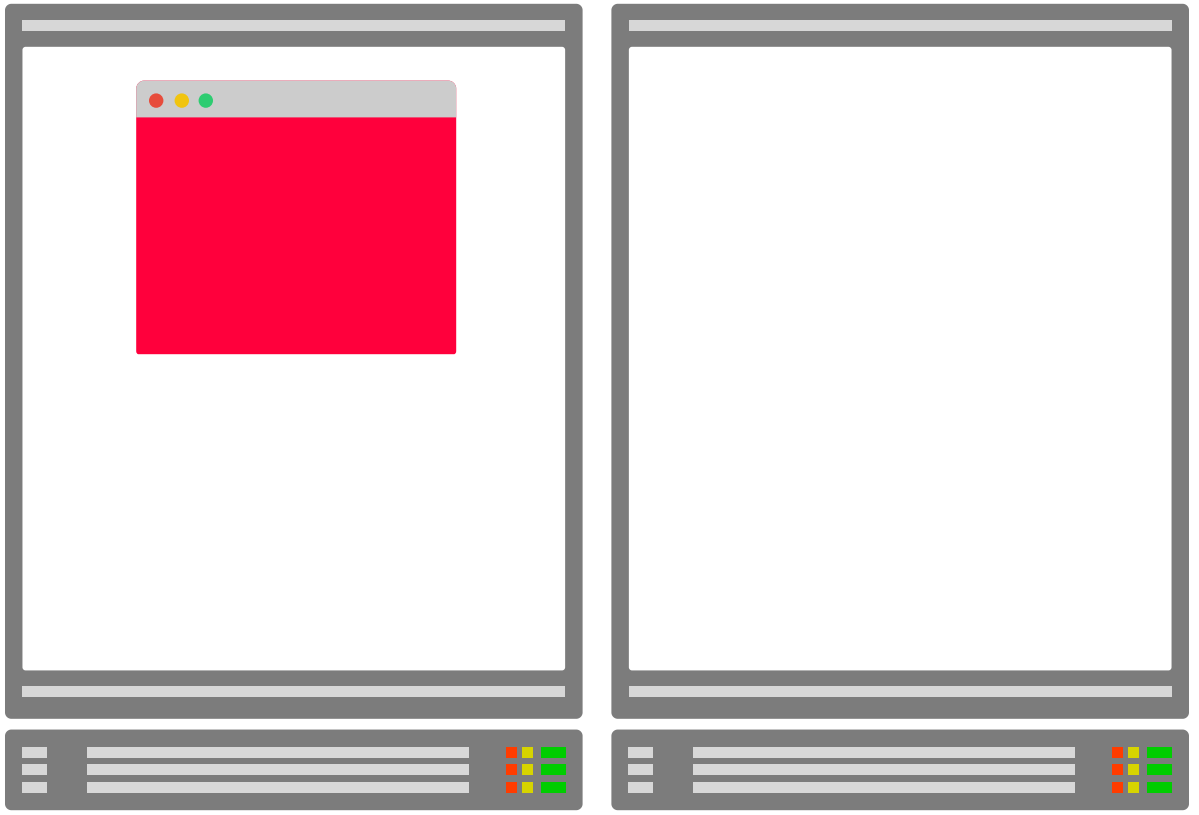


IAM



S3





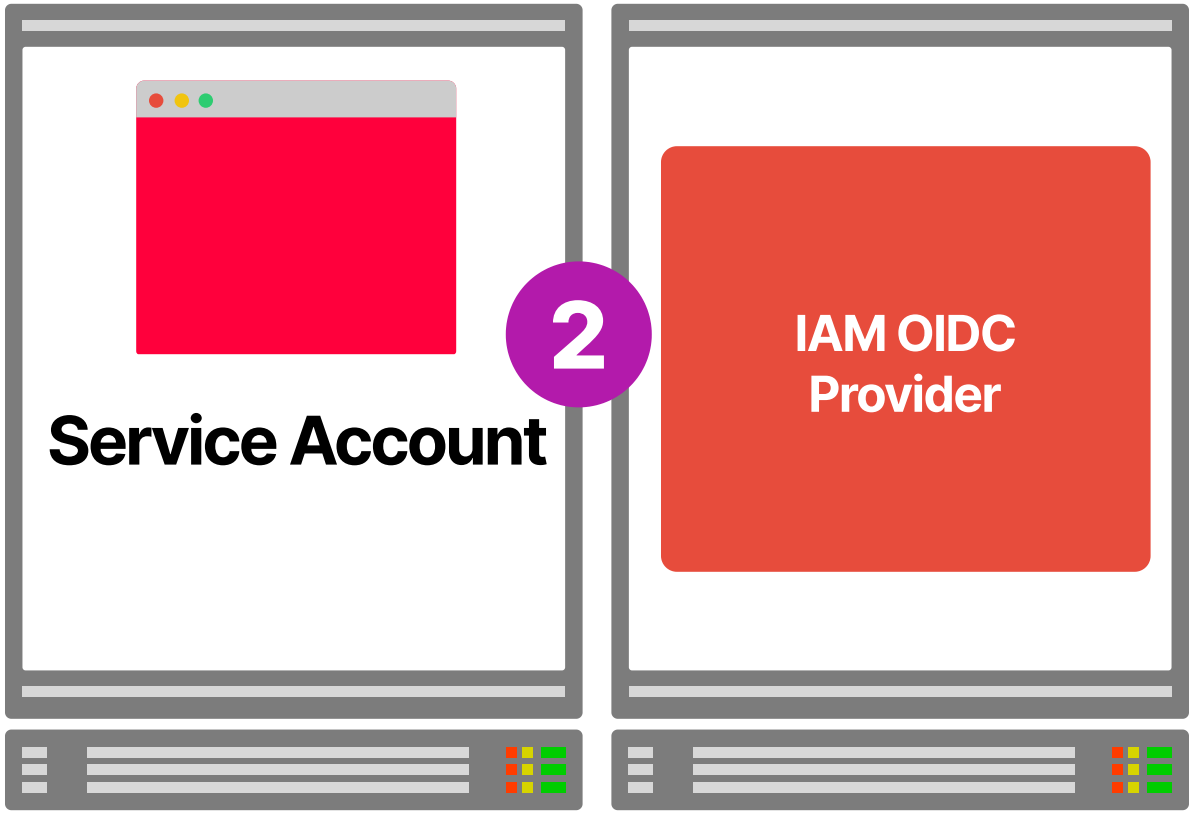
1 Create IAM Role + Policy

IAM

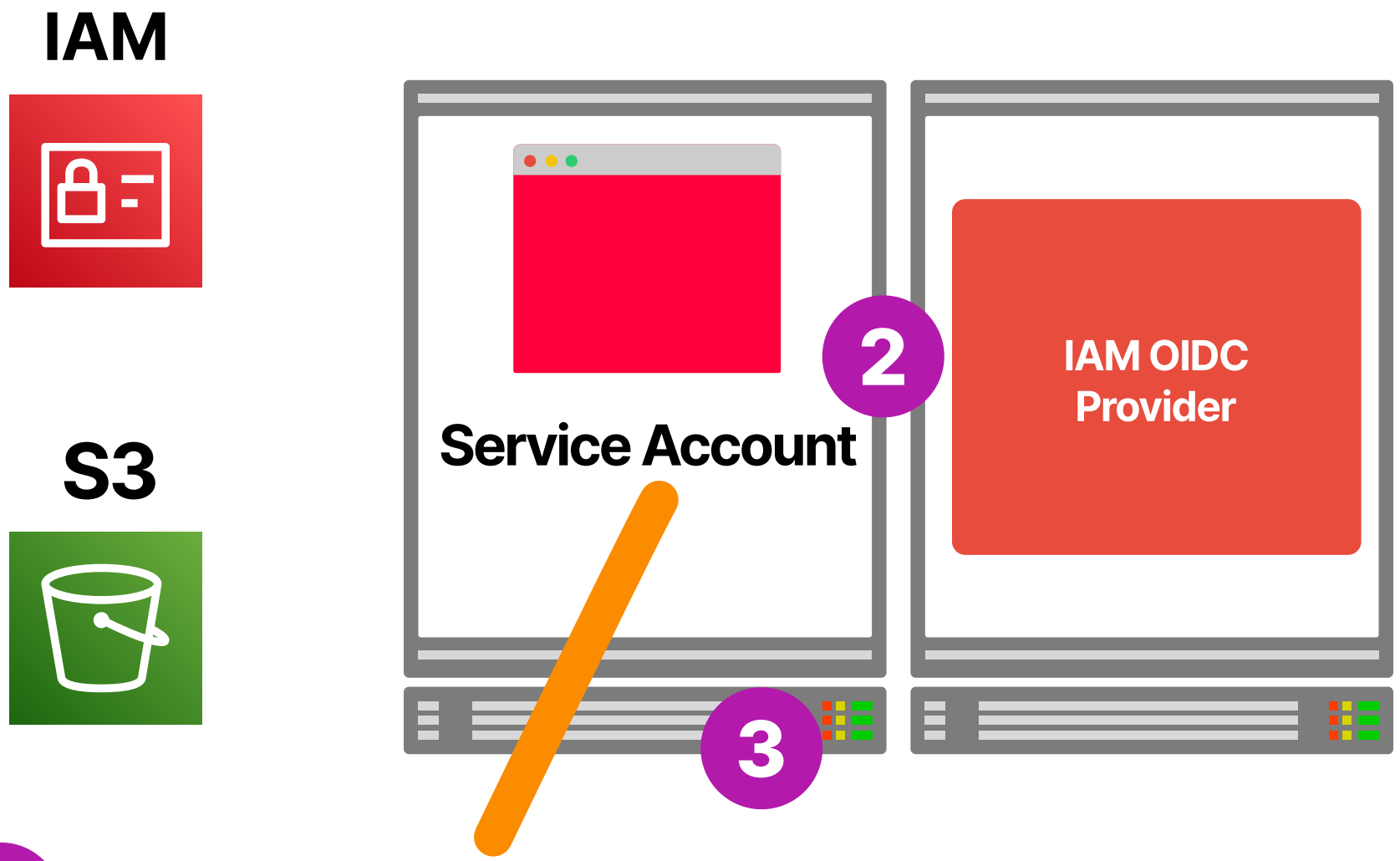


S3

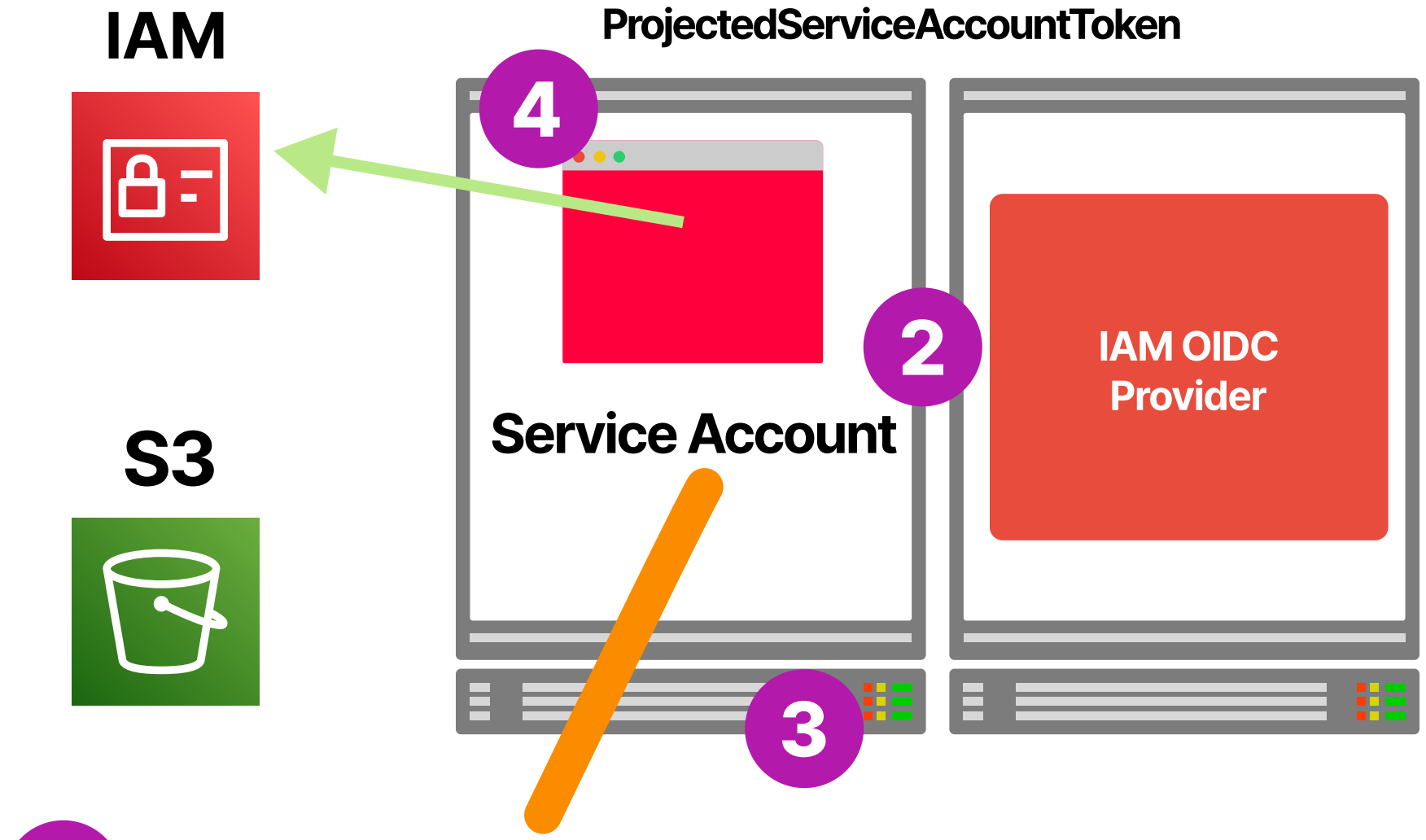




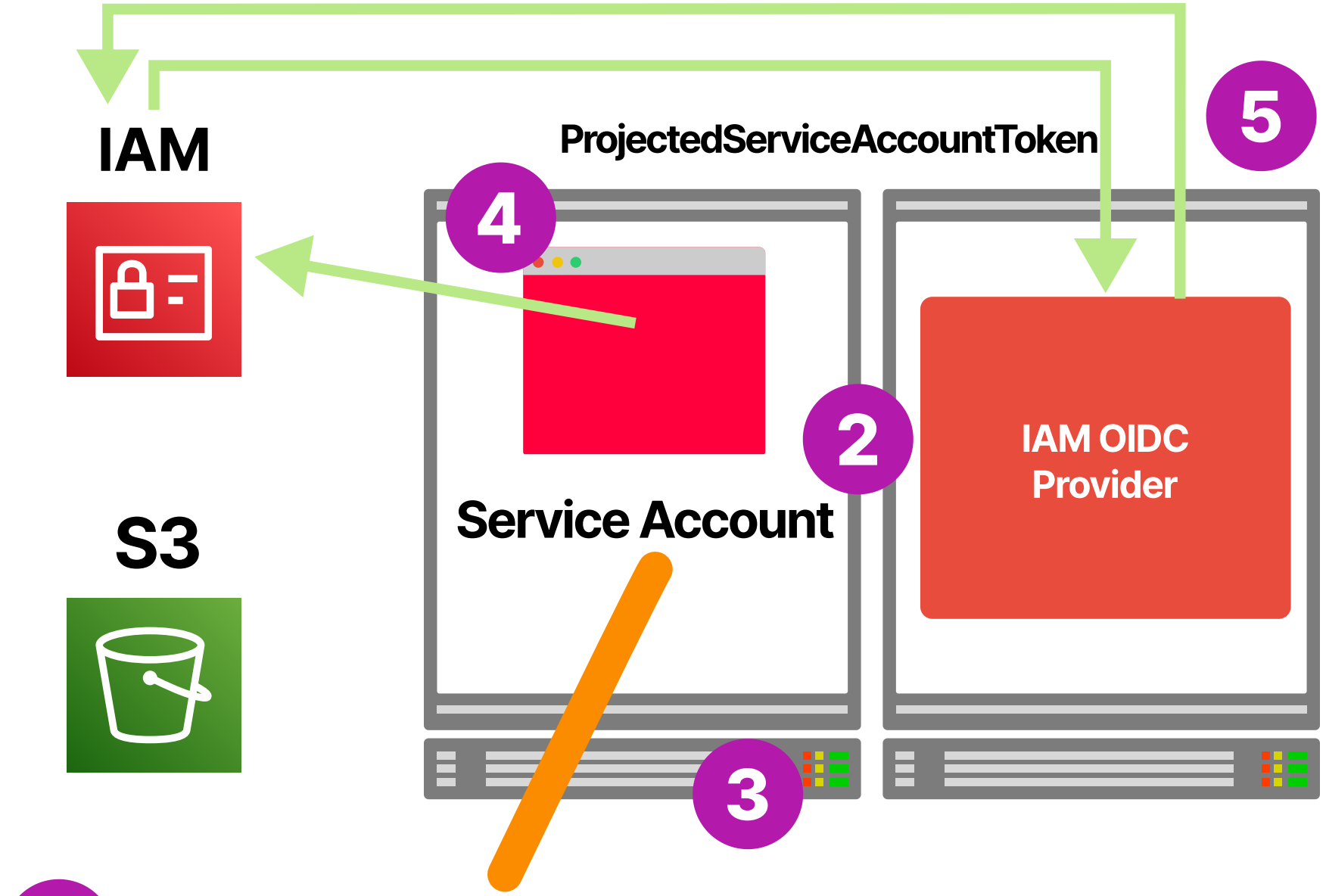
1 Create IAM Role + Policy

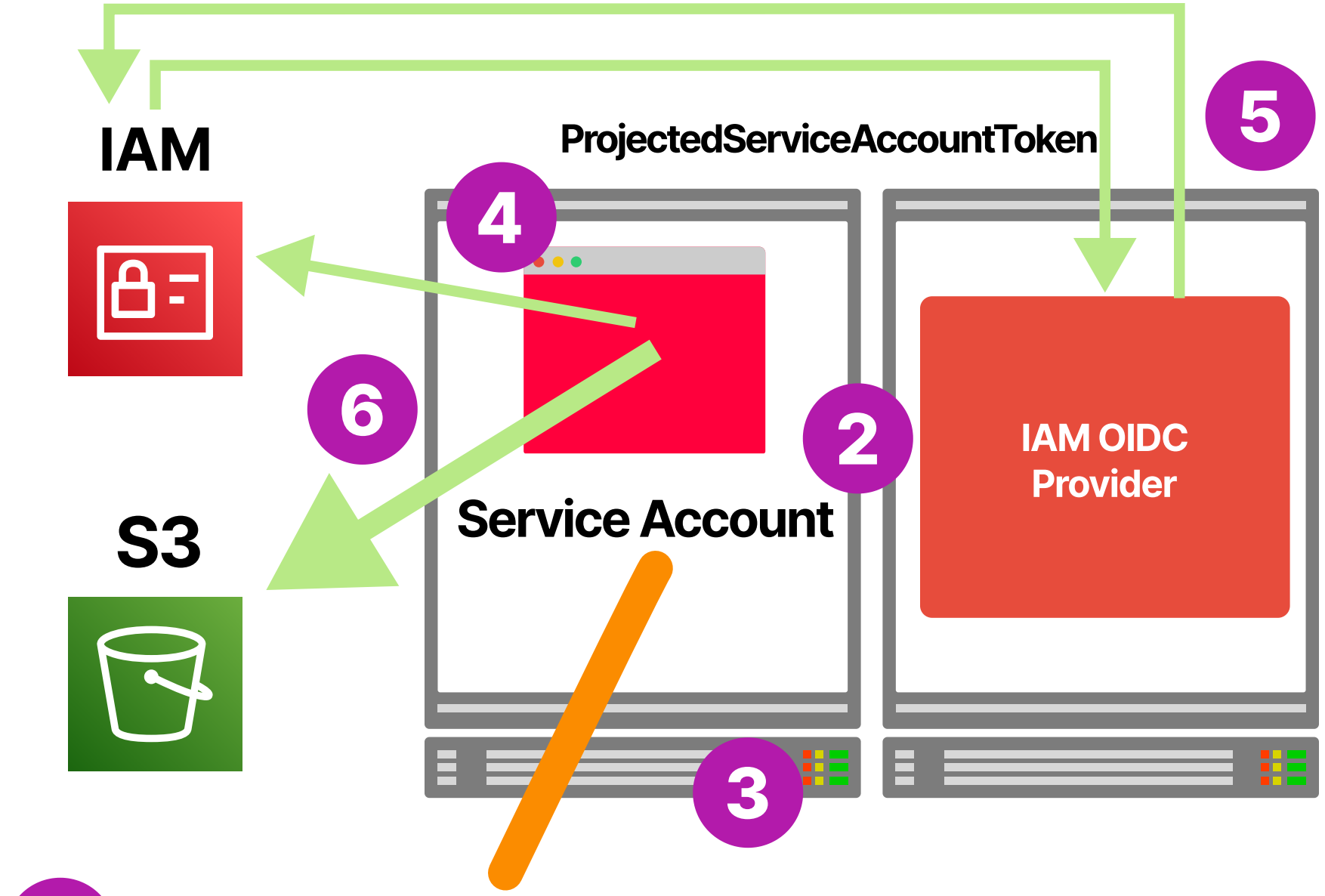


1 Create IAM Role + Policy



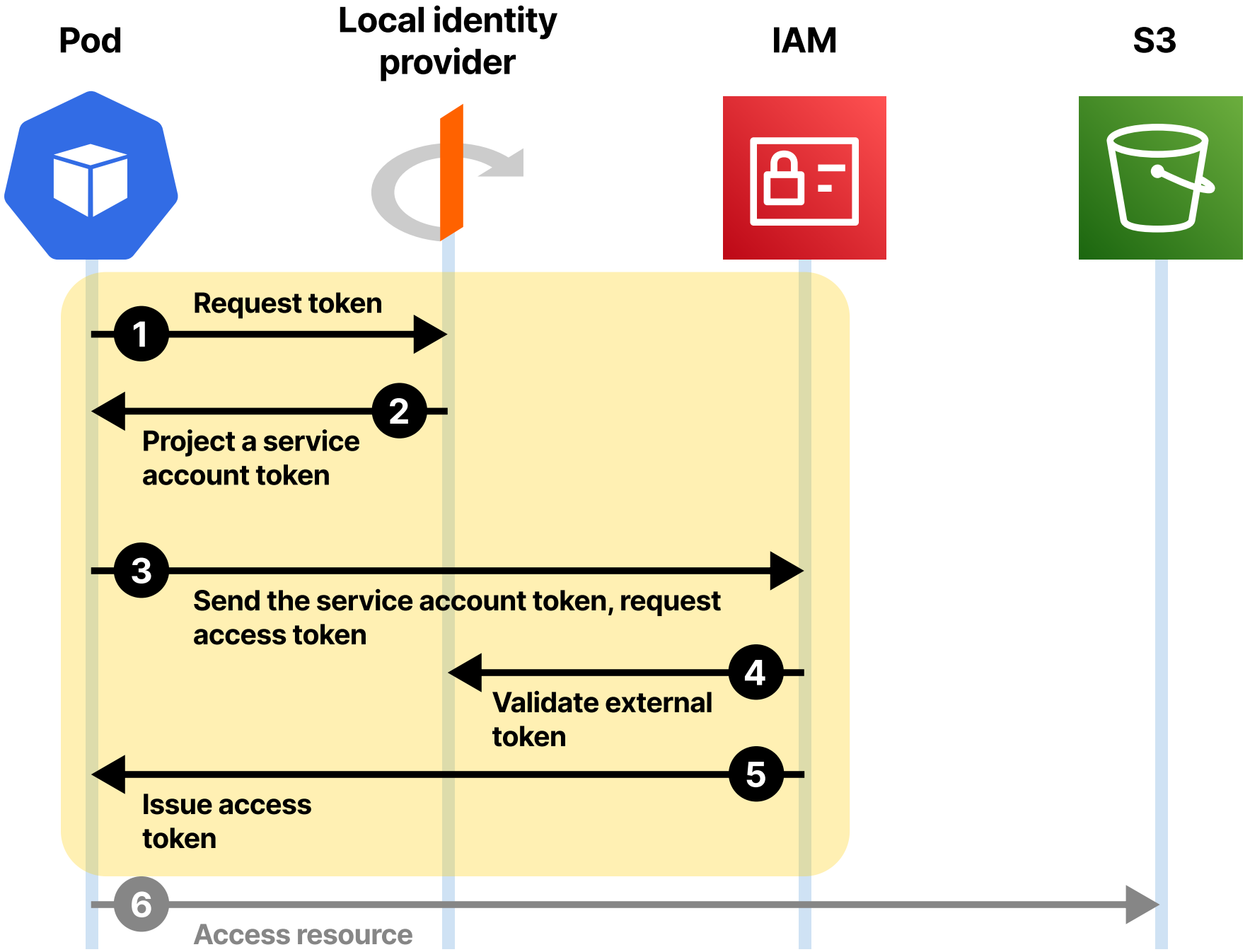
1 Create IAM Role + Policy

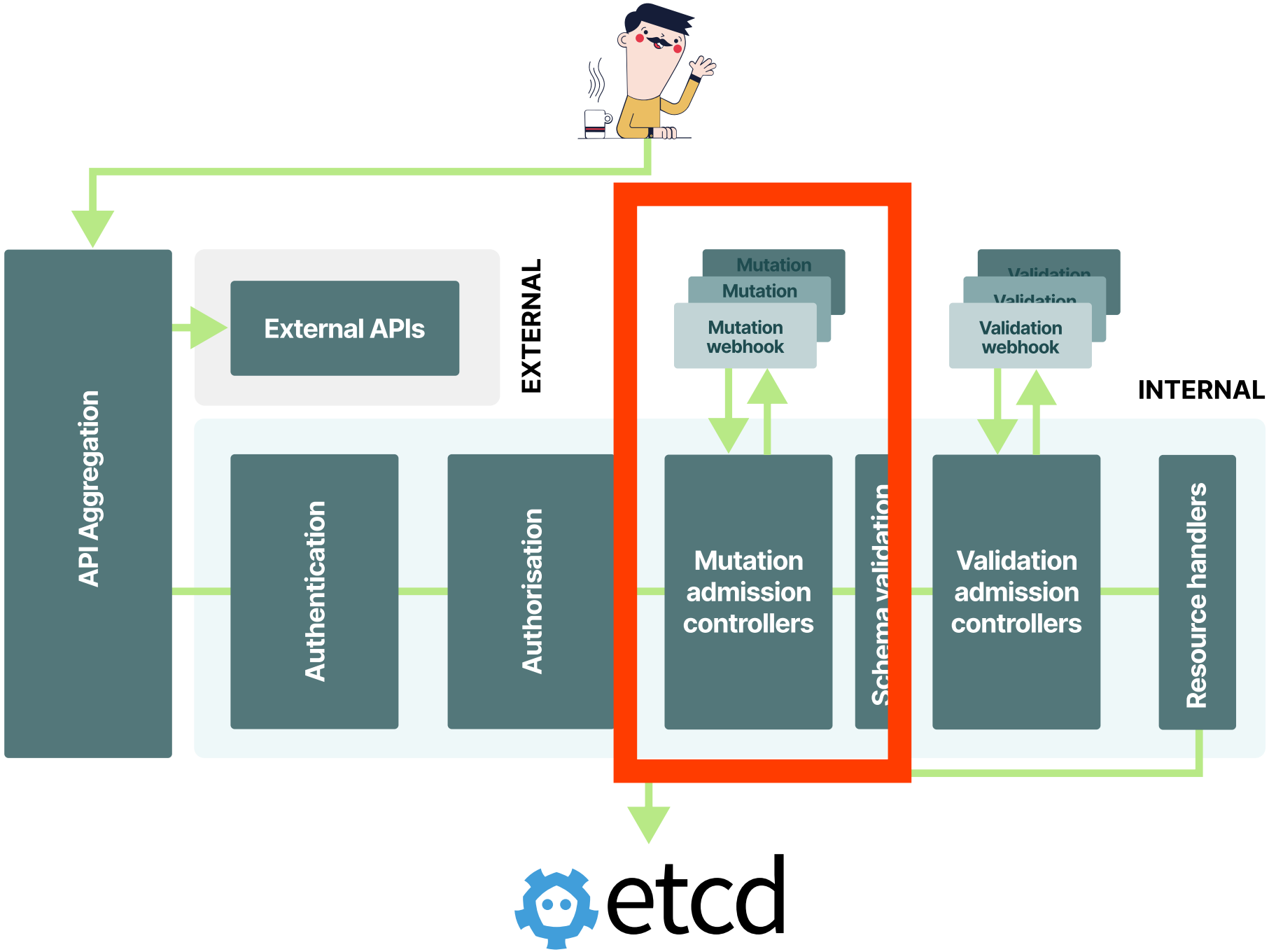

IAM
ProjectedServiceAccountToken
5
4
2
IAM OIDC
Provider
S3
Service Account
3
1
Create IAM Role + Policy


IAM
ProjectedServiceAccountToken
5
4
6
2
IAM OIDC
Provider
S3
Service Account
3
1
Create IAM Role + Policy

IAM roles for Service Accounts

Amazon EKS Pod Identity Webhook


Pod
Local identity provider
IAM
S3
Request token
1
2
Project a service account token
3
Send the service account token, request access token
4
Validate external token
5
Issue access token
6
Access resource


API Aggregation
External APIs
EXTERNAL
Mutation
Mutation
Mutation webhook
Validation
Validation
Validation webhook
INTERNAL
Authentication
Authorisation
Mutation admission controllers
Schema validation
Validation admission controllers
Resource handlers
etcd

```yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  name: my-serviceaccount
  namespace: default
  annotations:
    eks.amazonaws.com/role-arn: "arn:aws:iam::111122223333:role/s3-reader"
    # optional: Defaults to "sts.amazonaws.com" if not set
    eks.amazonaws.com/audience: "sts.amazonaws.com"
    # optional: When set to "true", adds AWS_STS_REGIONAL_ENDPOINTS env var
    #   to containers
    eks.amazonaws.com/sts-regional-endpoints: "true"
    # optional: Defaults to 86400 for expirationSeconds if not set
    #   Note: This value can be overwritten if specified in the pod
    #         annotation as shown in the next step.
    eks.amazonaws.com/token-expiration: "86400"
```

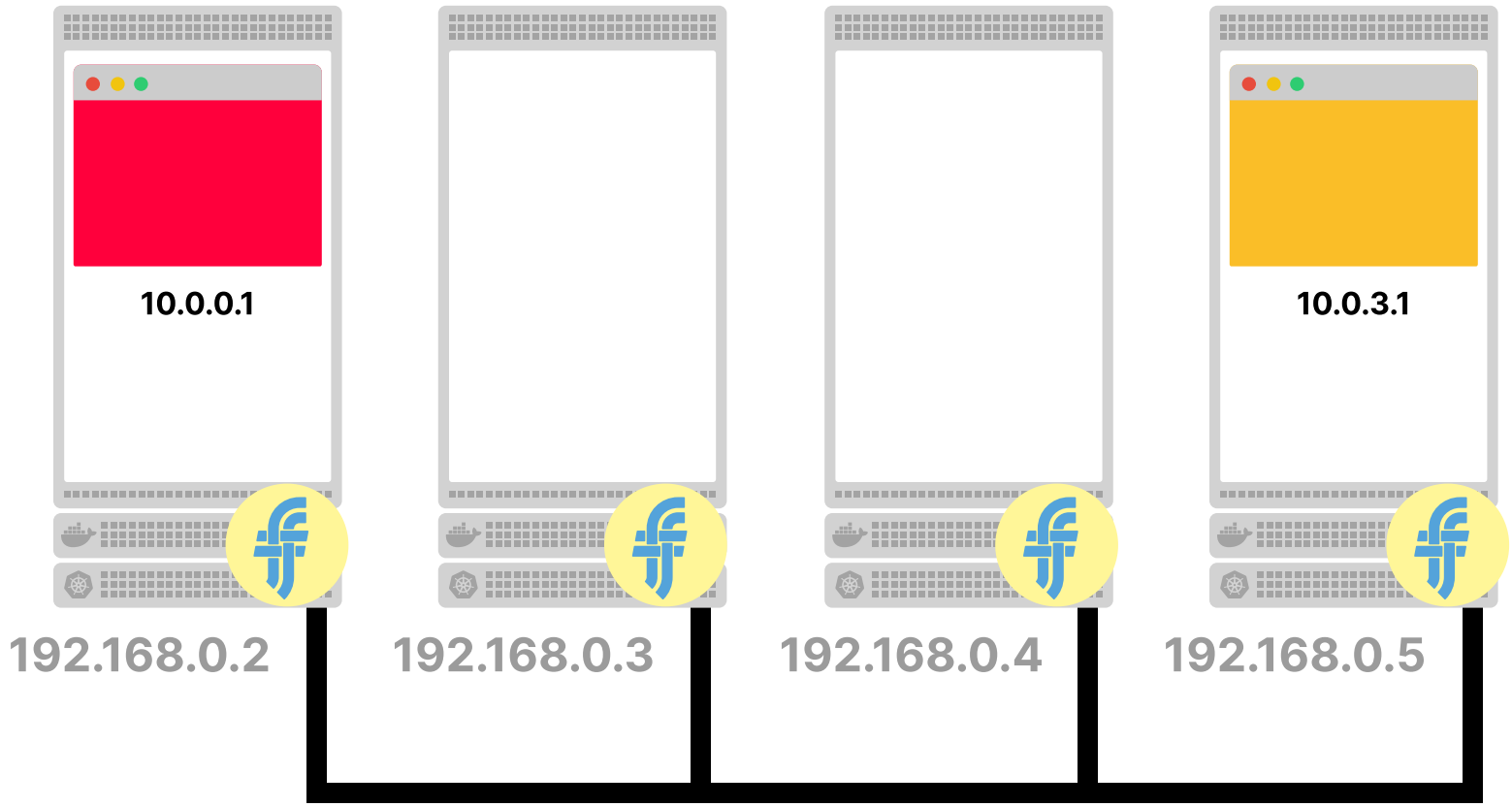
```yaml
apiVersion: v1
kind: Pod
metadata:
  name: my-pod
spec:
  serviceAccountName: my-serviceaccount
  containers:
  - name: container-name
    image: container-image:version
    env:
    - name: AWS_DEFAULT_REGION
      value: us-west-2
    - name: AWS_REGION
      value: us-west-2
    - name: AWS_ROLE_ARN
      value: "arn:aws:iam::111122223333:role/s3-reader"
    - name: AWS_WEB_IDENTITY_TOKEN_FILE
      value: "/var/run/secrets/eks.amazonaws.com/serviceaccount/token"
    - name: AWS_STS_REGIONAL_ENDPOINTS
      value: "regional"
    volumeMounts:
    - mountPath: "/var/run/secrets/eks.amazonaws.com/serviceaccount/"
      name: aws-token
  volumes:
  - name: aws-token
    projected:
      sources:
      - serviceAccountToken:
          audience: "sts.amazonaws.com"
          expirationSeconds: 86400
          path: token
```

INJECTED

aws/amazon-eks-pod-identity-webhook

Amazon EKS Pod Identity Webhook

Networking



FROM:

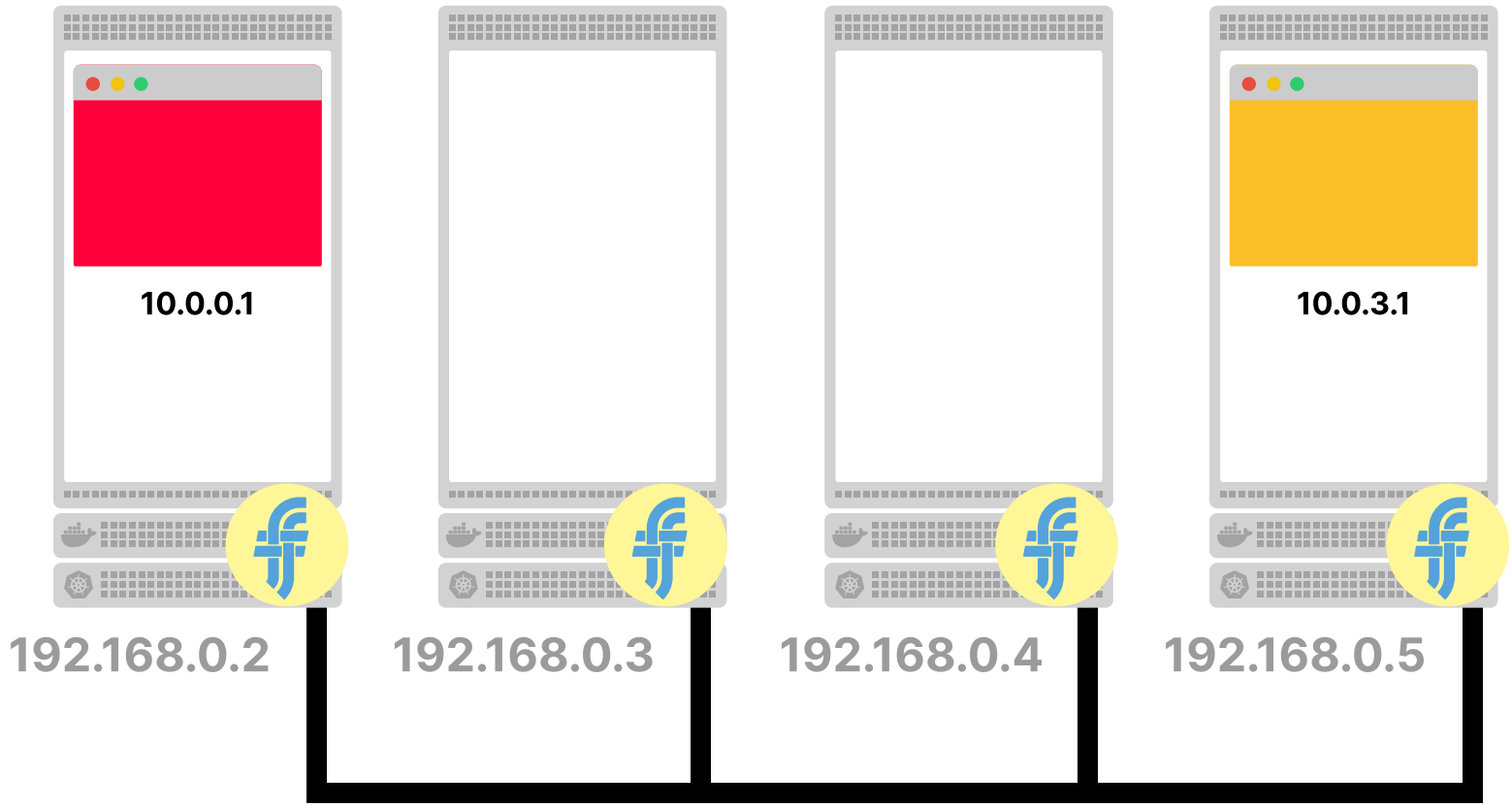
Pod ■ **(10.0.0.1)**

TO:

Pod ■ **(10.0.3.1)**

FROM: Pod1
TO: Pod2

Payload

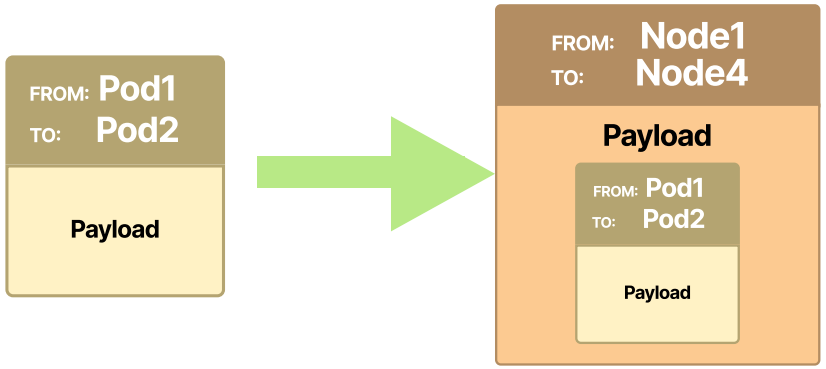


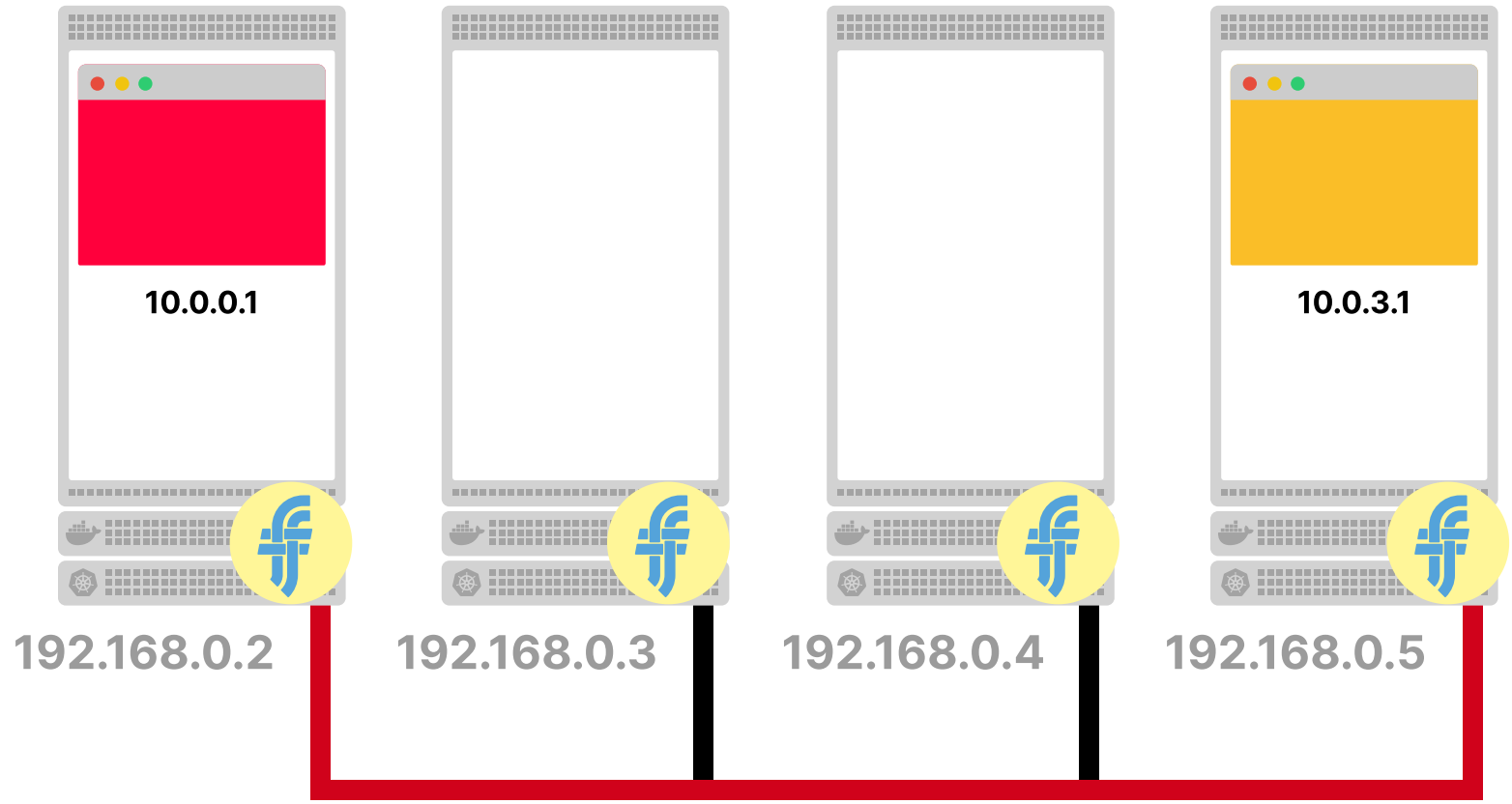
FROM:

Pod ■ **(10.0.0.1)**

TO:

Pod ■ **(10.0.3.1)**





FROM:

Pod (10.0.0.1)

TO:

Pod (10.0.3.1)

Overlay networks in AWS

Kubernetes network

Flannel network

VPC network

Overlay networks in AWS

Kubernetes network

Flannel network

VPC network

Overlay networks in AWS

Kubernetes network

Flannel network

VPC network

2 nested network overlays

Overlay networks in AWS

Overlay networks in AWS

VPC flow logs



Security groups

Extra point of failure

Overlay networks in AWS

VPC flow logs ✗

Security groups ✗

Extra point of failure

Overlay networks in AWS

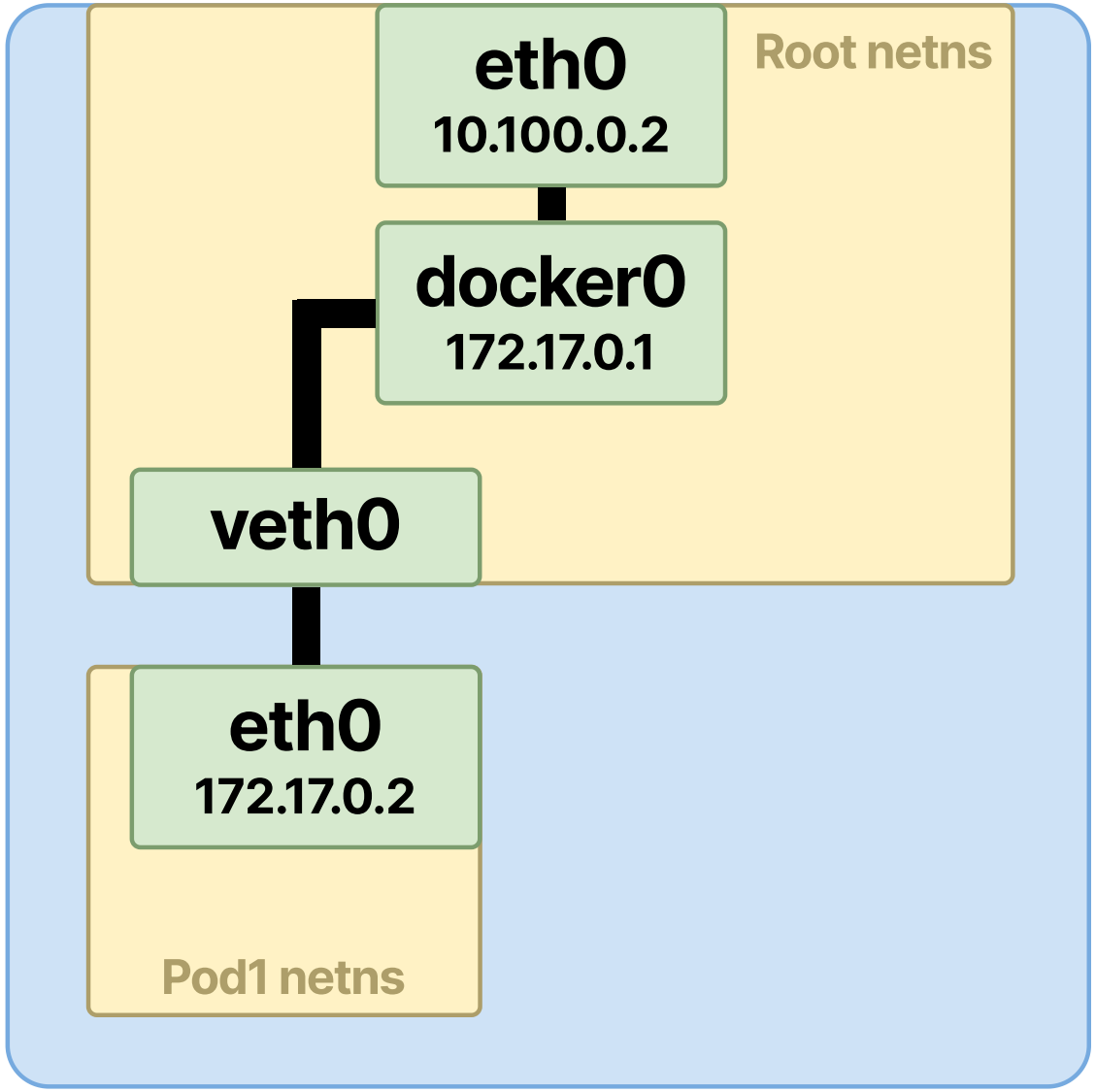
VPC flow logs 

Security groups 

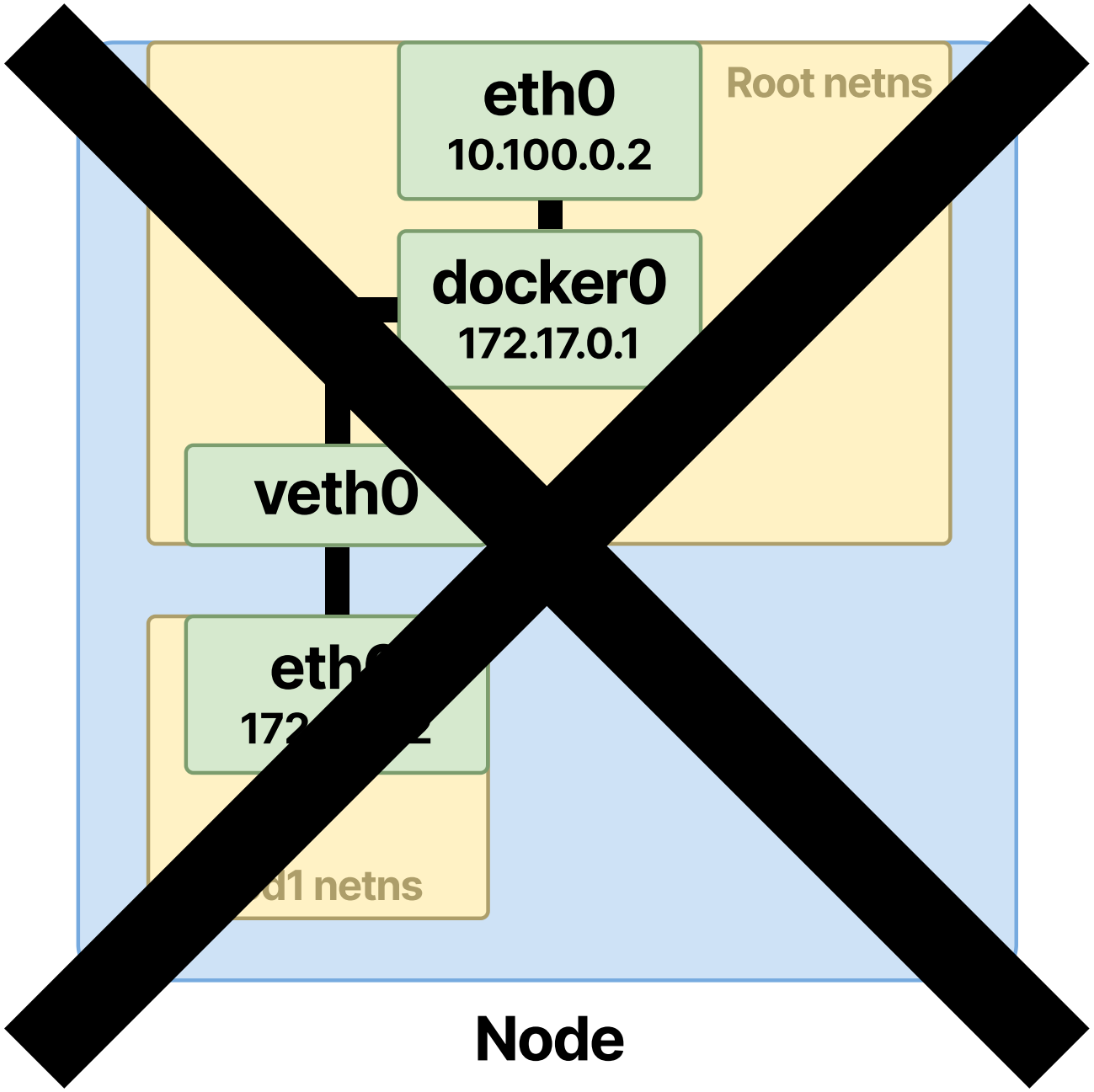
Extra point of failure 

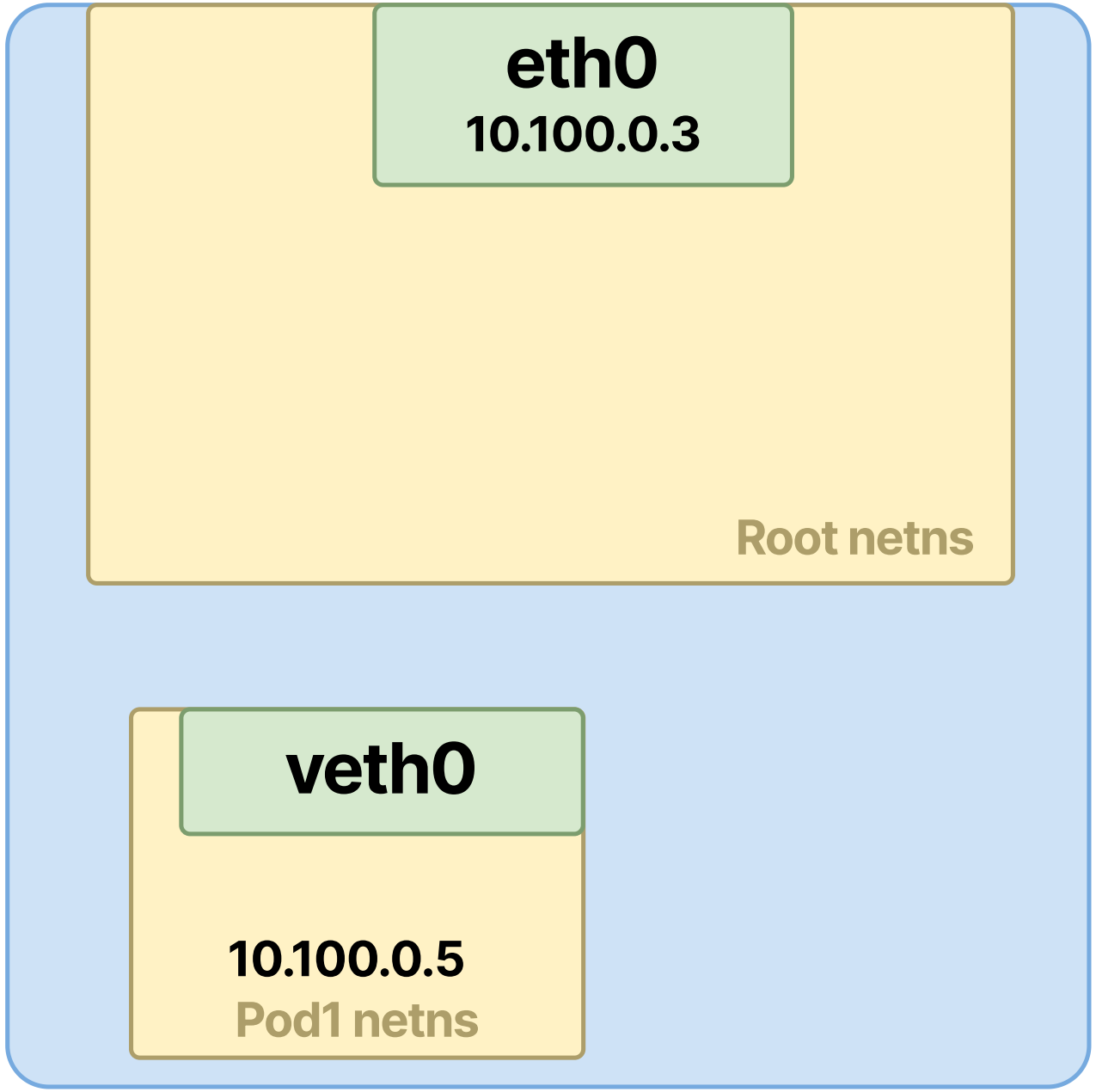
Networking

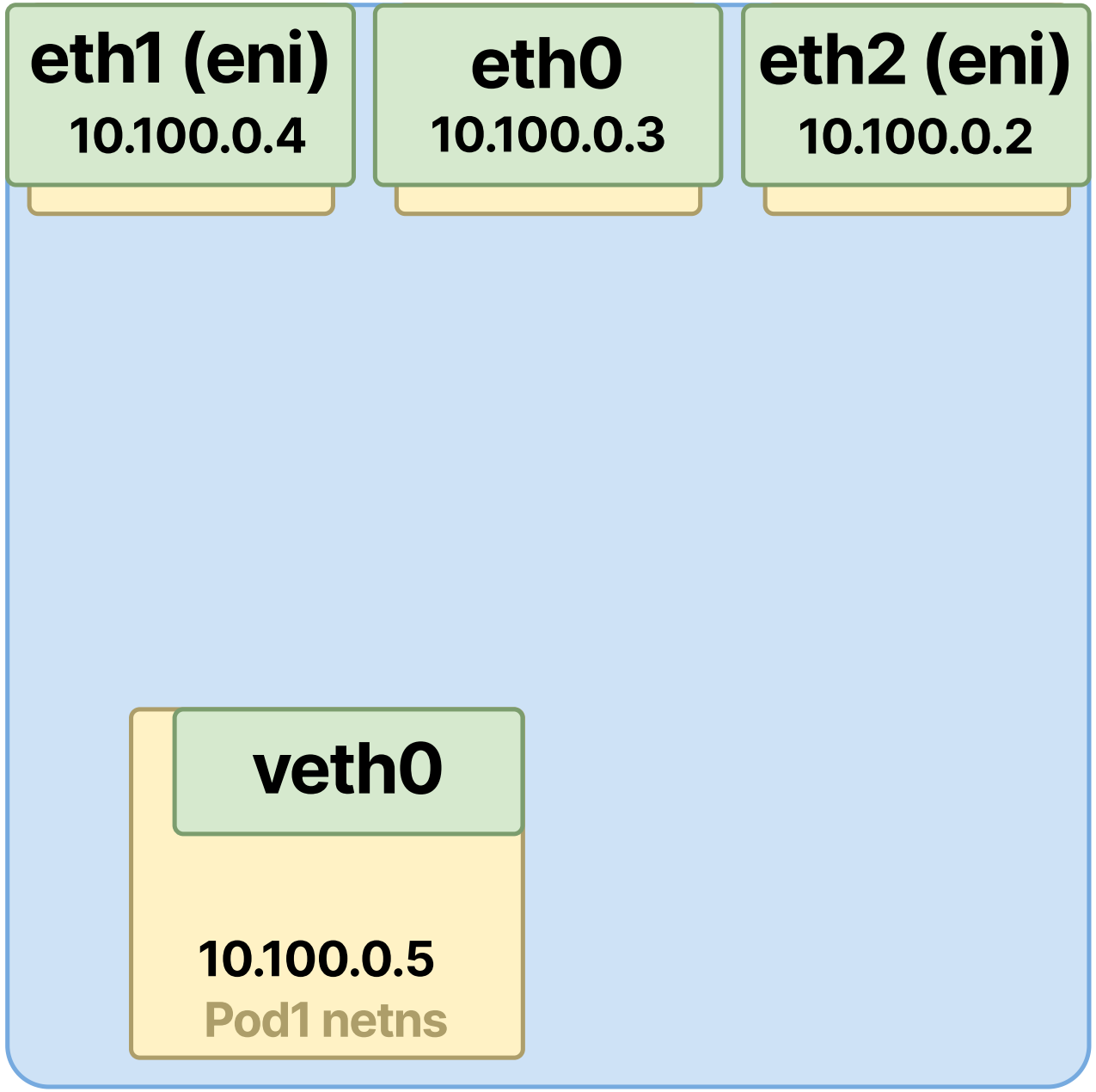
AWS CNI

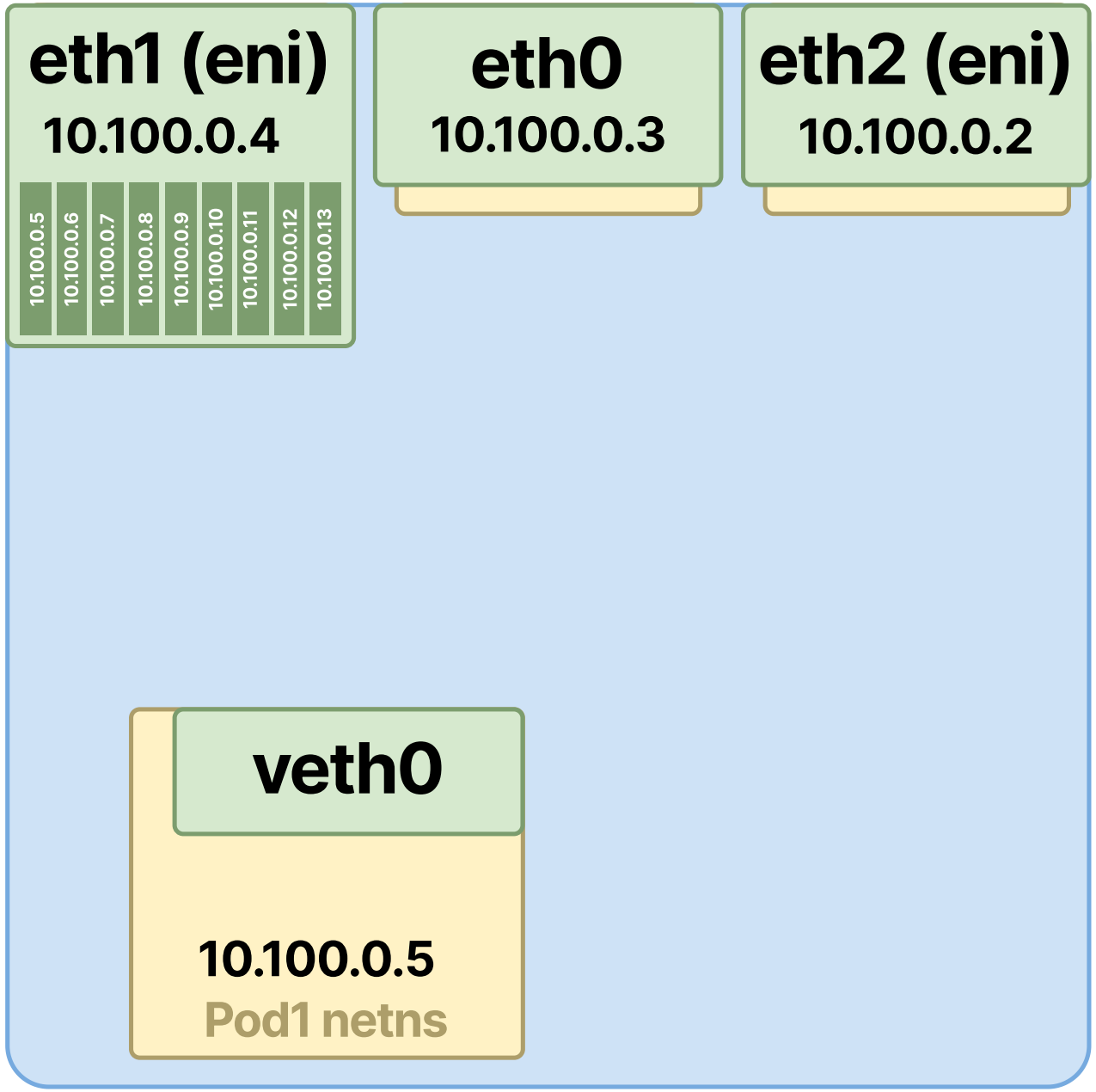

eth0
10.100.0.2
Root netns
docker0
172.17.0.1
veth0
eth0
172.17.0.2
Pod1 netns

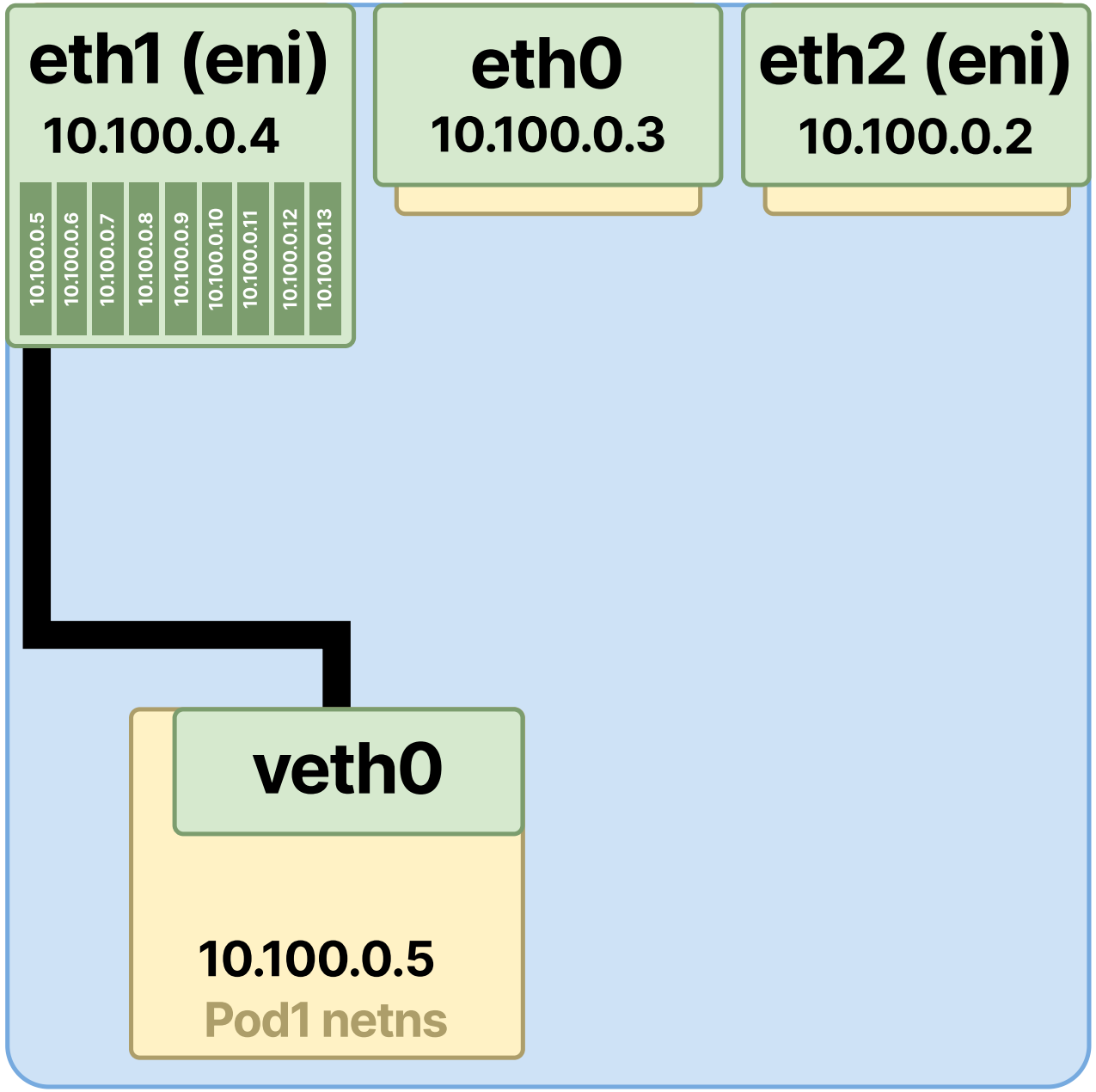
Node

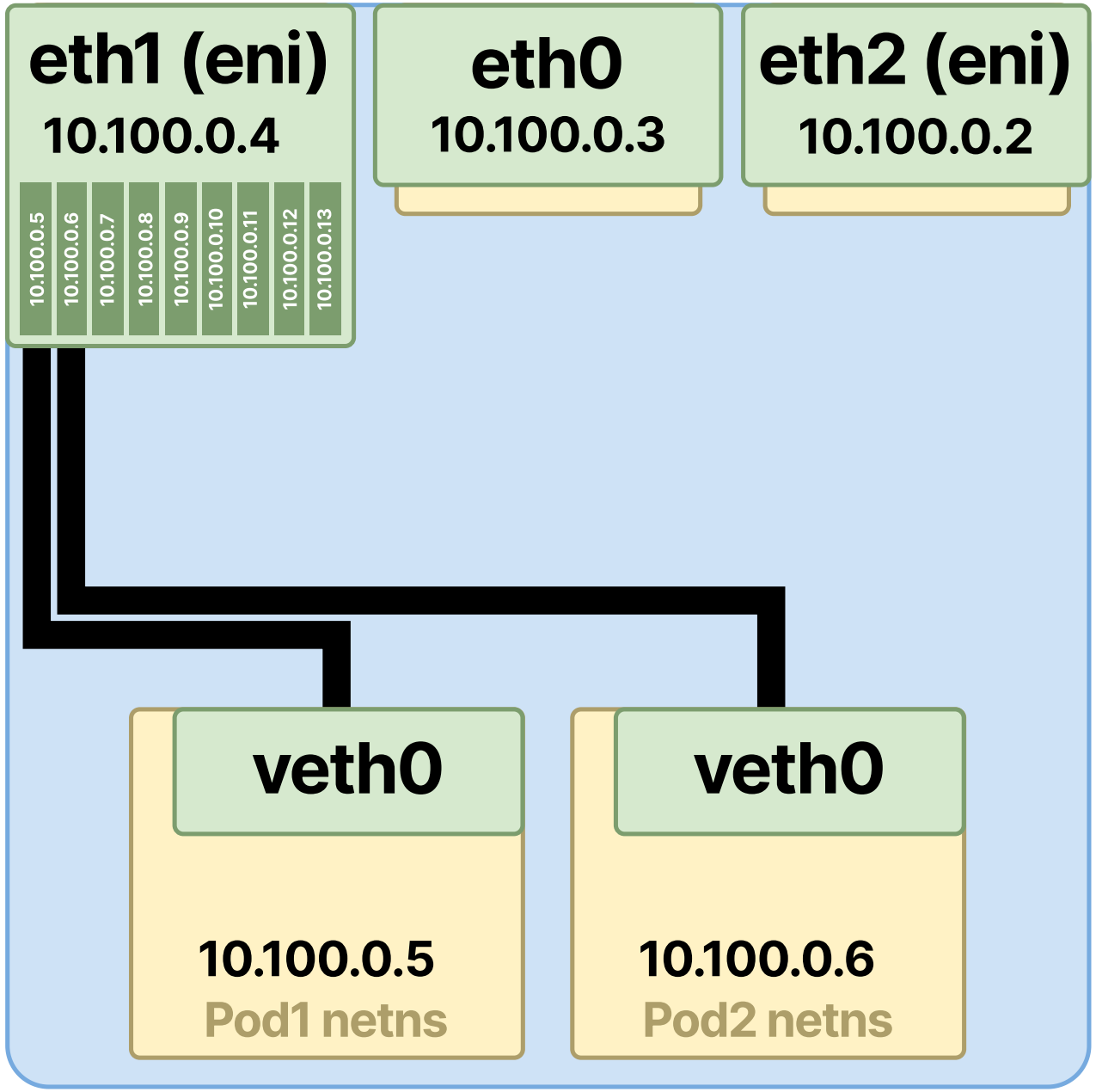

eth0
10.100.0.2
Root netns
docker0
172.17.0.1
veth0
Node

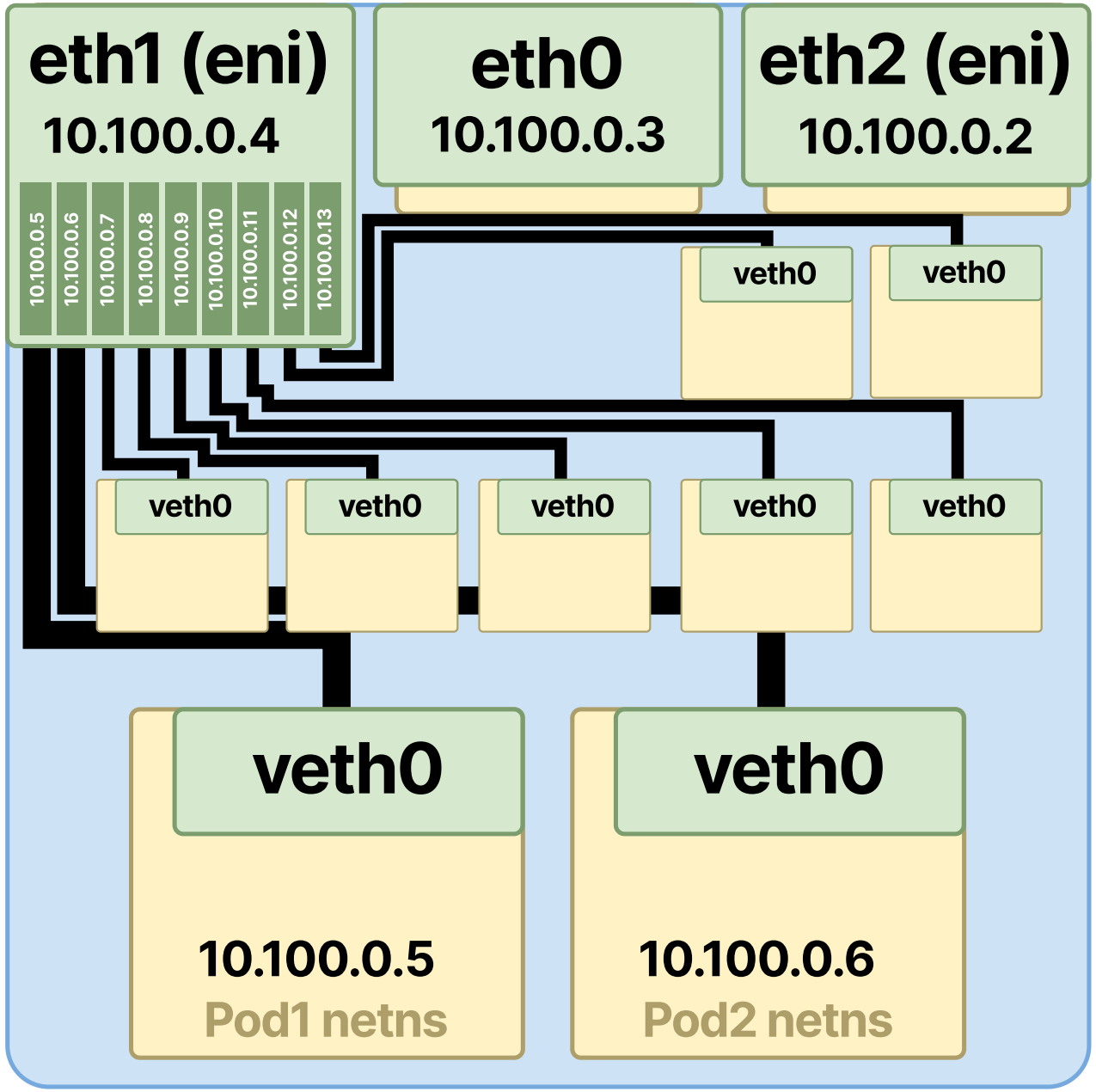

eth0
10.100.0.3
Root netns
veth0
10.100.0.5
Pod1 netns
m5.large


eth1 (eni)
10.100.0.4
eth0
10.100.0.3
eth2 (eni)
10.100.0.2
veth0
10.100.0.5
Pod1 netns
m5.large


eth1 (eni)
10.100.0.4
10.100.0.5
10.100.0.6
10.100.0.7
10.100.0.8
10.100.0.9
10.100.0.10
10.100.0.11
10.100.0.12
10.100.0.13
eth0
10.100.0.3
eth2 (eni)
10.100.0.2
veth0
10.100.0.5
Pod1 netns
m5.large


eth1 (eni)
10.100.0.4
10.100.0.5
10.100.0.6
10.100.0.7
10.100.0.8
10.100.0.9
10.100.0.10
10.100.0.11
10.100.0.12
10.100.0.13
eth0
10.100.0.3
eth2 (eni)
10.100.0.2
veth0
10.100.0.5
Pod1 netns
m5.large


eth1 (eni)
10.100.0.4
10.100.0.5
10.100.0.6
10.100.0.7
10.100.0.8
10.100.0.9
10.100.0.10
10.100.0.11
10.100.0.12
10.100.0.13
eth0
10.100.0.3
eth2 (eni)
10.100.0.2
veth0
10.100.0.5
Pod1 netns
veth0
10.100.0.6
Pod2 netns
m5.large


eth1 (eni)
10.100.0.4
10.100.0.5
10.100.0.6
10.100.0.7
10.100.0.8
10.100.0.9
10.100.0.10
10.100.0.11
10.100.0.12
10.100.0.13
eth0
10.100.0.3
eth2 (eni)
10.100.0.2
veth0
veth0
veth0
veth0
veth0
veth0
veth0
veth0
10.100.0.5
Pod1 netns
10.100.0.6
Pod2 netns

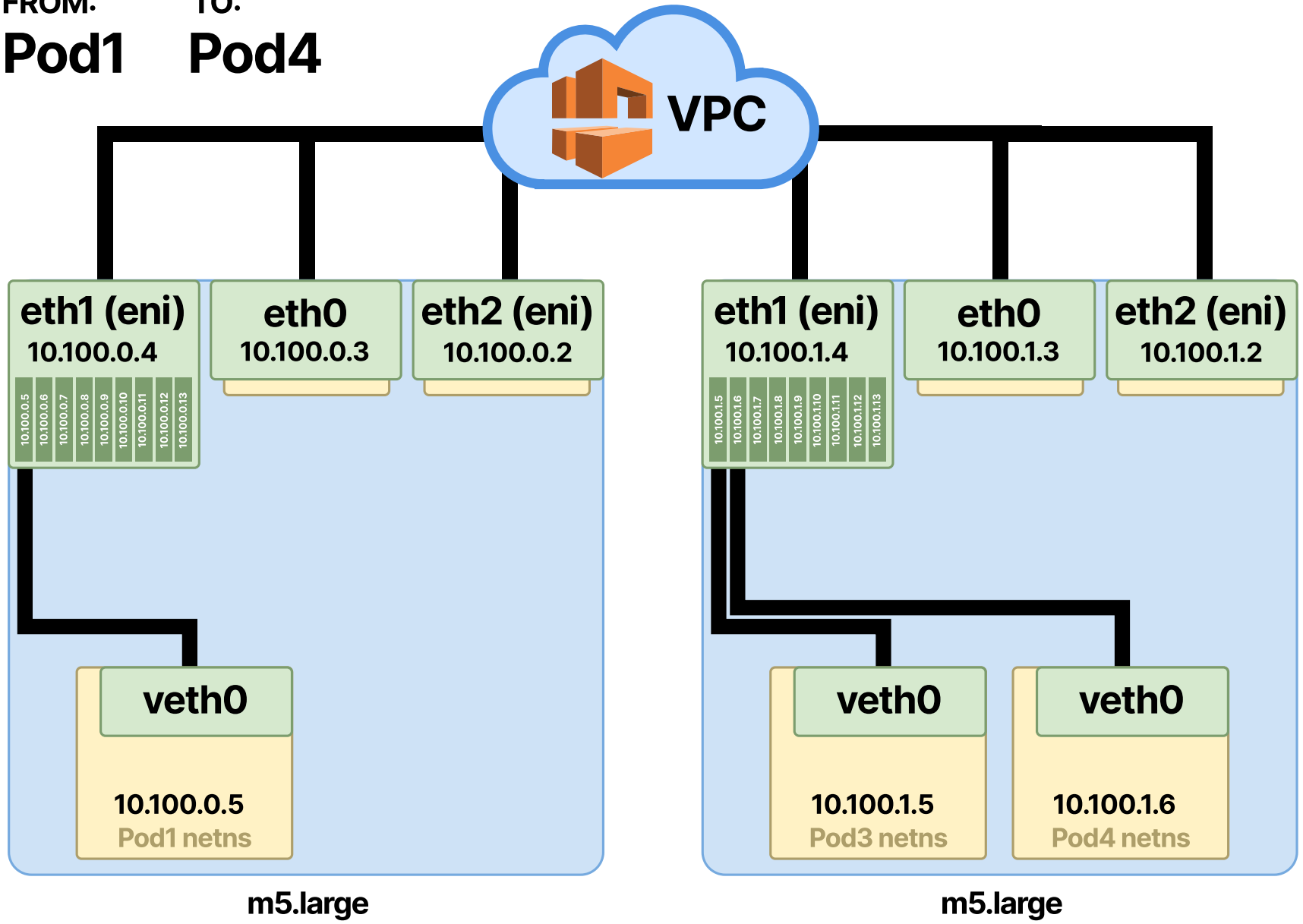
m5.large

AWS-CNI

Pod to Pod

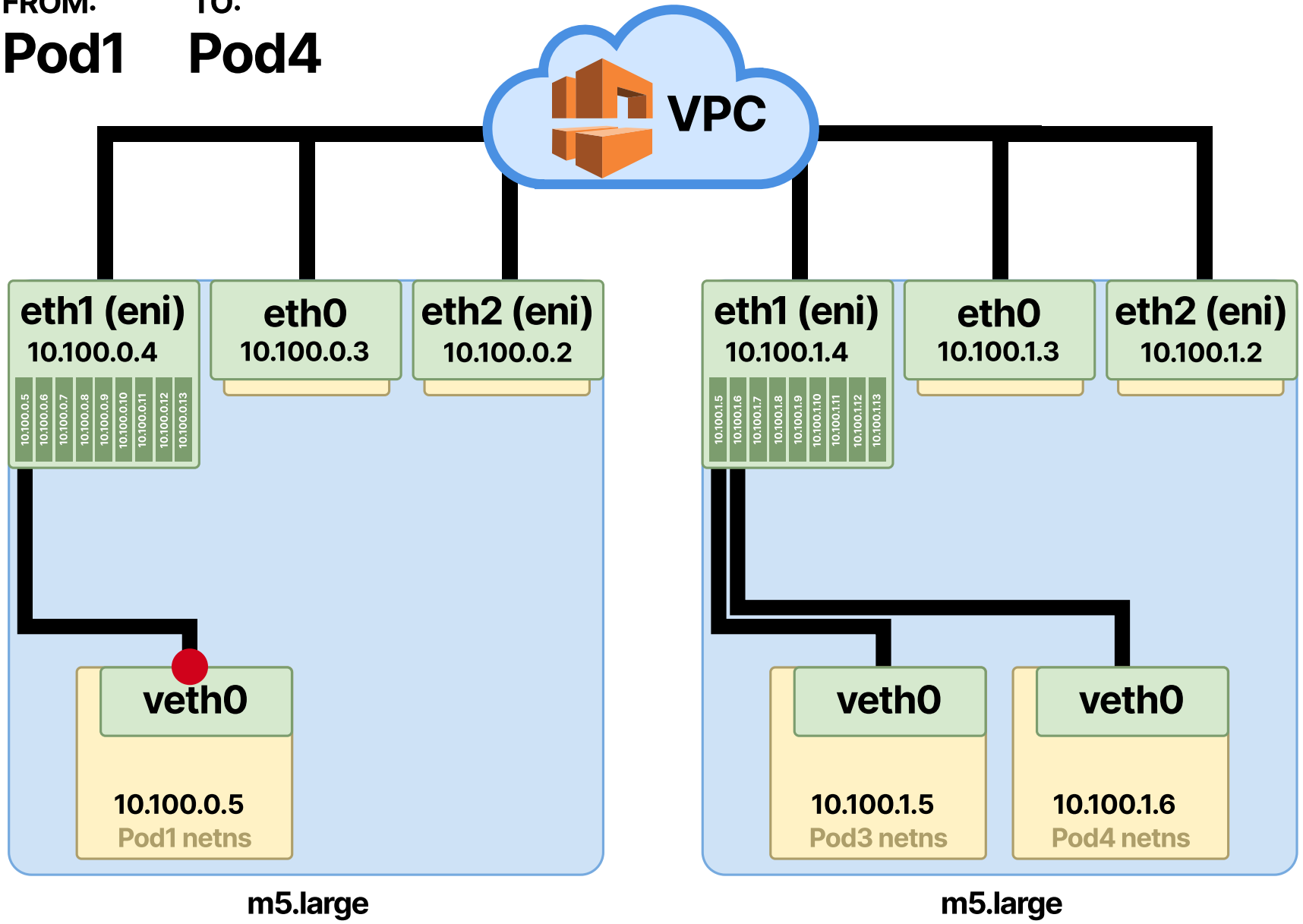
FROM: Pod1

TO: Pod4



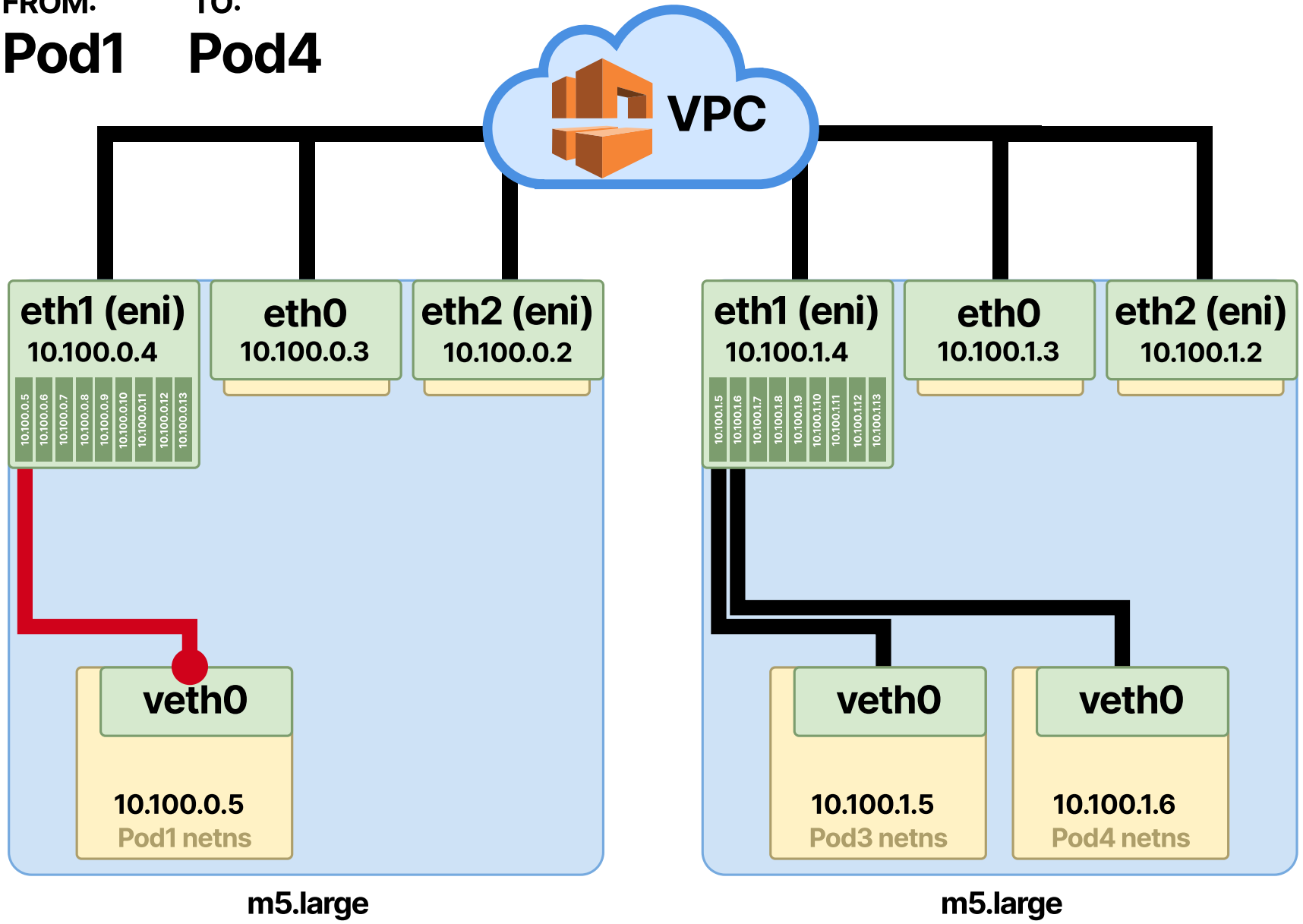
FROM: Pod1

TO: Pod4



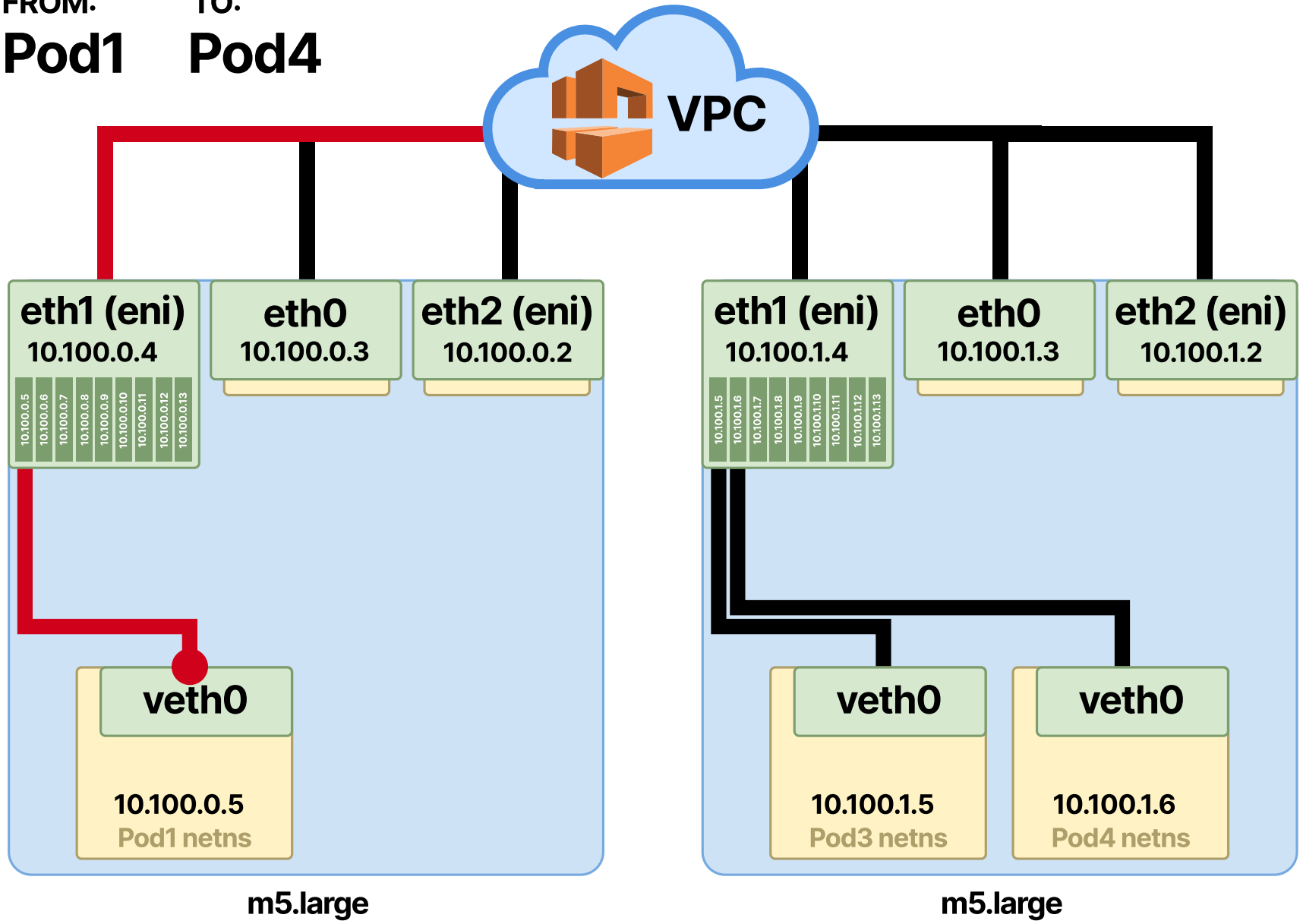
FROM: Pod1

TO: Pod4



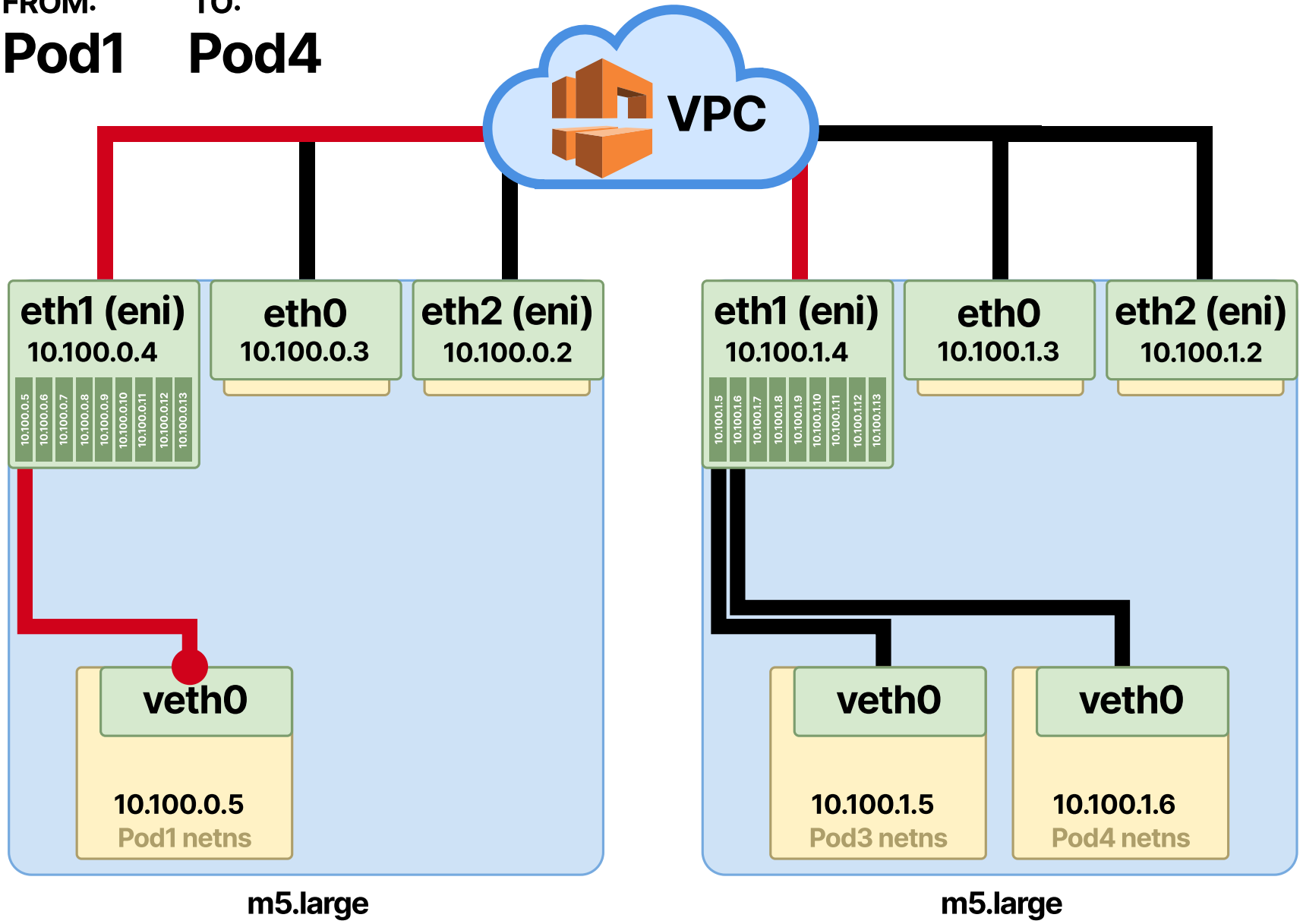
FROM: Pod1

TO: Pod4



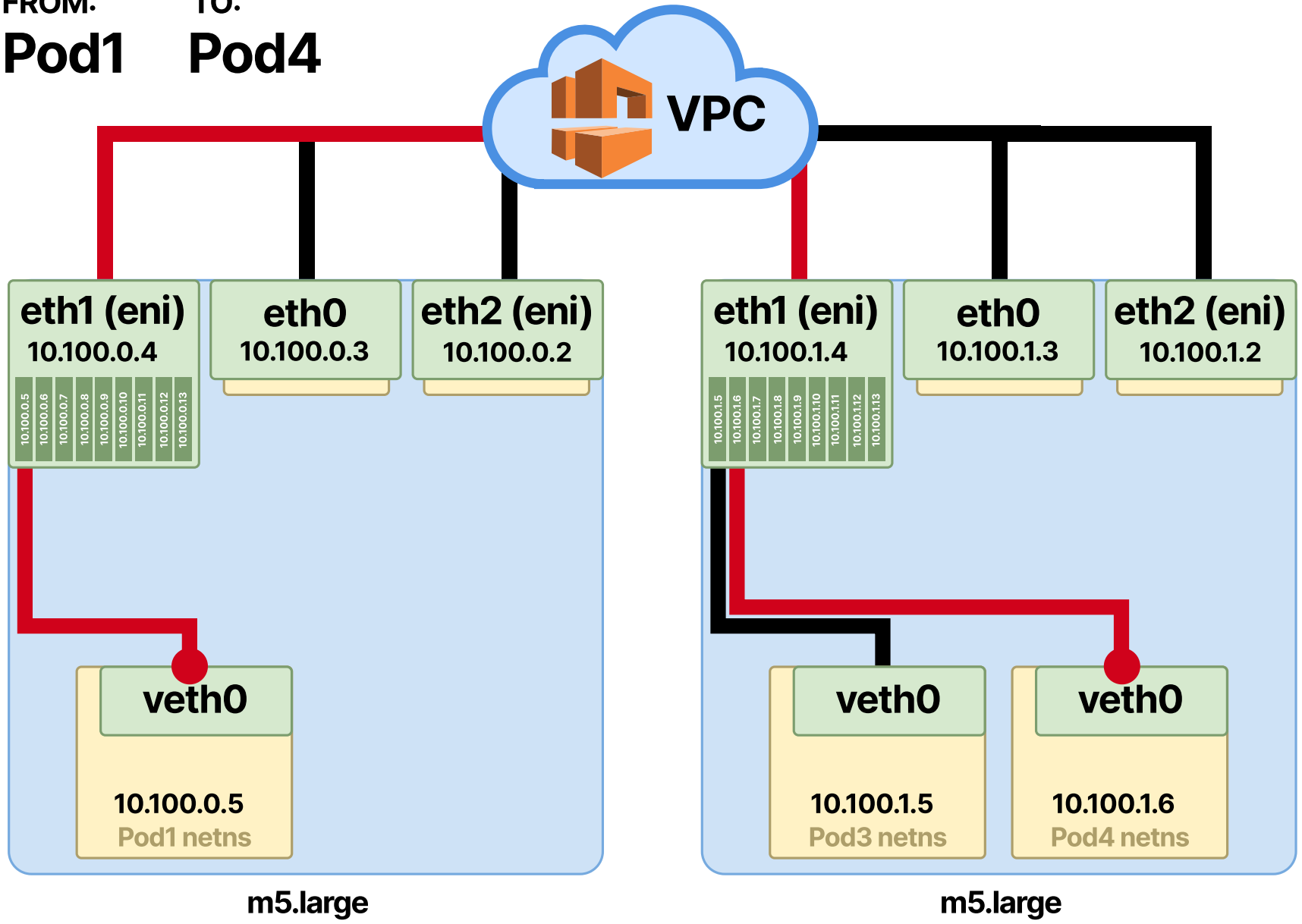
FROM: Pod1

TO: Pod4



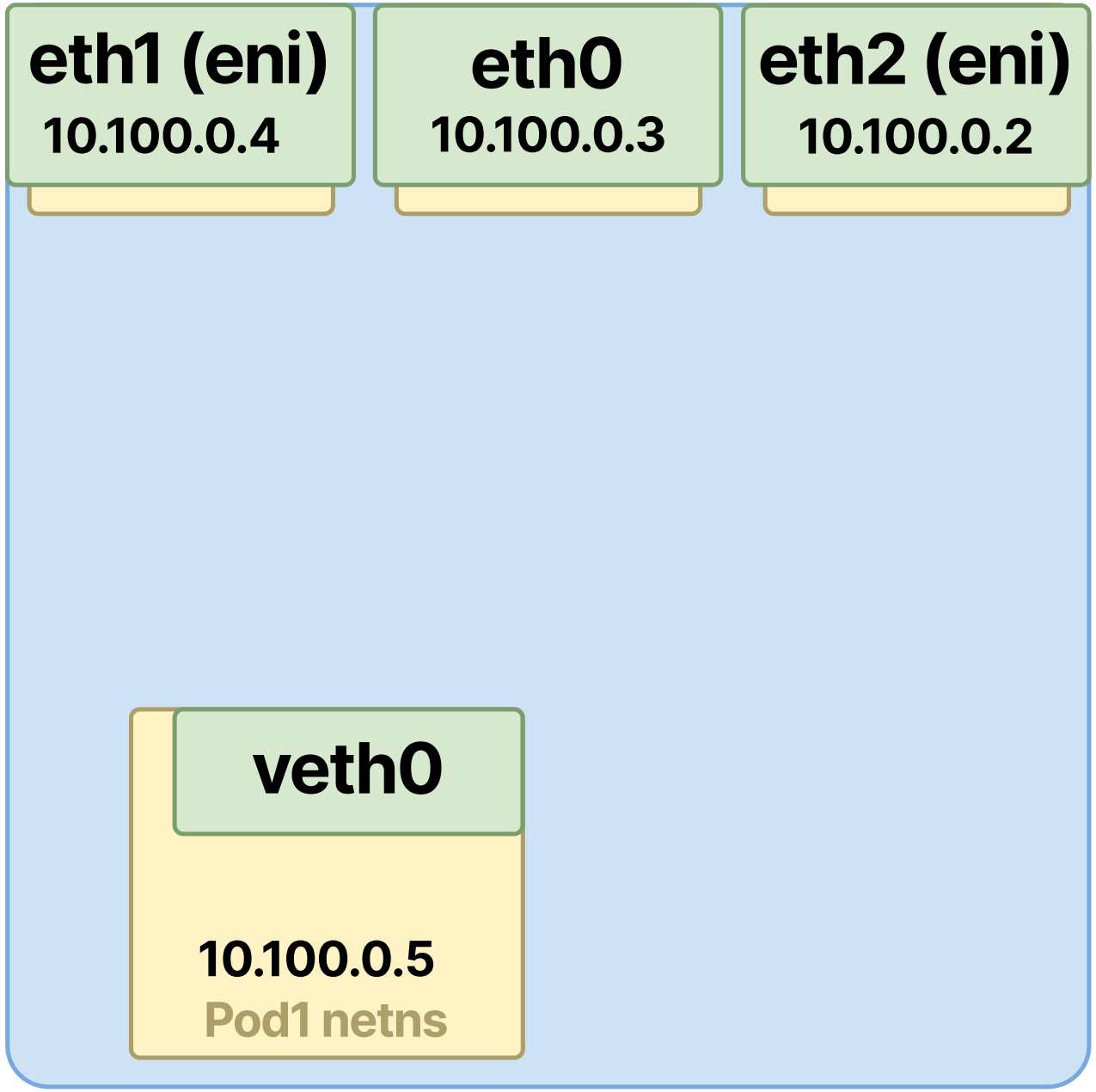
FROM: Pod1

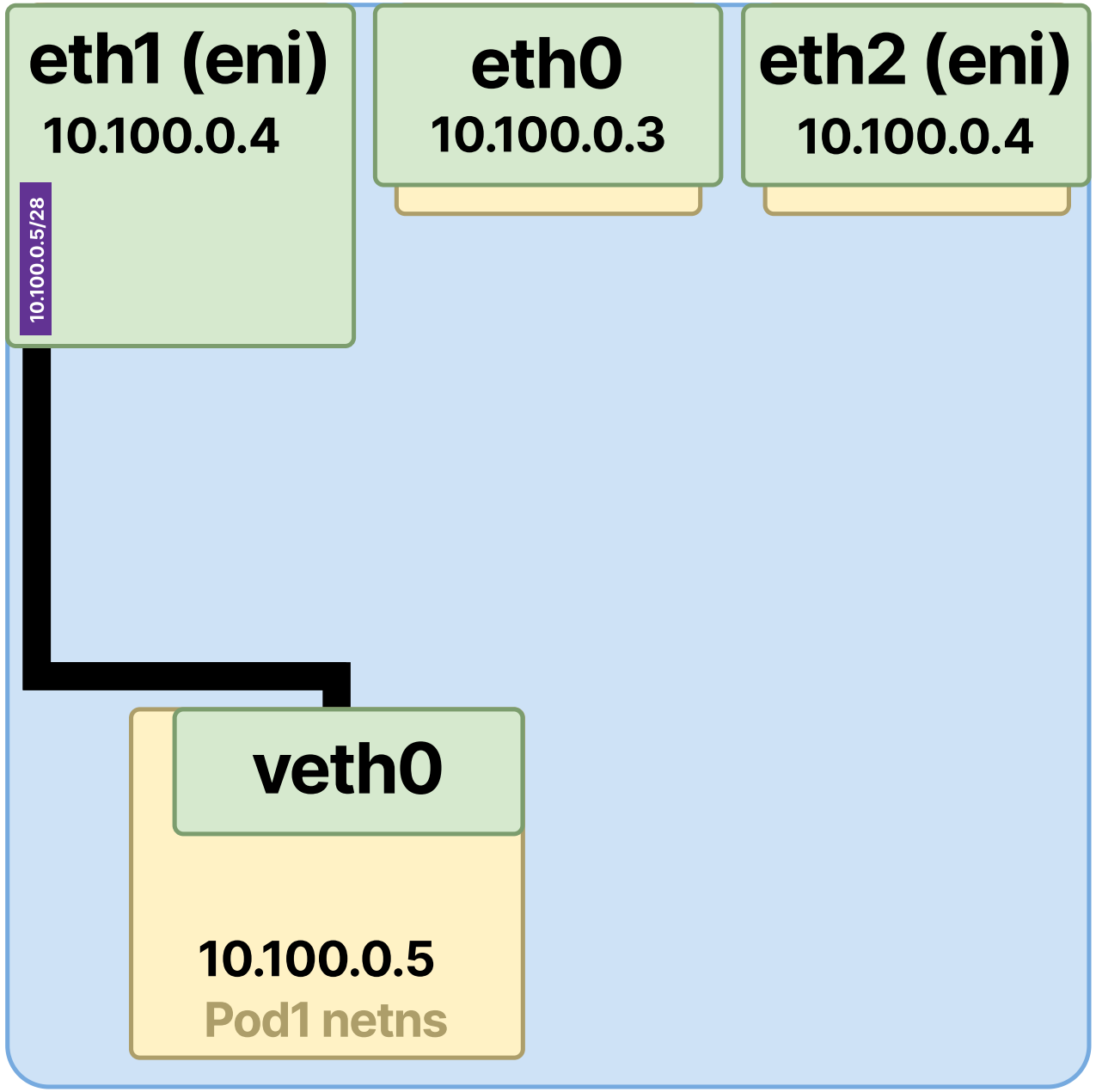
TO: Pod4

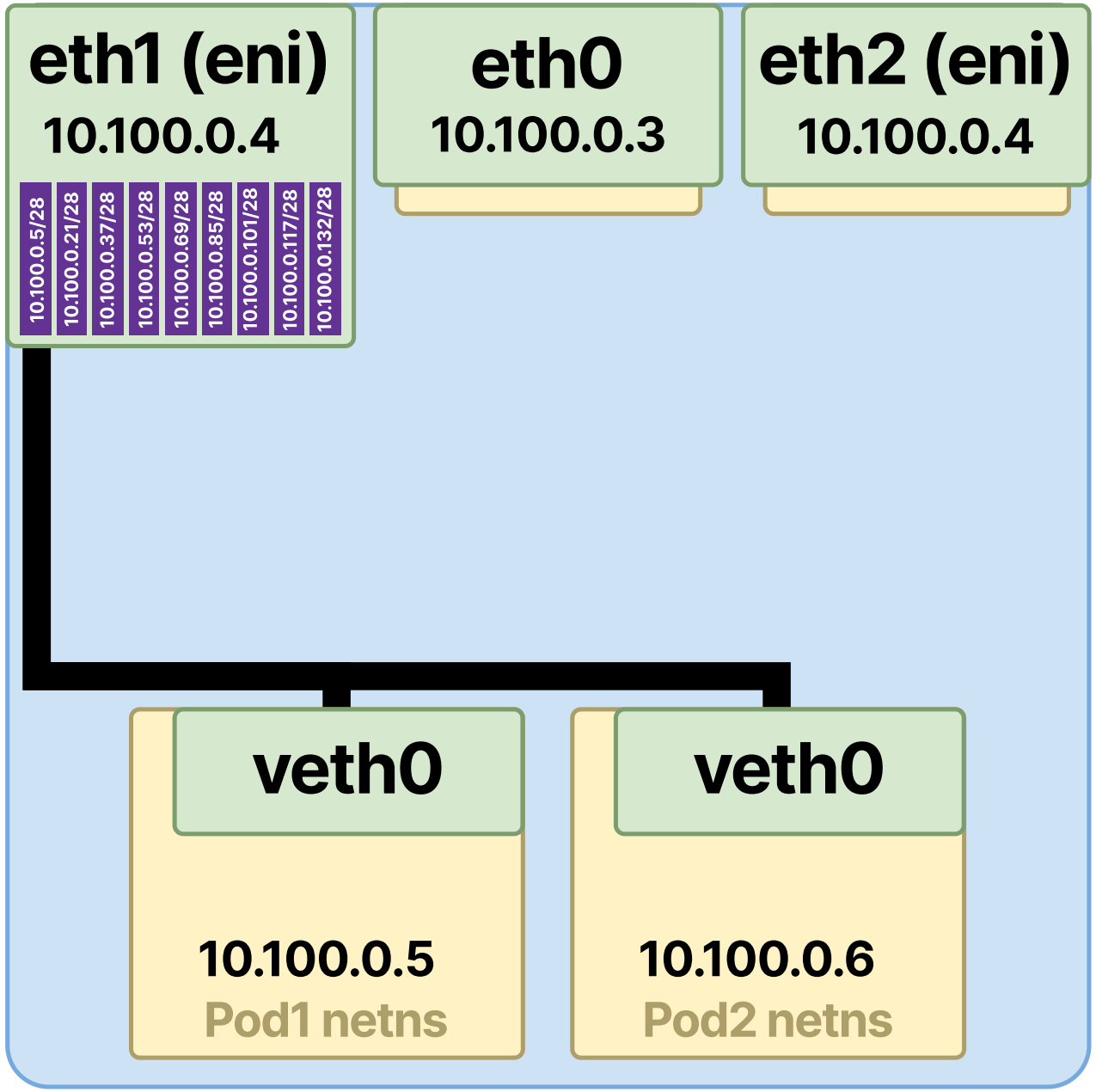


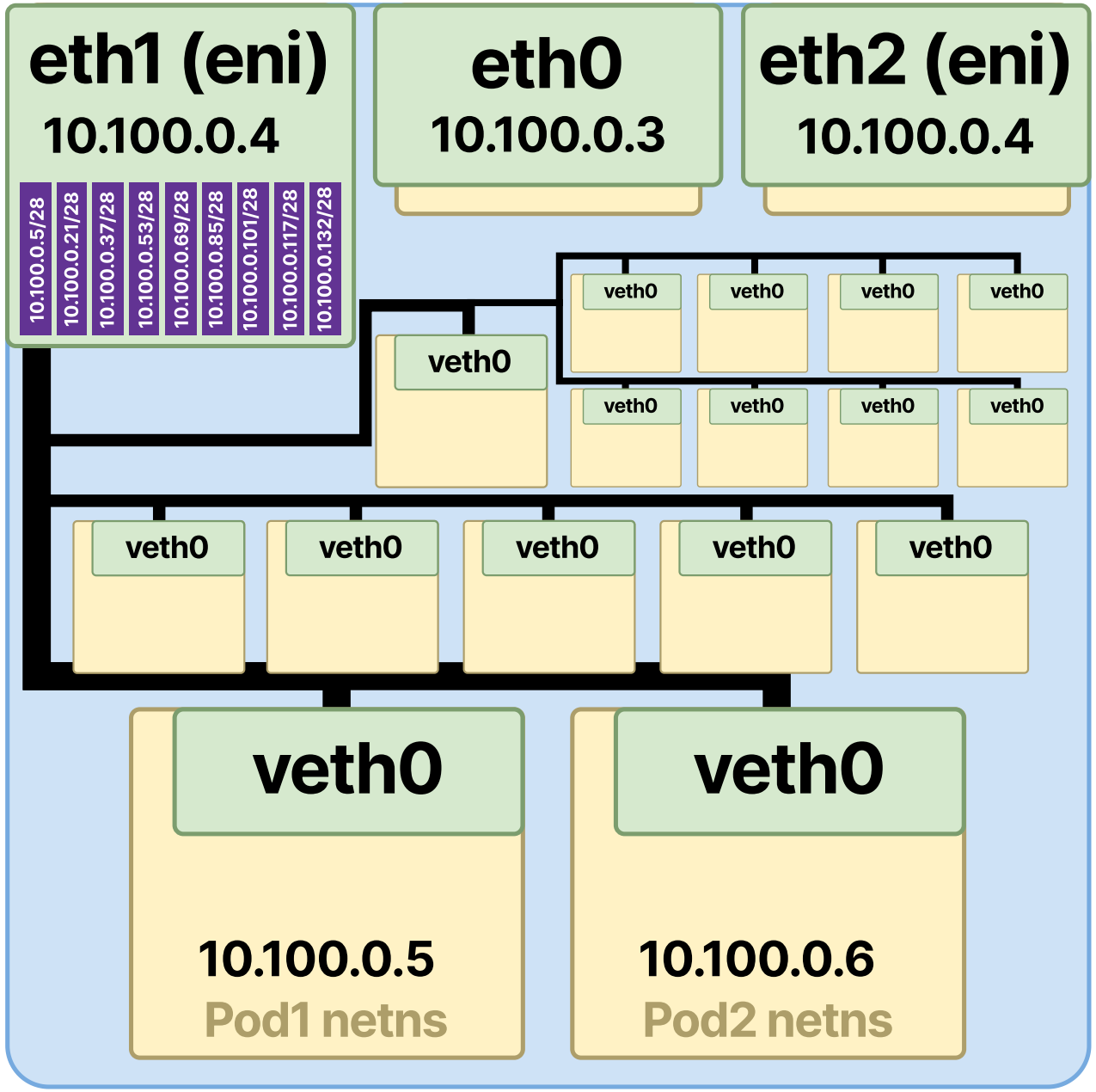
AWS-CNI

High density


eth1 (eni)
10.100.0.4
eth0
10.100.0.3
eth2 (eni)
10.100.0.2
veth0
10.100.0.5
Pod1 netns
m5.large


eth1 (eni)
10.100.0.4
10.100.0.5/28
eth0
10.100.0.3
eth2 (eni)
10.100.0.4
veth0
10.100.0.5
Pod1 netns
m5.large


eth1 (eni)
10.100.0.4
10.100.0.5/28
10.100.0.21/28
10.100.0.37/28
10.100.0.53/28
10.100.0.69/28
10.100.0.85/28
10.100.0.101/28
10.100.0.117/28
10.100.0.132/28
eth0
10.100.0.3
eth2 (eni)
10.100.0.4
veth0
10.100.0.5
Pod1 netns
veth0
10.100.0.6
Pod2 netns
m5.large


eth1 (eni)
10.100.0.4
10.100.0.5/28
10.100.0.21/28
10.100.0.37/28
10.100.0.53/28
10.100.0.69/28
10.100.0.85/28
10.100.0.101/28
10.100.0.117/28
10.100.0.132/28
eth0
10.100.0.3
eth2 (eni)
10.100.0.4
veth0
veth0
veth0
veth0
veth0
veth0
veth0
veth0
veth0
veth0
veth0
veth0
veth0
veth0
veth0
10.100.0.5
Pod1 netns
10.100.0.6
Pod2 netns

m5.large

AWS-CNI — High density

More pods per node

Prefix allocates 16 IPs

Nitro & >=1.9.0 only

Reserve IPs

Multiple CIDR ranges

AWS-CNI — High density

More pods per node

Prefix allocates 16 IPs

Nitro & >=1.9.0 only

Reserve IPs

Multiple CIDR ranges

AWS-CNI — High density

More pods per node

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Multiple CIDR ranges

AWS-CNI — High density

More pods per node

Prefix allocates 16 IPs

Nitro & >=1.9.0 only

Reserve IPs

Multiple CIDR ranges

AWS-CNI — High density

More pods per node

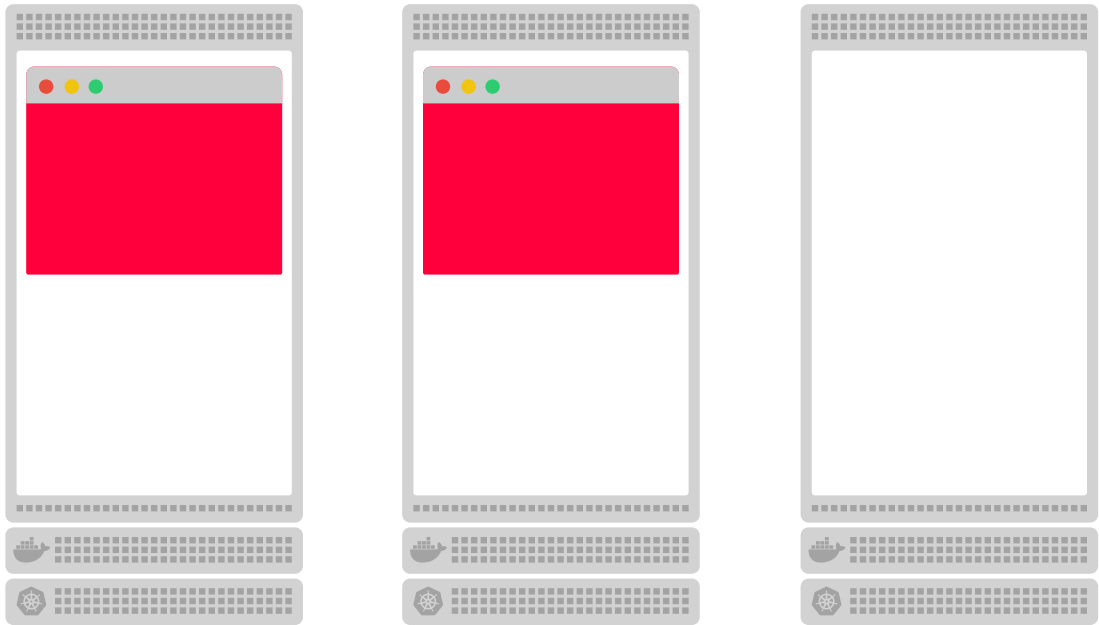
Prefix allocates 16 IPs

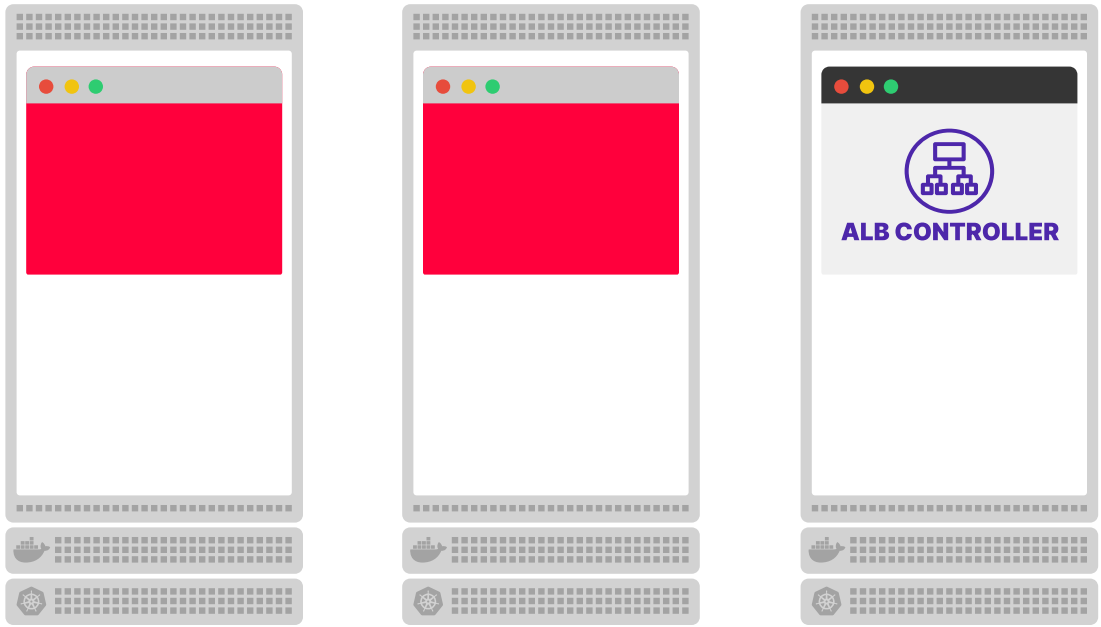
Nitro & >=1.9.0 only

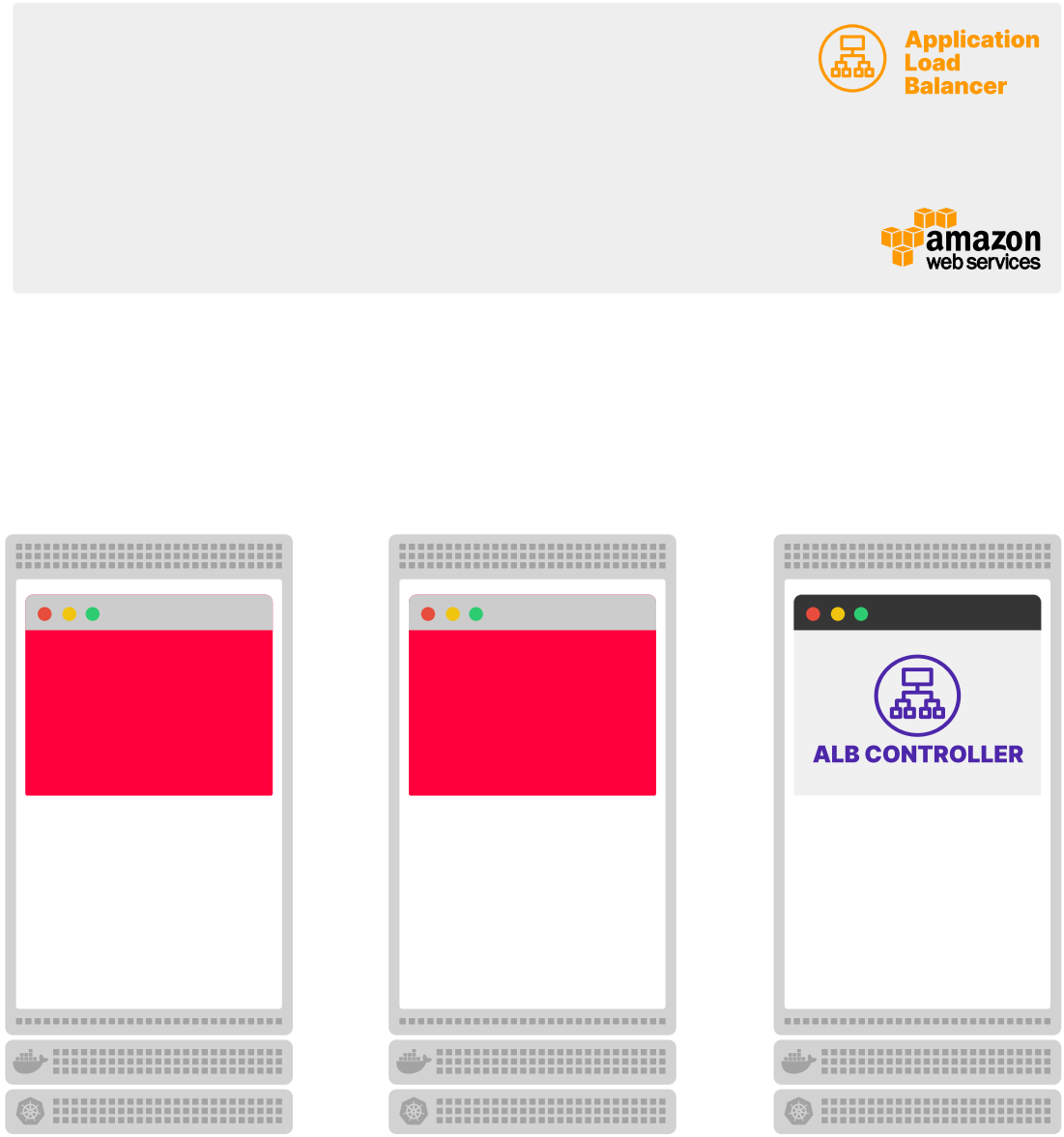
Reserve IPs

Multiple CIDR ranges

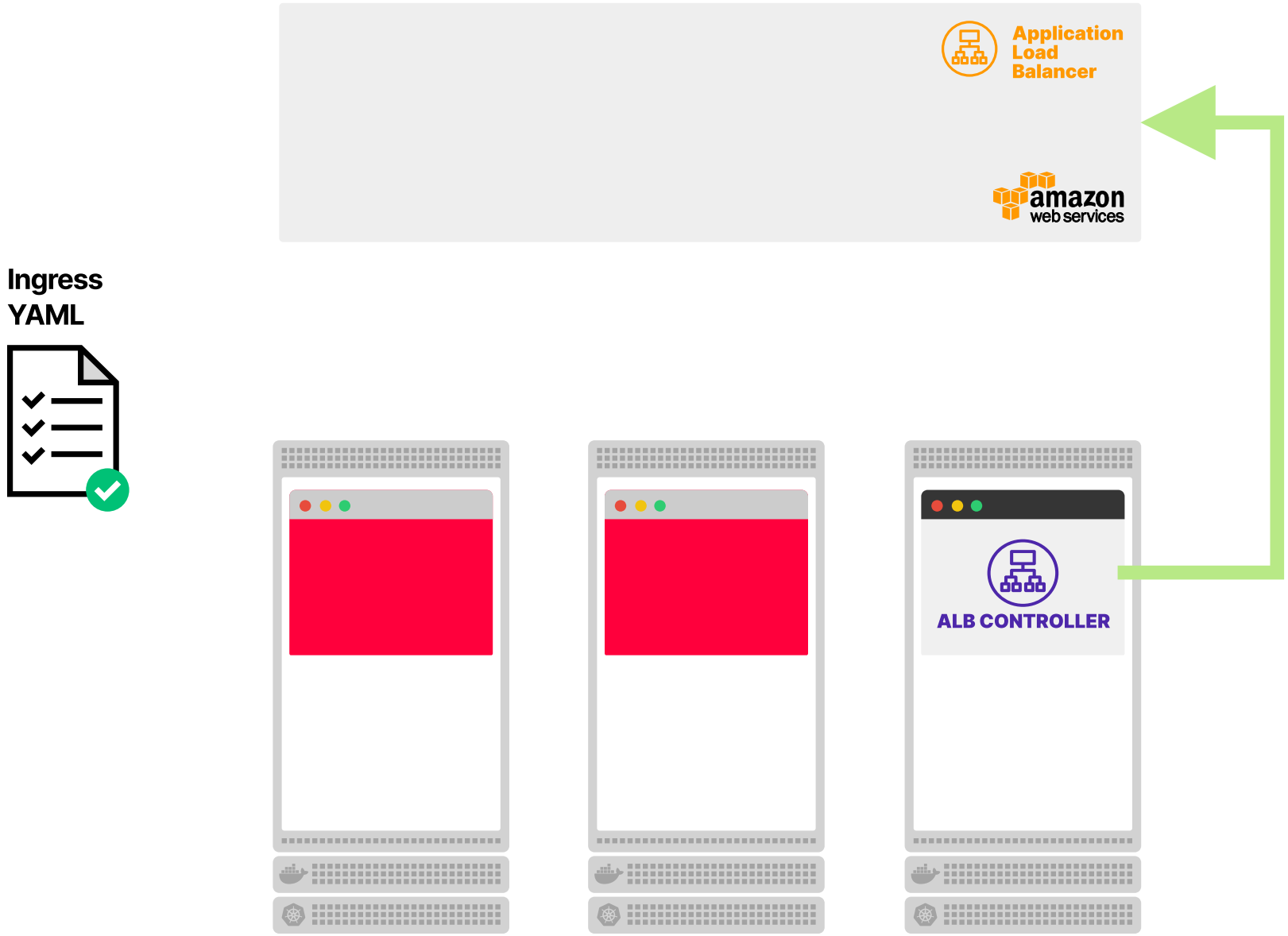
AWS Load Balancer Controller

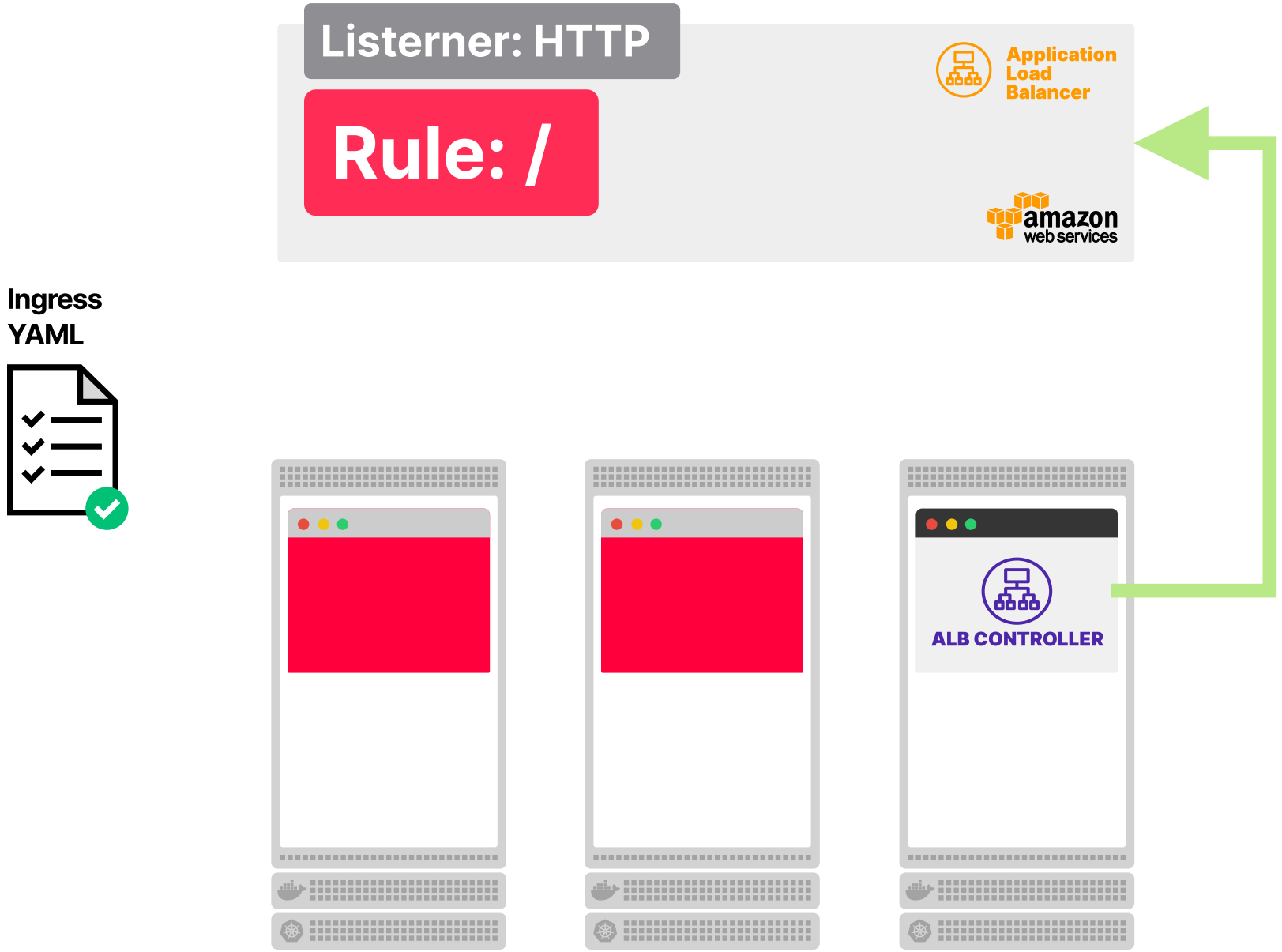


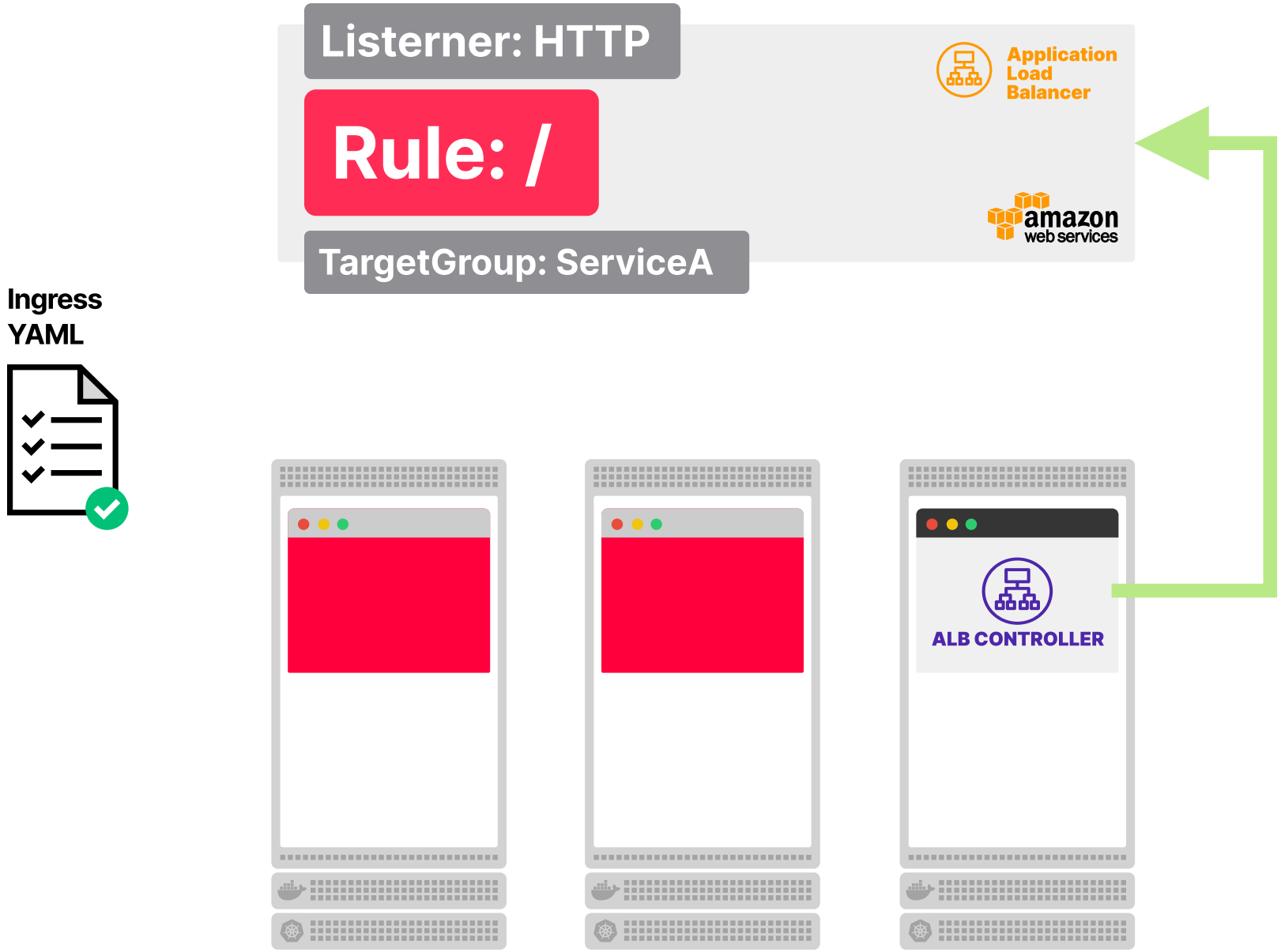

ALB CONTROLLER


Application
Load
Balancer
amazon
web services
ALB CONTROLLER

```
~$ cat ingress.yaml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: app
spec:
  rules:
    - http:
        paths:
          - backend:
              service:
                name: app
                port:
                  number: 80
            path: /
            pathType: Prefix
```

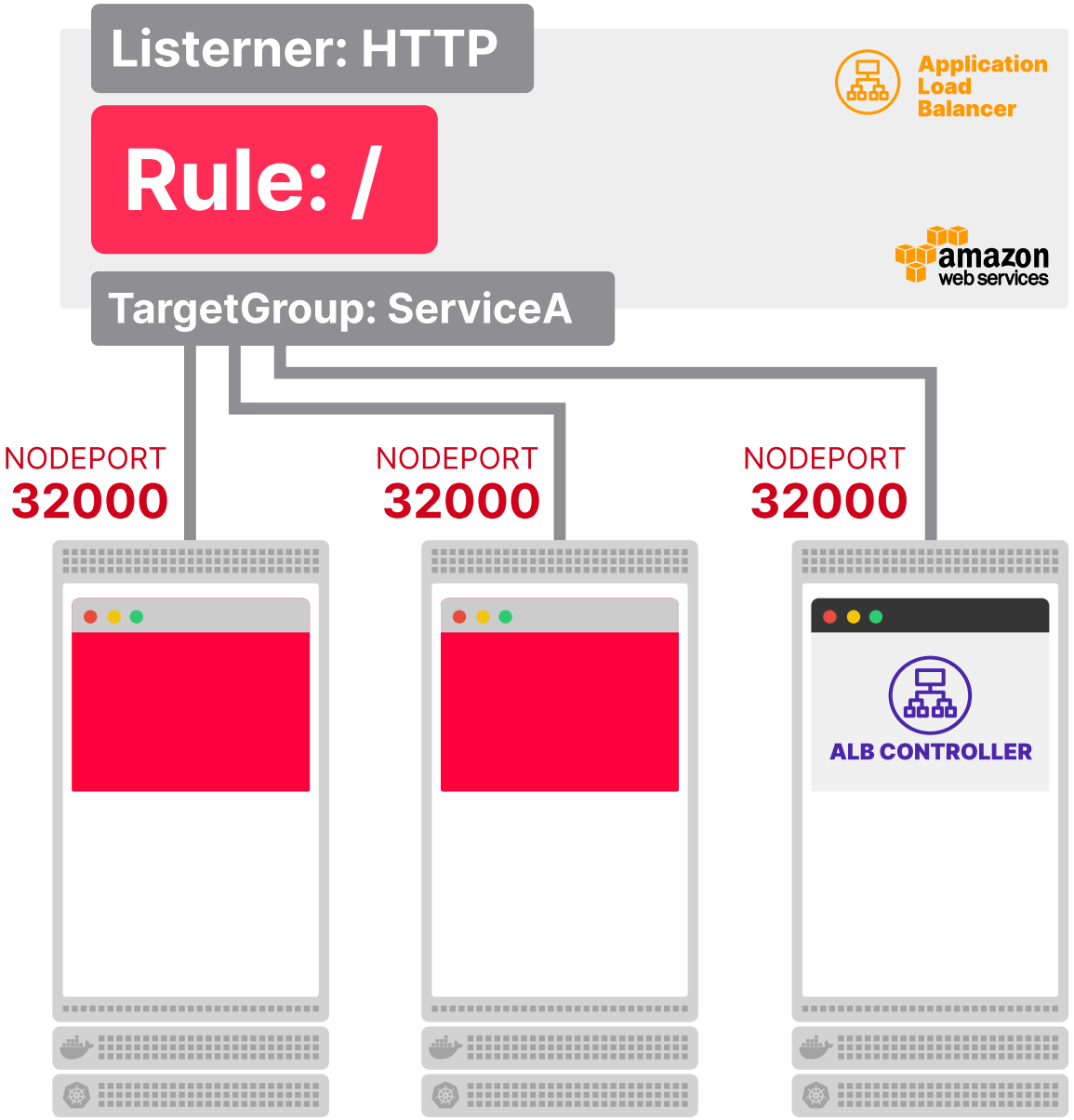

Application
Load
Balancer
amazon
web services
Ingress
YAML
ALB CONTROLLER


Listerner: HTTP
Rule: /
Application Load Balancer
amazon web services
Ingress YAML
ALB CONTROLLER


Listerner: HTTP
Rule: /
TargetGroup: ServiceA
Application Load Balancer
amazon web services
Ingress YAML
ALB CONTROLLER

1

INSTANCE MODE



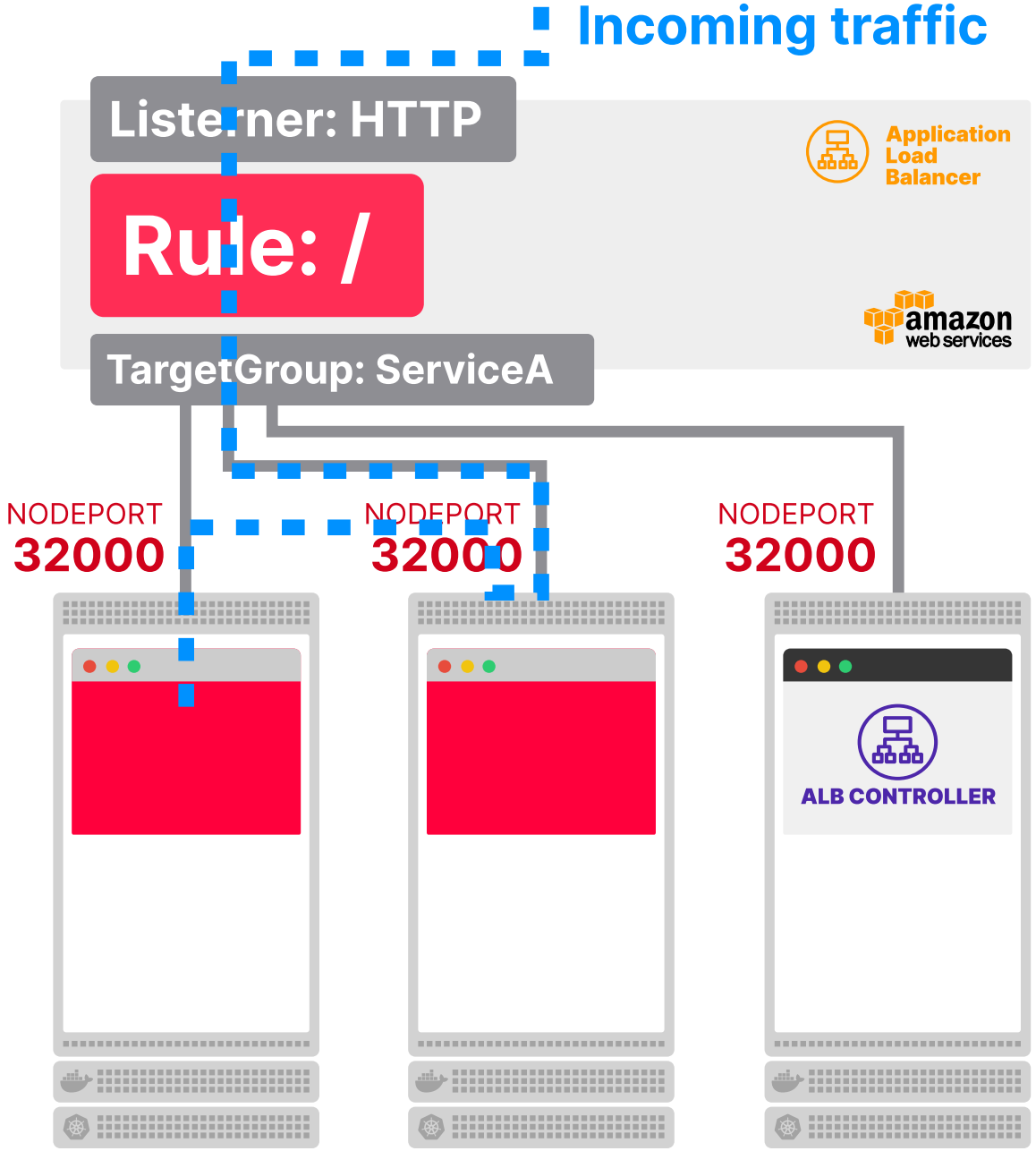
1

INSTANCE MODE



1

INSTANCE MODE



2

IP MODE

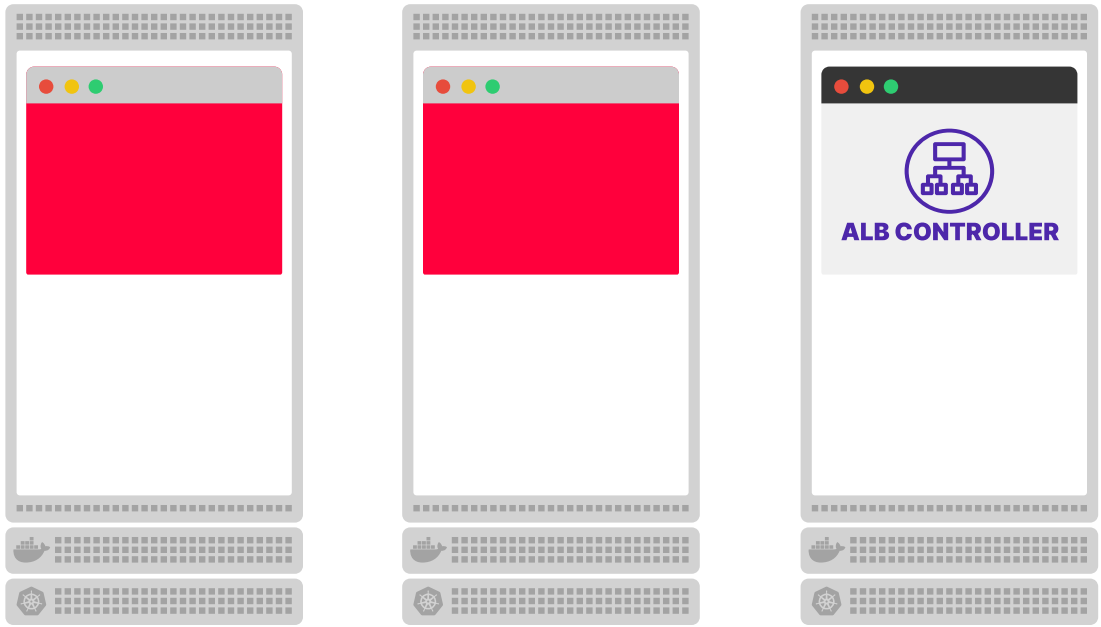
Listerner: HTTP

Rule: /

TargetGroup: ServiceA

Application Load Balancer

amazon web services



2

IP MODE

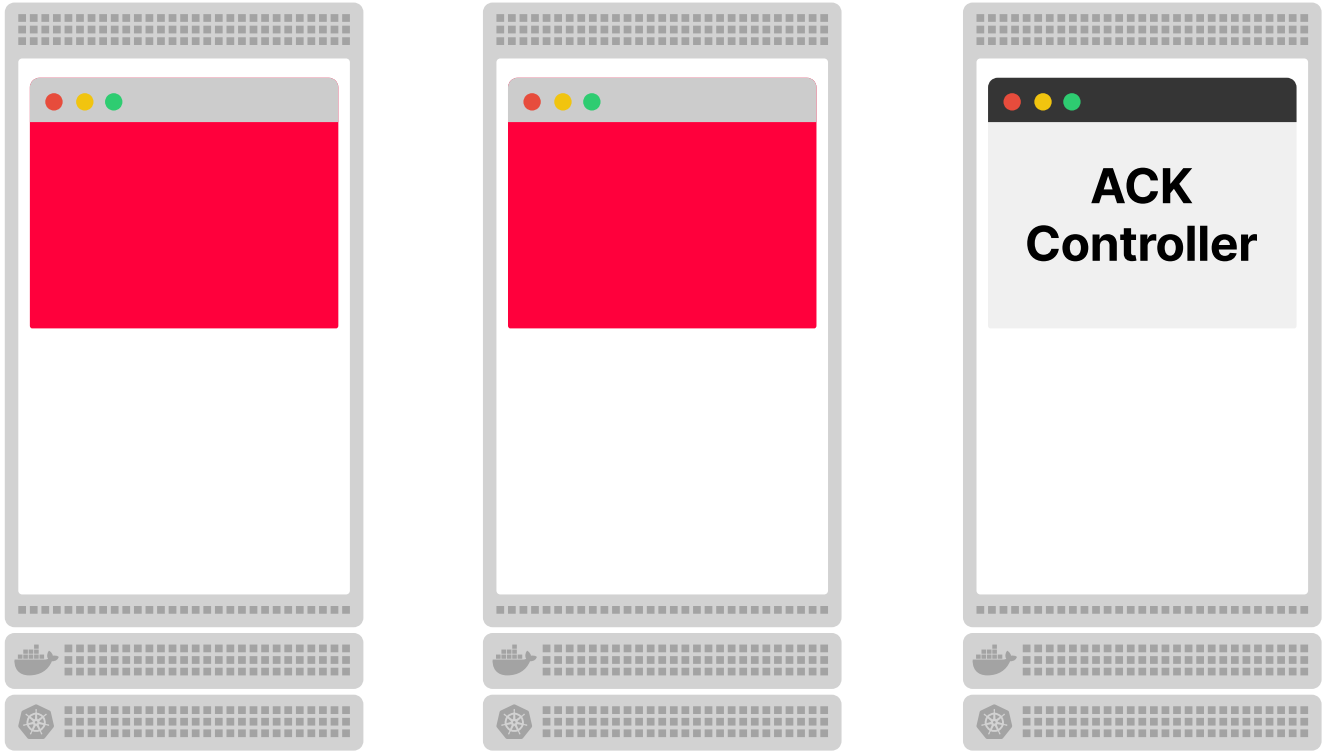

Listerner: HTTP
Rule: /
TargetGroup: ServiceA
Application Load Balancer
amazon web services
ALB CONTROLLER

AWS Controllers for Kubernetes

```
~$ cat s3.yaml
apiVersion: s3.services.k8s.aws/v1alpha1
kind: Bucket
metadata:
  name: my-bucket
spec:
  name: s3-bucket
  createBucketConfiguration:
    locationConstraint: ap-southeast-1
```

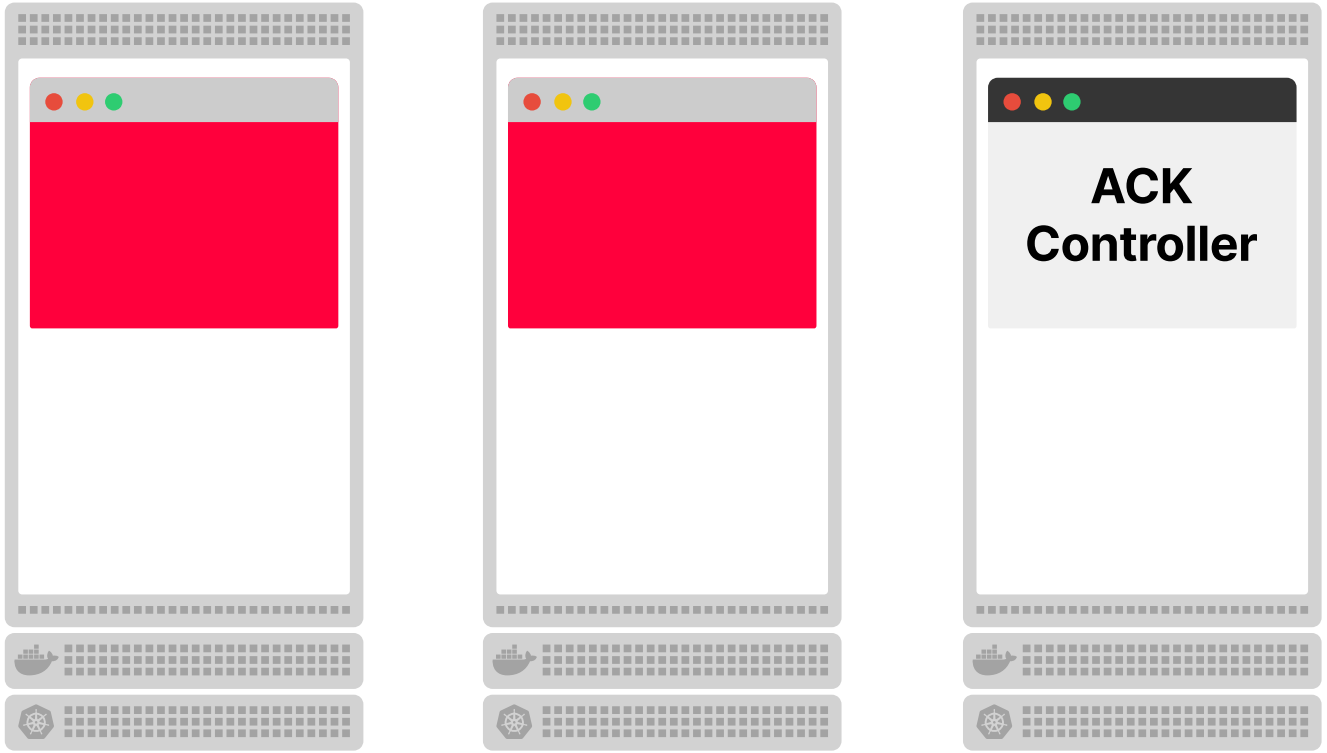
```
~$ cat dynamo.yaml
apiVersion: dynamodb.services.k8s.aws/v1alpha1
kind: Table
metadata:
  name: my-table
spec:
  tableName: my-table
  provisionedThroughput:
    readCapacityUnits: 1000
    writeCapacityUnits: 1000
```

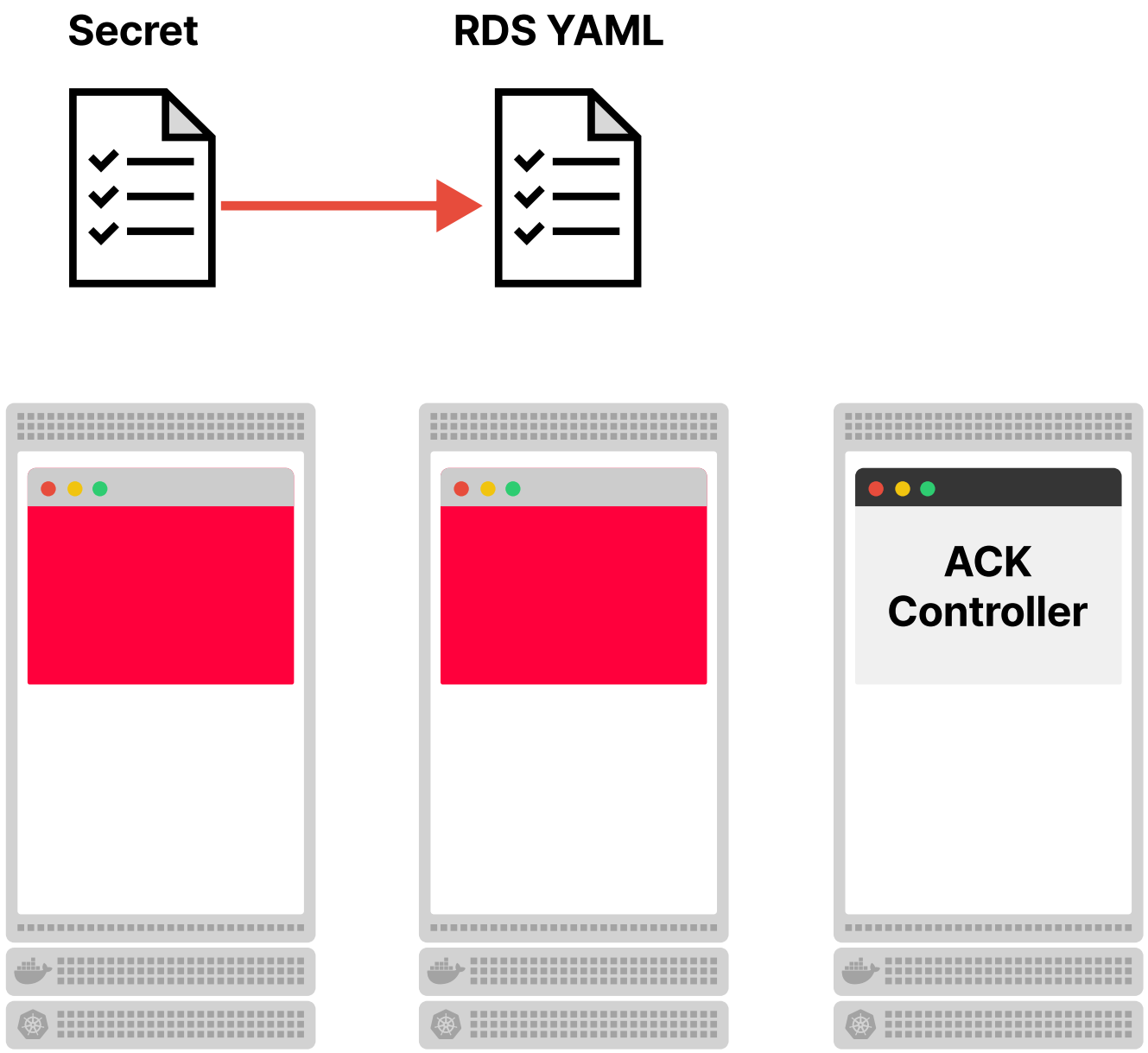
```
~$ cat rds.yaml
apiVersion: rds.services.k8s.aws/v1alpha1
kind: DBInstance
metadata:
  name: my-bucket
spec:
  allocatedStorage: 1
  autoMinorVersionUpgrade: false
  backupRetentionPeriod: 10
  characterSetName: utc-8
  copyTagsToSnapshot: true
  dbClusterIdentifier: RDS
  dbInstanceClass: MySQL
  dbInstanceIdentifier: MySQL
```

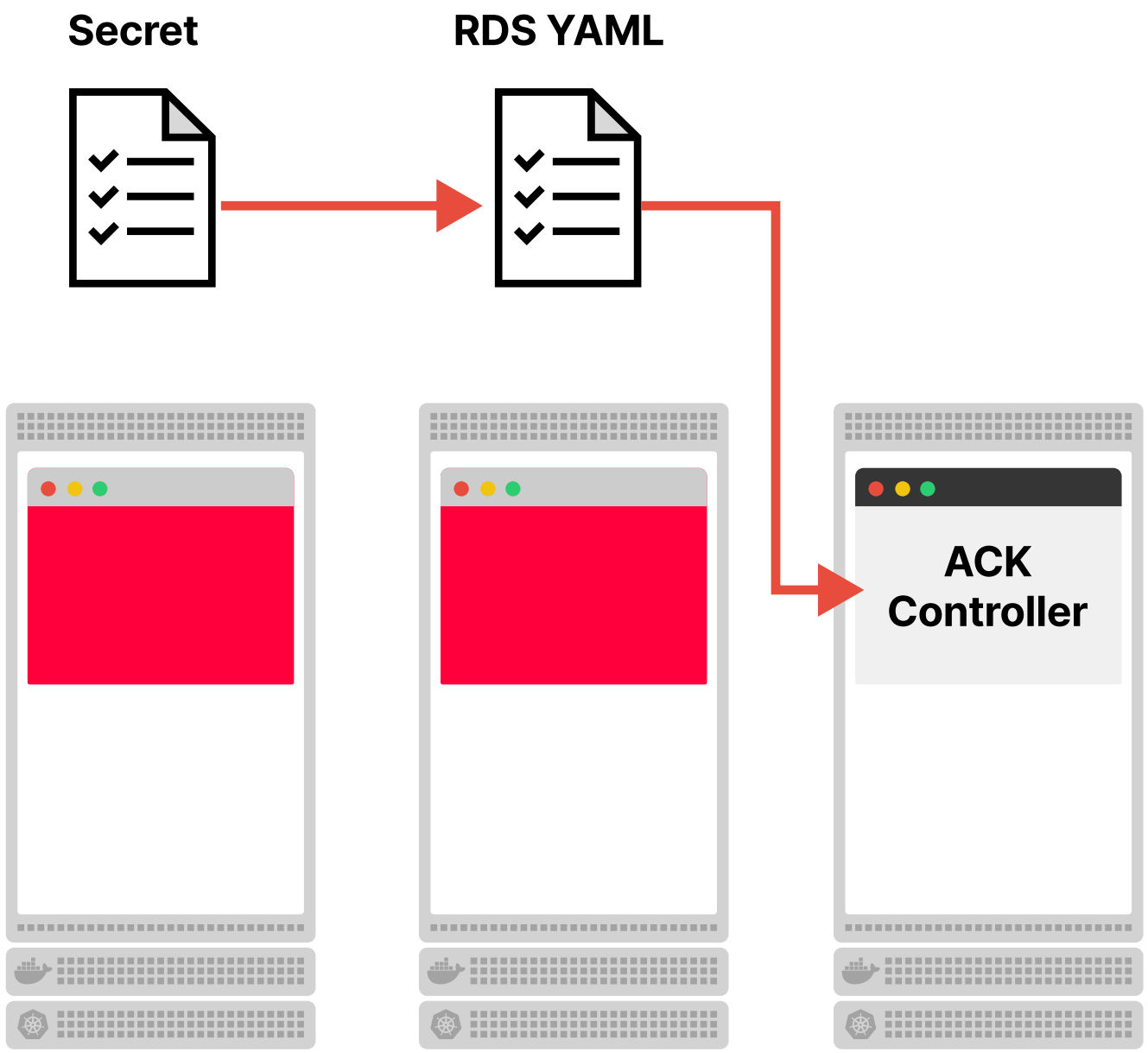

ACK
Controller

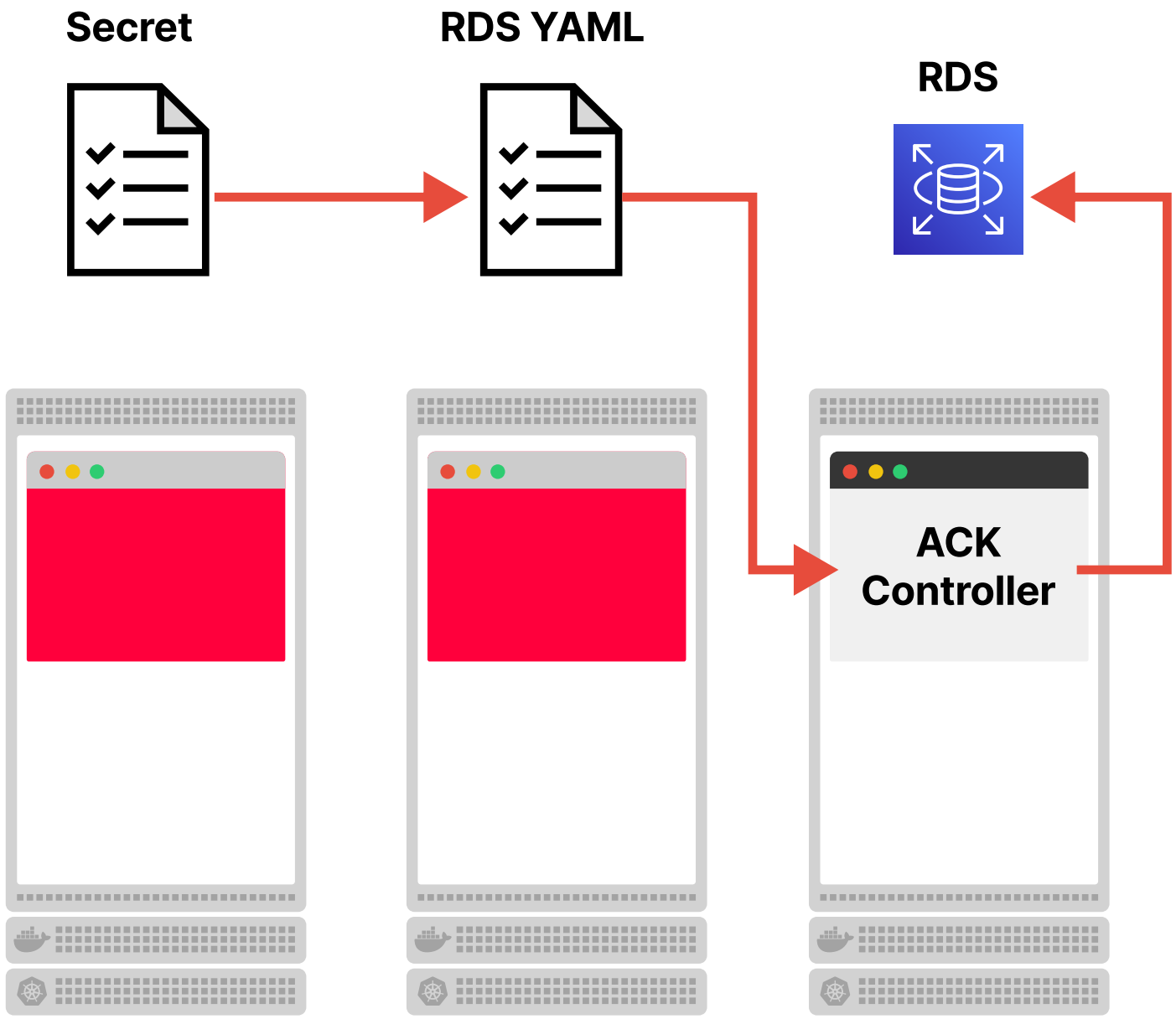
RDS YAML






Secret
RDS YAML
ACK
Controller


Secret
RDS YAML
ACK
Controller

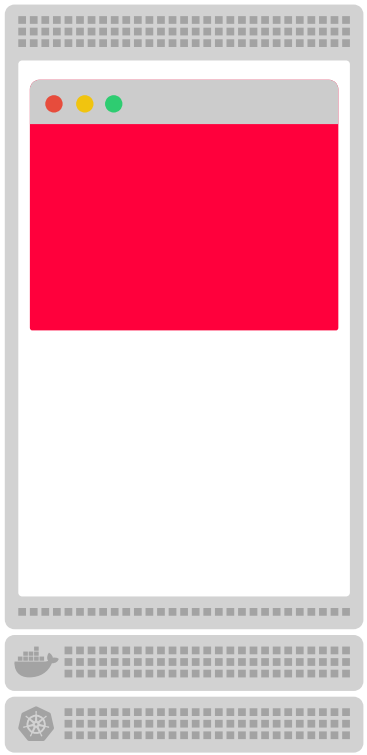

Secret
RDS YAML
RDS
ACK
Controller

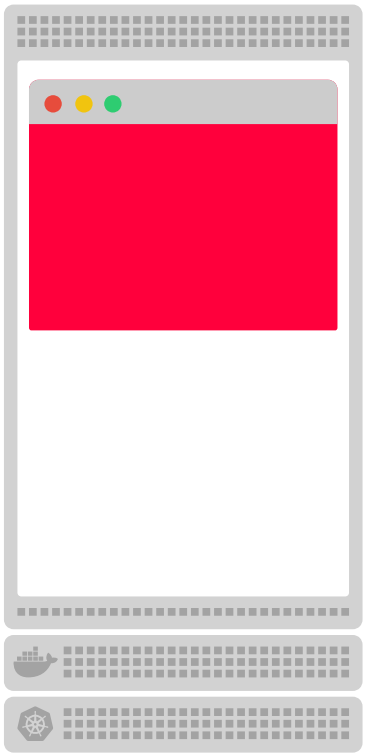
Secret



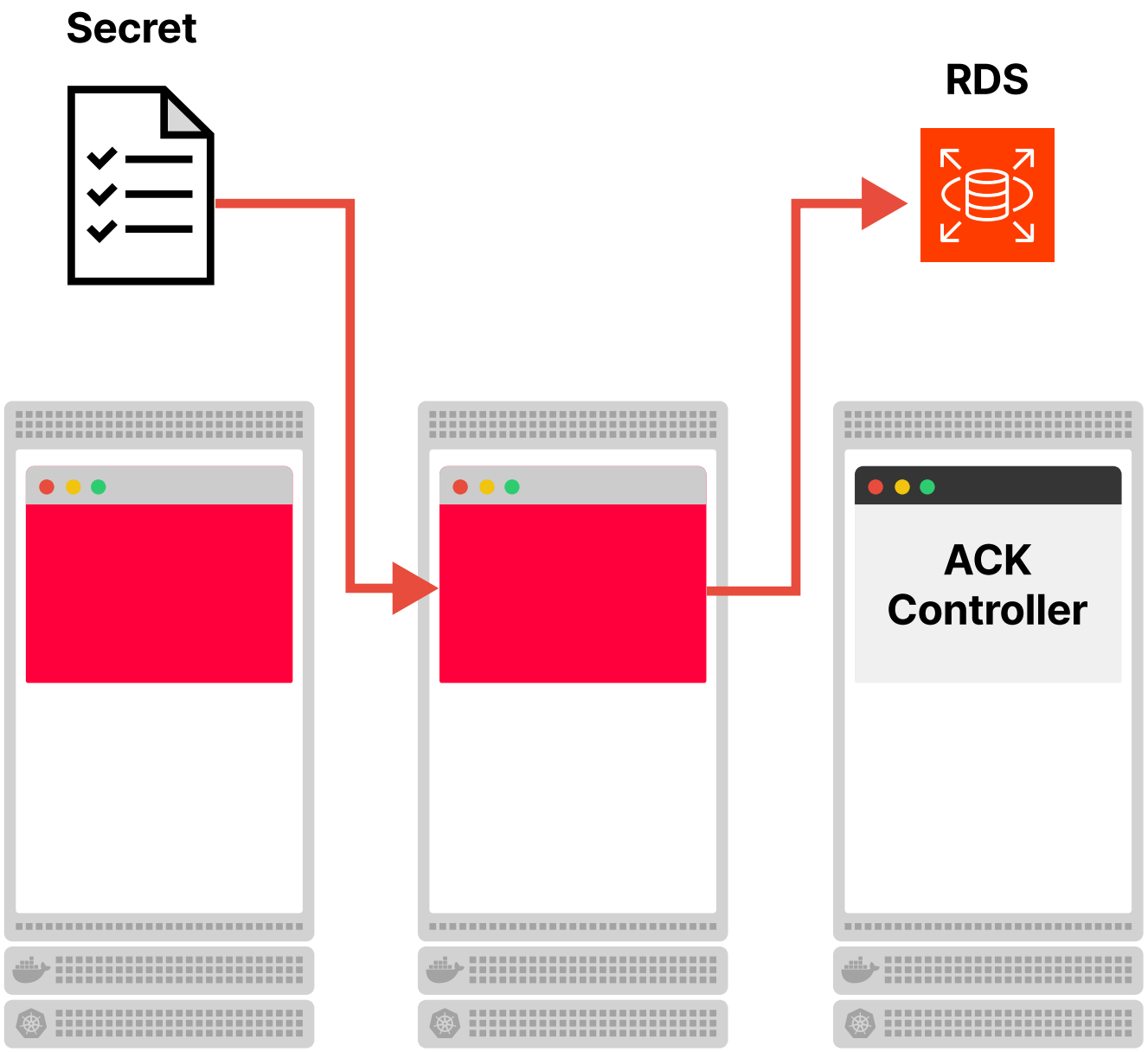
RDS










Secret
RDS
ACK
Controller

AWS controllers for Kubernetes

Read-only role → IAM
Write-only role → IAM

AWS Controllers for Kubernetes



Thanks.